

Group Representation Theory

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1 Introduction

§1.1 Examples Realizing Groups as Matrices

Example 1.1.1

Given a group G , the **trivial representation** sends every $g \in G$ the matrix $\begin{pmatrix} 1 \end{pmatrix}$.

Example 1.1.2

Let $G = C_n$, $\{1, g, \dots, g^{n-1}\}$. Given an n -th root of unity ζ , with $\zeta^n = 1$, we have a representation:

$$\rho(g^m) = \begin{pmatrix} \zeta^m \end{pmatrix}$$

This is well-defined, since

$$\rho(g^n) = \begin{pmatrix} \zeta^n \end{pmatrix} = \begin{pmatrix} 1 \end{pmatrix} = \rho(1).$$

Particularly, we get that this preserves multiplication:

$$\rho(g^i \cdot g^j) = \rho(g^{i+j}) = \begin{pmatrix} \zeta^{i+j} \end{pmatrix} = \begin{pmatrix} \zeta^i \end{pmatrix} \begin{pmatrix} \zeta^j \end{pmatrix} = \rho(g^i) \rho(g^j).$$

Example 1.1.3

Let $G = S_n$. Representations send $\sigma \in S_n$ to a permutation matrix $\rho(\sigma)$. For instance, if

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix},$$

its representation is

$$\rho(\sigma) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Example 1.1.4

Let $G = S_n$. We can attain a 1×1 -matrix representation for a permutation σ given by

$$\rho(\sigma) = \begin{pmatrix} \text{sign}(\sigma) \end{pmatrix}.$$

Example 1.1.5

Recall that

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

is a counter-clockwise rotation by θ . Also,

$$\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$$

is a reflection through a line at angle θ counter-clockwise from the x -axis. Using these, we can write down a natural representation of the dihedral group D_n . Note that elements of the dihedral group can be classified into rotations and reflections.

Example 1.1.6

Continuing from the above, a representation of D_4 is as follows: rotations give

$$I_4, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and reflections give

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

§1.2 Formal Definitions

Given a vector space V , we have $GL(V)$, the general linear group on V . It consists of the invertible linear transformations of V . Given a basis $v_1, v_2, \dots, v_n$ of V , we can put $GL(V)$ in correspondence with GL_n , the group of $n \times n$ invertible matrices.

Definition 1.2.1. A **representation** of a group G is a pair (V, ρ) , where V is a vector space and $\rho: G \rightarrow GL(V)$ is a group homomorphism.

Remark 1.2.2 — The point of this is that we can represent a group as a collection of invertible matrices.

Definition 1.2.3. A representation (V, ρ) is **faithful**, if ρ is injective.

Remark 1.2.4 — Recall that an action of G on X is a map $G \times X \rightarrow X$, denoted as $g \cdot x \mapsto g \cdot x$, such that $\forall x, e \cdot x = x$, and $\forall g, h, x$, we have $(gh) \cdot x = g \cdot (h \cdot x)$. We observe that a representation (V, ρ) of G provides an action of G on V : for $g \in G, v \in V$, $g \cdot v = \rho(g)v$. The resulting action is linear, in the sense that $\forall g, v \mapsto g \cdot v$ is a linear map.

Let (V, ρ) and (V', ρ') be representation of a group G .

Definition 1.2.5. A **homomorphism of representations** from (V, ρ) to (V', ρ') is a linear map $T: V \rightarrow V'$, such that $\forall g \in G, T \circ \rho(g) = \rho'(g) \circ T$. This gives the following commutative diagram:

$$\begin{array}{ccc} V & \xrightarrow{\rho(g)} & V \\ \downarrow T & & \downarrow T \\ V' & \xrightarrow{\rho'(g)} & V' \end{array}$$

An isomorphism of representations is when T is invertible, and, in this case, we say that (V, ρ) and (V', ρ') are isomorphic.

Now suppose we have bases $\mathcal{B}$ of V and $\mathcal{B}'$ of V' , and a linear map $f: V \rightarrow V'$, with notation $[f]_{\mathcal{B}, \mathcal{B}'}$ for matrix of f .

Notation 1.2.6. Given (V, ρ) with $\dim V = n$ and a basis $\mathcal{B}$ of V , let $\rho^\beta: G \rightarrow GL_n(\mathbb{C})$ be the map $\rho^\beta(g) = [\rho(g)]_{\mathcal{B}}$.

Lemma 1.2.7

Given bases $\mathcal{B}, \mathcal{B}'$ for V , $\rho^{\mathcal{B}'}(g) = P^{-1} \rho^\beta(g) P$, where $P = [I]_{\mathcal{B}', \mathcal{B}}$.

Proof. $\rho^{\mathcal{B}'}(g) = [\rho(g)]_{\mathcal{B}', \mathcal{B}'} = [I]_{\mathcal{B}, \mathcal{B}'} [\rho(g)]_{\mathcal{B}, \mathcal{B}} [I]_{\mathcal{B}', \mathcal{B}}$. We set $P = [I]_{\mathcal{B}', \mathcal{B}}$ and $P^{-1} = [I]_{\mathcal{B}, \mathcal{B}'}$, so

$$P^{-1} [\rho(g)]_{\mathcal{B}, \mathcal{B}} P = P^{-1} \rho^\beta(g) P.$$

□

Lemma 1.2.8

Let $(V, \rho), (V', \rho')$ be two representations of a group G , with bases $\mathcal{B}, \mathcal{B}'$, respectively. A linear map $T: V \rightarrow V'$ is a homomorphism of representations if and only if $\forall g \in G$,

$$[T]_{\mathcal{B}, \mathcal{B}'} \rho^\beta(g) = (\rho')^{\mathcal{B}'}(g) [T]_{\mathcal{B}, \mathcal{B}'}$$

Further, T is an isomorphism of representations if $[T]_{\mathcal{B}, \mathcal{B}'}$ is invertible, so that it conjugates $\rho^\beta(g)$ to $(\rho')^{\mathcal{B}'}(g)$:

$$(\rho')^{\mathcal{B}'}(g) = [T]_{\mathcal{B}, \mathcal{B}'}^{-1} \rho^\beta(g) [T]_{\mathcal{B}, \mathcal{B}'}$$

Proof. This follows immediately from the linear algebra rule for matrices in terms of bases. □

Definition 1.2.9. $\rho, \rho': G \rightarrow GL_n$ are **equivalent** if $\exists P \in GL_n, \forall g, \rho'(g) = P^{-1} \rho(g) P$.

Proposition 1.2.10

Let $(V, \rho), (V', \rho')$ be two finitely dimensional representations. They are isomorphic if and only if

- $\dim V = \dim V'$
- For all bases $\mathcal{B}$ of V and $\mathcal{B}'$ of V' , ρ^β and $(\rho')^{\mathcal{B}'}$ are equivalent.

Proof. First, it is obvious that the two representations can be isomorphic only if they have the same dimension. Now the statement follows from lemma 1.2.8 $\square$

Exercise 1.2.11. Representations of a cyclic group $G = \langle g \rangle$ are in bijection with

1. pairs (V, T) , where V is a vector space and $T \in GL(V)$ (G is infinite case, so $G \cong \mathbb{Z}$).
2. pairs (V, T) , where V is a vector space, $T \in GL(V)$ such that $T^n = I$. (G is finite case, so $G \cong C_n$).

Example 1.2.12 (Representation of C_4)

We have a representation of C_4 :

$$\begin{aligned}\rho(1) &= I \\ \rho(g) &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = T \\ \rho(g^2) &= \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = T^2 \\ \rho(g^3) &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = T^3\end{aligned}$$

§1.3 Representations from Group Actions

Let X be a finite set.

Notation 1.3.1. $\mathbb{C}[X]$ is the vector space of **formal linear combinations** of elements of X :

$$\sum_{x \in X} a_x x,$$

where for all x , $a_x \in \mathbb{C}$ and addition and scalar multiplication is done coefficient wise.

Example 1.3.2

For $X = \{x_1, x_2, x_3\}$,

$$2x_1 + 3x_3$$

means $\sum_{x \in X} a_x x$, where $a_{x_1} = 2, a_{x_2} = 0, a_{x_3} = 3$.

Remark 1.3.3 — This is just notation. $\mathbb{C}[X]$ is the same as $\mathbb{C}^X$ (i.e., the set of functions from $X \rightarrow \mathbb{C}$).

Definition 1.3.4. Given a group action of G on X , we have an **induced representation** of G on $\mathbb{C}[X]$ with, $\forall g \in G$ and for all formal linear combinations $\sum_{x \in X} a_x x \in \mathbb{C}[X]$,

$$\rho(g) \left(\sum_{x \in X} a_x x \right) = \sum_{x \in X} a_x g \circ x.$$

Remark 1.3.5 — We must check that this is linear, and that it has the appropriate property that $\rho(e) = id$ and $\rho(gh) = \rho(g) \circ \rho(h)$.

Example 1.3.6

Let $X = \{1, \dots, n\}$. S_n acts on X by permutation. This group action induces the permutation representation of S_n on $\mathbb{C}^n$.

§1.4 Regular Representation

Definition 1.4.1. The representation of G on $\mathbb{C}[G]$ induced by the action of G on itself by left-multiplication is called the **left-regular representation**.

Remark 1.4.2 — The left regular representation is an example of a faithful representation. It is faithful because $\rho(g)(e) = g$, which is distinct for all $g \in G$.

Example 1.4.3

Let $G = C_n$, the cyclic group of order n . We know that we can specify a representation of G by giving a matrix T with $T^n = 1$. For the regular representation of C_n , in the basis $\mathcal{B} = \{1, g, \dots, g^{n-1}\}$,

$$\rho^{\mathcal{B}}(g) = \begin{pmatrix} 0 & 0 & 0 & \dots & 1 \\ 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & \dots & & & 0 \end{pmatrix}$$

We need to check that $\rho^{\mathcal{B}}(g) \cdot g^m = \begin{cases} g^{m+1}, & m < n-1 \\ 1, & m = n-1 \end{cases}$.

Example 1.4.4

We check the above for $n = 3$:

$$\begin{aligned} \rho(1) &= I \\ \rho(g) &= \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \\ \rho(g^2) &= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \end{aligned}$$

Notation 1.4.5. $\text{Hom}_G(\mathbb{C}[G], V)$ denotes the set of homomorphisms of representations $\mathbb{C}[G] \rightarrow V$.

Exercise 1.4.6. Let (V, ρ_V) be a representation of G . Show that:

- For every vector $v \in V$, there is a unique homomorphism of representation $\mathbb{C}[G] \rightarrow V$ sending $e \in G \subseteq \mathbb{C}[G]$ to $v \in V$.
- Conversely, show that all homomorphisms are of this form.
- Deduce that there is a bijection of sets $\phi: \text{Hom}_G(\mathbb{C}[G], V) \xrightarrow{\sim} V$, $\phi(T) = T(e)$.
- Moreover, prove that this is linear, in the sense that

$$\phi(aS + bT) = a\phi(S) + b\phi(T), \quad \forall a, b \in \mathbb{C},$$

and all homomorphisms of representations $S, T: \mathbb{C}[G] \rightarrow V$.

Remark 1.4.7 — In fact, as we will explain later how $\text{Hom}_G(\mathbb{C}[G], V)$ is a vector space under addition and scalar multiplication. From the above, we've proven that ϕ is an isomorphism of vector spaces.

Proposition 1.4.8

Let V be a representation of G . Then, there is a bijection between $\text{Hom}(\mathbb{C}[G], V)$ and V given by $T \mapsto T(e)$ with inverse $T_v(g) := \rho_V(g)v$.

Proof. The preceding exercise leads to this result. □

§1.5 Subrepresentations

Definition 1.5.1. Let (V, ρ) be a representation of G . A **subrepresentation** (or a **G -invariant subspace**) of (V, ρ) is a vector subspace $W \subseteq V$, such that $\forall g \in G$, $\rho(g)W \subseteq W$.

Remark 1.5.2 — This has a natural structure as a representation of G : $(W, \rho|_W)$, defined by $\rho|_W(g) = \rho(g)|_W$. Here, we are not restricting the domain of ρ , but rather the domain of $\rho(g)$ for all $g \in G$.

Exercise 1.5.3. Prove that for W finite-dimensional, $\rho(g)(W) \subseteq W$ implies $\rho(g)W = W$.

Definition 1.5.4. A representation is **irreducible**, or **simple**, if it is nonzero and has no proper nonzero subrepresentation.

Definition 1.5.5. A representation is **reducible** if it has a proper nonzero subrepresentation.

Proposition 1.5.6

If G is finite and (V, ρ) is an irreducible representation of G , then V is finite dimensional.

Proof. Let $v \in V$ with $v \neq 0$. Consider the subrepresentation

$$W = \left\{ \sum_{g \in G} a_g \rho(g)v \mid \forall g, a_g \in \mathbb{C} \right\}.$$

We check that it's a subrepresentation. It is clear that it's a vector space and that operation on the identity provides the identity. Thus, we just need to show the multiplication property. $\forall h \in G$,

$$\rho(h) \sum_{g \in G} a_g \rho(g)v = \sum_{g \in G} a_g \rho(h)\rho(g)v = \sum_{g \in G} a_g \rho(hg)v$$

$W \neq (0)$, since $v \in W$. Since (V, ρ) is irreducible, $W = (V, \rho)$. W , by construction, is finite dimensional because it is the span of a finite number of vectors (namely, the places that v can be sent under each of the elements in the group). Therefore, it follows that V is finite dimensional as well. $\square$

Example 1.5.7

One-dimensional representations are irreducible. This is because there is no smaller nonzero dimension smaller.

Example 1.5.8

The two dimensional representations of D_n for $n \geq 3$ are irreducible.

Example 1.5.9 (Permutation Representation is not Irreducible)

Here are two subrepresentation of the permutation representation of S_n on $\mathbb{C}^n$.

1. $\text{Span}(1, 1, \dots, 1) = \{(a, a, \dots, a) \mid a \in \mathbb{C}\} \subseteq \mathbb{C}^n$. This is isomorphic to the trivial representation of S_n .
2. The **reflection representation** of S_n , $\{(a_1, a_2, \dots, a_n) \mid a_1 + a_2 + \dots + a_n = 0\} \subset \mathbb{C}^n$ ($n - 1$ -dimensional).

Proposition 1.5.10

Let $T: (V, \rho) \rightarrow (V', \rho')$ be a homomorphism of G -representations. Then, $\ker(T)$ and $\text{Im}(T)$ are subrepresentations.

Proof. Let $v \in \ker(T)$. Let $g \in G$.

$$T(\rho(g)v) = \rho'(g)T(v) = \rho'(g)0 = 0,$$

so $\rho(g)v \in \ker(T)$. Next, let $v \in \text{Im}(T)$. Then, $\exists w \in V$ such that $v = T(w)$. Let $g \in G$. Then,

$$\rho'(g)v = \rho'(g)T(w) = T(\rho(g)w) \in \text{Im}(T).$$

$\square$

Remark 1.5.11 — An alternative proof to proposition 1.5.6 is to construct the desired subrepresentations as the image of $\mathbb{C}[G]$ under one of the homomorphisms constructed in (example 1.4.6).

Example 1.5.12

Let $(\mathbb{C}^n, \rho)$ be the permutation representation of S_n . Define $T: (\mathbb{C}^n, \rho) \rightarrow (\mathbb{C}^n, \rho)$, where

$$T(a_1, a_2, \dots, a_n) = \left(\sum_{i=1}^n a_i \right) (1, 1, \dots, 1).$$

This is a homomorphism of representations. Let $\sigma \in S_n$. Then,

$$\begin{aligned} T(\rho(\sigma)(a_1, \dots, a_n)) &= T(a_{\sigma(1)}, \dots, a_{\sigma(n)}) \\ &= \left(\sum_{i=1}^n a_{\sigma(i)} \right) (1, 1, \dots, 1) \\ &= \left(\sum_{i=1}^n a_i \right) (1, 1, \dots, 1) = T(a_1, \dots, a_n) \\ &= \rho(\sigma)T(a_1, \dots, a_n). \end{aligned}$$

This show that T is, indeed, a homomorphism of representations. Then, we have the following:

$$\begin{aligned} \text{Im}(T) &= \{(a, a, \dots, a) \mid a \in \mathbb{C}\} = \text{Span}(1, 1, \dots, 1) \\ \ker(T) &= \{(a_1, a_2, \dots, a_n) \mid a_1 + a_2 + \dots + a_n = 0\} \end{aligned}$$

Definition 1.5.13. Let (V, ρ) be a representation of G and $W \subseteq V$ be a subrepresentation. V/W has a natural structure as a representation of G , the **quotient representation** $(V/W, \rho_{V/W})$, defined by

$$\rho_{V/W}(g)(v + W) = \rho_V(g)v + W.$$

We check that the above is well defined. For $g \in G, v \in V, w \in W$, we have

$$\begin{aligned} \rho_{V/W}(g)(v + w + W) &= \rho_V(g)(v + w) + W = \rho_V(g)(v) + \underbrace{\rho_V(g)w}_{\in W} + W \\ &= \rho_V(g)(v) + W = \rho_{V/W}(g)v. \end{aligned}$$

We also check that this is a representation. For $g, h \in G, v \in V$, we have

$$\begin{aligned} \rho_{V/W}(gh)(v + W) &= \rho_V(gh)v + W = \rho_V(g)\rho_V(h)v + W \\ &= \rho_{V/W}(g)[\rho_V(h)v + W] = \rho_{V/W}(g)[\rho_{V/W}(h)[v + W]]. \end{aligned}$$

Proposition 1.5.14

Given $T: (V, \rho) \rightarrow (V', \rho')$ a homomorphism of representations, we have a canonical isomorphism $\bar{T}: (V/\ker(T), \rho_{V/\ker(T)}) \xrightarrow{\sim} \text{Im}(T)$, defined by

$$\bar{T}(v + \ker(T)) := T(v).$$

Proof. To check this, $\bar{T}$ is a homomorphism of representations, let $g \in G$, $v \in V$. Then,

$$\begin{aligned}\bar{T}(\rho_{V/\ker(T)}(g)[v + \ker(T)]) &= \bar{T}(\rho_V(g)(v) + \ker(T)) \\ &= T(\rho_V(g)(v)) = \rho'(g)T(v) \\ &= \rho'(g)\bar{T}(v + \ker(T)).\end{aligned}$$

□

Definition 1.5.15. Let (V, W) be representations of G . A linear map $T: V \rightarrow W$ is **G -linear**, if it is a homomorphism of representations.

Notation 1.5.16. For vector space V, W , $\text{Hom}(V, W)$ denotes linear maps from V to W , considered as a vector space.

Notation 1.5.17. For vector spaces V, W , $\text{End}(V)$ denotes $\text{Hom}(V, V)$.

Lemma 1.5.18

For $T \in \text{End}(V)$, the following are equivalent:

- $T^2 = T$
- $\forall w \in \text{Im}(T), T(w) = w$.

Further, in this case we have $V = \ker(T) \oplus \text{Im}(T)$

Proof. Suppose $T^2 = T$. Let $w \in \text{Im}(T)$. Then, $\exists v$ such that $w = T(v)$. Thus,

$$T(w) = T(T(v)) = T(v) = w.$$

For the converse, suppose $\forall w \in \text{Im}(T), T(w) = w$. Then, let $v \in V$. Then,

$$T(\underbrace{T(v)}_{\in \text{Im}(T)}) = T(v),$$

so $T^2 = T$.

We now prove the second part of the lemma. We first need to show $\ker(T) \cap \text{Im}(T) = (0)$. Let $v \in \ker(T) \cap \text{Im}(T)$. Then, $\exists w$ such that $v = T(w)$. Then, as $v \in \ker(T)$, we have

$$0 = T(v) = T(T(w)) = T(w) = v.$$

Next, we need to show that $\ker(T) + \text{Im}(T) = V$. Let $v \in V$. Then,

$$T(v - T(v)) = T(v) - T(T(v)) = T(v) - T(v) = 0,$$

implying that $v - T(v) \in \ker(T)$. Therefore,

$$v = \underbrace{(v - T(v))}_{\in \ker(T)} + \underbrace{T(v)}_{\in \text{Im}(T)}.$$

□

Definition 1.5.19. A linear map T satisfying either of the two equivalent conditions of lemma 1.5.18 is called a **projection operator**.

Corollary 1.5.20

Let $T: (V, \rho) \rightarrow (V, \rho)$ be a G -linear projection operator. Then,

$$(V, \rho) = \ker(T) \oplus \operatorname{Im}(T).$$

§1.6 Direct Sums

Recall that there are usually two direct sums, the external direct sum and the internal direct sum. The external direct sum is the Cartesian product $V \oplus W = V \times W$ equipped with addition and scalar multiplication. The internal direct sum $V = V_1 \oplus V_2$ is a vector space V where $V_1, V_2 \subset V$ such that $V_1 \cap V_2 = \{0\}$ and $V_1 + V_2 = V$.

Notation 1.6.1. We may denote the external direct sum as $\oplus_{\text{ext}}$ and the internal direct sum as $\oplus_{\text{int}}$.

Remark 1.6.2 — The direct sum appearing in the previous lemma is the internal direct sum.

Remark 1.6.3 — Given $V_1, V_2 \subseteq V$ subspaces, we have a canonical linear map $V_1 \oplus_{\text{ext}} V_2 \rightarrow V, (v_1, v_2) \mapsto v_1 + v_2$. This canonical linear map is an isomorphism if and only if $V = V_1 \oplus_{\text{int}} V_2$.

Definition 1.6.4. Given representation (V_1, ρ_1) and (V_2, ρ_2) of G , the **external direct sum** is the representation of G on $V_1 \oplus_{\text{ext}} V_2$ defined by

$$\rho(g)(v_1, v_2) = (\rho(g)v_1, \rho(g)v_2).$$

We denote this by $(V_1, \rho_1) \oplus_{\text{ext}} (V_2, \rho_2)$ or $(V_1, \rho_1) \oplus (V_2, \rho_2)$

Definition 1.6.5. Given a representation (V, ρ) of G , and subrepresentation $V_1, V_2 \subseteq (V, \rho)$, we say that (V, ρ) is the **internal direct sum** of $(V_1, \rho|_{V_1})$ and $(V_2, \rho|_{V_2})$ if this is the case as vector spaces. We denote this by $V_1 \oplus_{\text{int}} V_2$ or $V_1 \oplus V_2$.

Definition 1.6.6. A nonzero representation of G is **decomposable**, if it is a direct sum of two proper nonzero subrepresentations, and **indecomposable**, if it is not.

Remark 1.6.7 — To be decomposable is a stronger property than reducibility. For example, we may have reducible representations that are indecomposable.

Remark 1.6.8 — The zero representation is neither decomposable nor indecomposable.

Example 1.6.9

We go back to the example of the subrepresentations of the permutation representation $(\mathbb{C}^n, \rho)$ of S_n (example 1.5.12). Since

$$\begin{aligned} W &:= \text{Im}(T) = \{(a, a, \dots, a) \mid a \in \mathbb{C}\} = \text{Span}(1, 1, \dots, 1) \\ U &:= \ker(T) = \{(a_1, a_2, \dots, a_n) \mid a_1 + a_2 + \dots + a_n = 0\} \end{aligned}$$

it follows that $(\mathbb{C}^n, \rho) = W \oplus U$.

Definition 1.6.10. A representation is **semisimple**, or **completely reducible**, if it is an internal direct sum of (some finite number of) irreducible representations.

Example 1.6.11

Let V be an indecomposable representation. Then by definition, we see that V is semisimple if and only if it is actually irreducible. Indeed, if $V = \oplus_{i=1}^m V_i$ is a direct sum expression with the V_i irreducible, then by indecomposability, $m = 1$ and $V = V_1$ is irreducible.

Remark 1.6.12 — As we will see shortly, by Maschke's theorem, when G is finite and V finite dimensional (working over $\mathbb{C}$ as always), then V is always semisimple. Hence in the case of main interest, indecomposable is equivalent to irreducible. As the next example shows, this is not true without these assumptions.

Example 1.6.13

Consider the representation of $\mathbb{Z}$ on $\mathbb{C}^2$ defined by $\rho(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$. This is, indeed, a representation, as $0 \mapsto I$ and $\rho(x+y) = \rho(x)\rho(y)$. This is reducible since the x -axis (or $\mathbb{C} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \text{Span}\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}\}$) is a subrepresentation. However, this is the only subrepresentation: any nonzero subrepresentation is a line fixed by all of these matrices, i.e., an eigenline. But the only eigenline of $\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$ is the x -axis. This shows that there cannot be a decomposition into irreducible subrepresentations, so ρ is indecomposable.

Notation 1.6.14. We denote by $\mathbb{C}v$ as the span of v .

§1.7 Maschke's Theorem

In this section finally the assumption that we are working over $\mathbb{C}$, as well as the finiteness of G , becomes very important.

Definition 1.7.1. Let (V, ρ) be a representation of G and $W \subseteq V$ a subrepresentation. A **complementary subrepresentation** is a subrepresentation $U \subseteq V$ such that $V = W \oplus U$.

Theorem 1.7.2 (Maschke's Theorem)

Let (V, ρ) be a finite-dimensional representation of a finite group G . Let $W \subseteq V$ be any subrepresentation. Then there exists a complementary subrepresentation $U \subseteq V$.

Proof. The idea of the proof is to construct a homomorphism $T: V \rightarrow V$ of representations which is a projection operator with image W , and set $U = \ker(T)$. Then we apply corollary 1.5.20.

First observe that, via linear algebra, there exists a complementary subspace $U \subseteq V$, i.e., a vector subspace such that $V = U \oplus W$ (we do this by extending a basis $(w_1, \dots, w_k)$ of W to a basis $(w_1, \dots, w_n)$ of V , and set U to be the span of $(w_{k+1}, \dots, w_n)$). As a result, every vector $v \in V$ is uniquely expressible as a sum $v = w + u$ for $w \in W, u \in U$. Due to this, we consider the map $T: V \rightarrow V$ defined by $T(w + u) = w$ for $w \in W, u \in U$. It's immediate that $T^2 = T$ and that T is a projection. By definition $\text{Im}(T) = W$. If T were a homomorphism of representations, we would be done by corollary 1.5.20. However, it need not be.

The idea is to now *average* T over the group G , in order to turn T into a homomorphism of representations. Namely, define

$$\tilde{T} := \frac{1}{|G|} \sum_{g \in G} \rho(g) \circ T \circ \rho(g)^{-1}.$$

We first check that this is a homomorphism of representations (G -linear):

$$\begin{aligned} \tilde{T}(\rho(h)v) &= \frac{1}{|G|} \sum_{g \in G} \rho(g) \circ T \circ \rho(g)^{-1} \rho(h)v \\ &= \frac{1}{|G|} \sum_{g' = h^{-1}g \in G} \rho(h)\rho(g') \circ T \circ \rho(g')^{-1}v \\ &= \rho(h) \circ \tilde{T}(v). \end{aligned}$$

It remains to check that $\tilde{T}(w)$ is still a projection with image W . We first show that $\tilde{T}(w) = w$ for all $w \in W$:

$$\begin{aligned} \tilde{T}(w) &= \frac{1}{|G|} \sum_{g \in G} \rho(g) \circ T(\rho(g)^{-1}(w)) \\ &= \frac{1}{|G|} \sum_{g \in G} \rho(g)\rho(g)^{-1}(w) = w. \end{aligned}$$

The second equality comes from the fact that $\rho(g)^{-1}(w) = \rho(g^{-1})(w) \in W$, and $T(w') = w'$ for all $w' \in W$ (in particular $w' = \rho(g^{-1})(w)$).

Since $\text{Im}(T) = W$, and $\rho(g)(W) \subseteq W$ for all $g \in G$ (because W is a subrepresentation), the definition of $\tilde{T}$ implies that $\text{Im}(\tilde{T}) \subseteq W$. Since $\tilde{T}(w) = w$ for all $w \in W$, we obtain that $\text{Im}(\tilde{T}) = W$. Therefore $\tilde{T}$ is a projection by lemma 1.5.18. Now setting $U := \ker(\tilde{T})$ we get a complementary subrepresentation to W by corollary 1.5.20. $\square$

Remark 1.7.3 — The point that we work over $\mathbb{C}$ (moreover, characteristic 0 fields) and that G is finite is key here.

Remark 1.7.4 — Here, there is a clever formula in the proof which turns a general linear map $T: V \rightarrow V$ into a homomorphism of representations.

A way to rephrase the Maschke's theorem is as follows:

Theorem 1.7.5 (Rephrasing of Maschke's Theorem)

If (V, ρ) is reducible, then it is decomposable.

The contrapositive of this statement is the corollary below.

Corollary 1.7.6

If (V, ρ) is a finite-dimensional indecomposable representation of a finite group, then V is irreducible.

Remark 1.7.7 — Actually the finite-dimensional hypothesis is not necessary, as every indecomposable representation of a finite group is also irreducible.

Corollary 1.7.8

If (V, ρ) is a finite-dimensional representation of a finite group, then it is a direct sum of some irreducible subrepresentations. That is, (V, ρ) is semisimple (completely reducible).

Proof. As stated, we proceed by strong induction on the dimension of V . (V, ρ) is either decomposable or not. If it is decomposable, then we write it as $V = W \oplus U$, and apply the inductive hypothesis to W and U (which are necessarily of lower dimension). If not, then V is either zero (direct sum of zero irreducible subrepresentations) or indecomposable. Thus, by corollary 1.7.6, it is irreducible; therefore, semisimple. $\square$

Remark 1.7.9 (Non-Examinable) — A counter example to Maschke's theorem over the field $\mathbb{Z}/2$. Consider the representation of $C_2 = \{1, g\}$ over $\mathbb{Z}/2$ defined by

$$\rho(g) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Note that $\rho(g)^2 = I$, so it is indeed a representation. Proper nonzero subrepresentations are of dimension 1; these are spans of eigenvectors of $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. 1 is the only eigenvalue of this matrix, and the eigenvectors are $\text{Span} \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} \right)$. There's only one

1-dimensional subrepresentation; therefore, this representation is indecomposable, but reducible.

We now know that finite-dimensional representations of finite groups admit decomposition as a direct sum of irreducible subrepresentations. The next natural question is to ask whether this decomposition is unique. Not literally, but there are reasonable practical results, which we'll now show.

Example 1.7.10

Given a vector space V and a group G , it has the trivial representation: $\forall g \in G, \rho(g) = id_V$. Any basis $(v_1, \dots, v_n)$ of V gives rise to a decomposition $V = \mathbb{C}V_1 \oplus \dots \oplus \mathbb{C}V_n$ of V as a direct sum of n copies of the trivial representation.

Exercise 1.7.11. Does it ever happen in this example that two different bases $(v_1, \dots, v_n)$ and $(w_1, \dots, w_n)$ give rise to the same direct sum decomposition, i.e.

$$\{\mathbb{C}v_1, \dots, \mathbb{C}v_n\} = \{\mathbb{C}w_1, \dots, \mathbb{C}w_n\}$$

Example 1.7.12

Let $C_m = \{1, g, \dots, g^{m-1}\}$ be the cyclic group of order m . Let its representation $(\mathbb{C}^m, \rho)$ is given by

$$\rho(g) = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & \zeta & 0 & & \vdots \\ 0 & 0 & \zeta^2 & & \\ \vdots & & & \ddots & \\ 0 & & & & \zeta^{m-1} \end{pmatrix}, \quad \zeta = e^{\frac{2\pi i}{m}}.$$

Thus, $\rho(g)e_i = \zeta^i e_i$. Let us show that this representation has a *unique* decomposition as a direct sum of irreducibles, $\mathbb{C}^n = \mathbb{C}e_0 \oplus \mathbb{C}e_1 \oplus \dots \oplus \mathbb{C}e_{m-1}$.

It suffices to show that for any nonzero subrepresentation $W \subseteq (\mathbb{C}^m, \rho)$, there exists $i \in \{0, \dots, m-1\}$ such that $\mathbb{C}e_i \subseteq W$. Indeed, let $W \subseteq (\mathbb{C}^m, \rho)$ be a nonzero subrepresentation, and let $w = (a_0, \dots, a_{m-1}) \neq 0$ with $w \in W$. Let i be such that $a_i \neq 0$. Then,

$$e_i = \frac{1}{ma_i} \left(w + \frac{Tw}{\zeta^i} + \frac{T^2w}{\zeta^{2i}} + \dots + \frac{T^{m-1}w}{\zeta^{(m-1)i}} \right) \in W.$$

We need to check this formula above.

Example 1.7.13

Consider the regular representation of $C_m = \{1, g, \dots, g^{m-1}\}$. For any ξ , a root of unity,

$$\begin{aligned}\rho(g) \left(1 + \xi g + \xi^2 g^2 + \dots + \xi^{m-1} g^{m-1} \right) &= g + \xi g^2 + \xi^2 g^3 + \dots + \xi^{m-1} g^m \\ &= \xi^{m-1} + g + \xi g^2 + \dots + \xi^{m-2} g^{m-1} \\ &= \xi^{-1} \left(1 + \xi g + \xi^2 g^2 + \dots + \xi^{m-1} g^{m-1} \right)\end{aligned}$$

Then, $v = 1 + \xi g + \xi^2 g^2 + \dots + \xi^{m-1} g^{m-1}$ has the property that $\rho(g)v = \xi^{-1}v$. Therefore,

$$\begin{aligned}v_0 &= 1 + g + \dots + g^{m-1} \\ v_1 &= 1 + \zeta^{-1}g + \zeta^{-2}g^2 + \dots + \zeta^{-(m-1)}g^{m-1} \\ &\vdots\end{aligned}$$

and in general,

$$v_i = 1 + \zeta^{-i}g + \zeta^{-2i}g^2 + \dots + \zeta^{-i(m-1)}g^{m-1}, \quad \zeta = e^{\frac{2\pi i}{m}}$$

is a ζ_i -generator of $\rho(g)$. Thus, in the basis $v_0, v_1, \dots, v_{m-1}$, we have that

$$\rho(g) = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & \zeta & 0 & & \vdots \\ 0 & 0 & \zeta^2 & & \\ \vdots & & & \ddots & \\ 0 & & & & \zeta^{m-1} \end{pmatrix}$$

Thus, $(\mathbb{C}[C_m], \rho)$ is isomorphic to the representation in the previous example (example 1.7.12), so the same analysis of unique decomposition holds.

§1.8 Schur's Lemma**Lemma 1.8.1**

Let (V, ρ_V) and (W, ρ_W) be irreducible representations of a group G .

1. If $T: V \rightarrow W$ is a G -linear map, then T is either an isomorphism or the zero map.
2. Suppose V is finite-dimensional. If $T: V \rightarrow V$ is G -linear, then $T = \lambda I$, for some $\lambda \in \mathbb{C}$.

Proof. (1): $\text{Im}(T)$ is a subrepresentation of W . Hence, since W is irreducible, $\text{Im}(T) = (0)$ or $\text{Im}(T) = W$. $\ker(T)$ is a subrepresentation of V ; hence, since V is irreducible, $\ker(T) = (0)$ or $\ker(T) = V$. If $\text{Im}(T) = (0)$, then $T = 0$. If $\text{Im}(T) = W$ and $\ker(T) = (0)$, then T is an isomorphism. If $\text{Im}(T) = W$ and $\ker(T) = V$, then $W = \text{Im}(T) = (0)$, which is a contradiction since the zero representation is not irreducible.

(2): Let $v \in V$ be a nonzero λ -eigenvector of T of λ , which we know exists because $\mathbb{C}$ is algebraically closed. Then, $T - \lambda \text{id}_V$ is also a homomorphism of representations, so by the first statement, $T - \lambda \text{id}_V$ is either an isomorphism or zero. However, it can’t be an isomorphism, since $(T - \lambda I_V)(v) = Tv - \lambda v = 0$, so this implies that $T - \lambda I_V = 0$, so $T = \lambda \text{id}_V$. $\square$

Remark 1.8.2 — This is a feature of irreducible representations, and this is certainly not the case for reducible representations. In general, if (V, ρ) is a reducible representation of G , they have G -linear maps $T: V \rightarrow V$ which are neither isomorphisms or zero. For example, take $W \subseteq V$, a proper nonzero subrepresentation. Then, by the proof of Maschke’s theorem, there exists a G -linear projection $T: V \rightarrow V$ such that $\text{Im}(T) = W$. This is neither zero nor an isomorphism.

Example 1.8.3

Take $G = S_n$ and $(\mathbb{C}^n, \rho)$ to be the permutation representation. Then, we attain G -linear map $T: \mathbb{C}^n \rightarrow \mathbb{C}^n$ which is not a multiple of the identity: $T(a_1, \dots, a_n) = (a, \dots, a)$, where $a = \frac{1}{n} \sum_i a_i$ is the average of $a_1, \dots, a_n$ (this is the projection of $\mathbb{C}^n$ onto the trivial subrepresentation). Thus, this representation is not irreducible by the contrapositive of Schur’s lemma.

Example 1.8.4

Take $C_n = \{1, g, \dots, g^{n-1}\}$ and the rotation representation of C_n on $\mathbb{C}^2$. Let

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$$

and let $\rho(g^i) = R_{\frac{2\pi i}{n}}$. Then, for any $\theta \in [0, 2\pi)$ and any $i \in \{0, \dots, n-1\}$,

$$R_\theta \circ \rho(g^i) = R_{\theta + \frac{2\pi i}{n}} = \rho(g^i) \circ R_\theta.$$

Therefore, (for $\theta \neq 0, \pi$) R_θ is a C_n -linear isomorphism from $\mathbb{C}^2$ to $\mathbb{C}^2$, not of the form $\lambda I_{\mathbb{C}^2}$ for some λ .

Remark 1.8.5 — This doesn’t break Schur’s lemma because this is not an irreducible representation. In particular, it is isomorphic to the representation sending g^i to

$$\begin{pmatrix} e^{\frac{2\pi i}{n}} & 0 \\ 0 & e^{-\frac{2\pi i}{n}} \end{pmatrix}.$$

which is the direct sum of two one-dimensional irreducible representations.

Example 1.8.6

Take $C_n = \{1, g, \dots, g^{n-1}\}$ and the rotation representation of C_n on $\mathbb{R}^2$. Now, the *rotation representation* is irreducible. Again we note that R_θ is a C_n -linear isomorphism from $\mathbb{R}^2$ to $\mathbb{R}^2$, not of the form $\lambda id_{\mathbb{R}^2}$ for any λ .

Remark 1.8.7 — This doesn't break Schur's lemma because we're working over $\mathbb{R}$, not $\mathbb{C}$. We highlight that the second part to Schur's lemma really does rely over the fact that we're working over $\mathbb{C}$.

Example 1.8.8 (Non-examinable)

Let K be a nontrivial extension field of $\mathbb{C}$. The group $K^\times = K \setminus \{0\}$ (with the operation of multiplication) has an irreducible representation over $\mathbb{C}$ on K , considered as a vector space over $\mathbb{C}$: $\rho(k) = k$. Fix $k \in K \setminus \mathbb{C}$, define an endomorphism of representations of $K^\times$, $T: K \rightarrow K$ by $T(x) = kx$. This is not a $\mathbb{C}$ -scalar multiple of id_K .

Remark 1.8.9 — This doesn't break Schur's lemma because we're working over an infinite dimensional vector space K .

Theorem 1.8.10

Let V be a representation of a group G . Then, any two direct sum decomposition into irreducibles $V \cong V_1 \oplus \dots \oplus V_n$, $V \cong V'_1 \oplus \dots \oplus V'_m$ are the same up to isomorphism and relabeling: $n = m$ and, after reordering, $V_i \cong V'_i$ for all i .

Proof. By induction on n , we write $T: V_1 \oplus \dots \oplus V_n \xrightarrow{\sim} V'_1 \oplus \dots \oplus V'_m$, as (T_{ij}) , $1 \leq i \leq n$, $1 \leq j \leq m$, with $T_{ij}: V_i \rightarrow V'_j$. If $n \geq 1$. If $n \geq 1$, then (since T is an isomorphism) there exist i, j such that $T_{ij} \neq 0$; in particular, $m \geq 1$ as well. By Schur's lemma, we must have that $T_{ij}: V_i \xrightarrow{\sim} V'_j$. Reorder then so that $i = j = 1$.

Claim 1.8.11. $T(v_1) \cap (V'_1 \oplus \dots \oplus V'_m) = (0)$.

Proof. If $v_1 \in V_1$, and $T(v_1) \in V'_2 \oplus \dots \oplus V'_m$, then by the definition of (T_{ij}) , this means $T_{11}(v_1) = 0$. Since T_{11} is an isomorphism, $v_1 = 0$. $\square$

Claim 1.8.12. $T(v_1) + V'_2 + \dots + V'_m = V'_1 \oplus \dots \oplus V'_m$.

Proof. Given $v'_1 \in V'_1$, then there exists $v_1 \in V_1$ such that $T_{11}(v_1) = v'_1$. Then, $T(v_1) - v'_1 \in V'_2 + \dots + V'_m$, implying that $v'_1 = T(v_1) + v$, where v is an element in $V'_2 + \dots + V'_m$. $\square$

Combining claim 1.8.11 and 1.8.12, $T(V_1) \oplus V'_2 \oplus \dots \oplus V'_m$ is still a direct sum decomposition of $V'_1 \oplus \dots \oplus V'_m$. T now descends to be an isomorphism $\tilde{T}: V_2 \oplus \dots \oplus V_n \xrightarrow{\sim} V'_2 \oplus \dots \oplus V'_m$. Now, we apply the inductive hypothesis. $\square$

Theorem 1.8.13

Suppose that $V = V_1 \oplus \cdots \oplus V_n$ is a decomposition into irreducible subrepresentations which are pairwise non-isomorphic. Then the V_i are the unique irreducible subrepresentations of V , so in particular this is the unique decomposition into irreducible subrepresentations.

Proof. Suppose that $V = V_1 \oplus \cdots \oplus V_n$ is a decomposition into irreducible subrepresentations which are pairwise non-isomorphic. Then the V_i are the unique irreducible subrepresentations of V , so in particular this is the unique decomposition into irreducible subrepresentations. $\square$

Remark 1.8.14 — This proof will again not require anything about the group (which can be infinite) or the representations (which can be infinite-dimensional).

Remark 1.8.15 — By Maschke’s theorem, it follows from this that, if G is finite (hence V is finite dimensional), then every subrepresentation is of the form $V_{i_1} \oplus \cdots \oplus V_{i_m}$, for some $1 \leq i_1 < \cdots < i_m \leq n$. We will prove this for general G and V (and even for modules over arbitrary algebras) later.

§1.9 Representations of Abelian Groups

By Schur’s lemma, we can understand finite-dimensional irreducible representations of abelian groups up to isomorphism (and also by Maschke’s theorem, finite-dimensional representations of finite abelian groups).

Lemma 1.9.1

Let (V, ρ_V) be a representation of a group G . Let $z \in G$ be central (commutes with all elements in G). Then, $\rho_V(z): V \rightarrow V$ is G -linear.

Proof. We have

$$\rho_V(z) \circ \rho_V(g) = \rho_V(zg) = \rho_V(gz) = \rho_V(g) \circ \rho_V(z).$$

$\square$

Proposition 1.9.2

Let G be an abelian group (not necessarily finite). Then every finite dimensional irreducible representation of G is one-dimensional.

Proof. Assume G is abelian. Let (V, ρ_V) be a finite-dimensional irreducible representation. Since every $g \in G$ is central, by lemma 1.9.1, $\rho_V(g)$ is G -linear. By lemma 1.8.1, $\rho_V(g)$ is actually a multiple of the identity matrix. Now let $v \in V$ be any nonzero vector. Let $V_1 := \mathbb{C} \cdot v$, the span of v , which is one-dimensional. We have that V_1 is a subrepresentation, since $\rho_V(g)(v)$ is always a multiple of v . By irreducibility, $V = V_1$. $\square$

Remark 1.9.3 — The hypothesis of finite-dimensionality is required, since the proof invokes Schur’s lemma (1.8.1).

Example 1.9.4

Let K be a nontrivial extension field of $\mathbb{C}$. We’ve shown that $K^\times = K \setminus \{0\}$ (which is an abelian group) has an irreducible representation on K itself, considering K as an (infinite-dimensional) vector space over $\mathbb{C}$.

Example 1.9.5

In example 1.6.13, we constructed an indecomposable representation $(\mathbb{C}^2, \rho)$ of $\mathbb{Z}$, given by

$$\rho(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}.$$

However, this is not irreducible.

Similar, the same formula gives an indecomposable (but not irreducible) two dimensional representation $(\mathbb{C}^2, \rho)$ of $\mathbb{C}$ (considered as an additive group):

$$\rho(z) = \begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix}.$$

Corollary 1.9.6

Let G be a finite abelian group. Then, every finite-dimensional representation is a direct sum of one-dimensional representations.

Proof. This follows by Maschke’s theorem and proposition 1.9.2. □

Corollary 1.9.7

For $G = C_n$, there are exactly n irreducible representations, up to isomorphism, which are one-dimensional and are the ones given in example 1.1.2:

$$\rho(g^i) = (\zeta^i),$$

where ζ ranges over all the n -th roots of unity.

Proof. Since G is finite, all irreducible representations are finite-dimensional by proposition 1.5.6. By proposition 1.9.2, all irreducible representations are therefore one-dimensional. By proposition 1.2.10, up to isomorphism, they are given by representations $G \rightarrow GL_1(\mathbb{C}) = \mathbb{C}^\times$, i.e., multiplicative assignments of G to nonzero complex numbers. But the generator $g \in G$ has to map to an n -th root of unity, call it ζ . Then we obtain the representation ρ . Note that two one-dimensional matrix representations are conjugate if and only if they are identical, since one-by-one matrices commute. □

Exercise 1.9.8. Classify all irreducible representations of finite abelian groups. Namely, from group theory, recall that every finite abelian group is isomorphic to a product $G = C_{n_1} \times \cdots \times C_{n_k}$.

Solution. If we let ζ_i be an n_i -th root of unity, we have

$$\rho_{\zeta_1, \dots, \zeta_k}(g_1^{r_1}, \dots, g_k^{r_k}) = \zeta_1^{r_1} \cdots \zeta_k^{r_k}.$$

Indeed, we have $n_1 \cdots n_k$ representations, which are nonisomorphic. $\square$

Example 1.9.9

Let $G = \{\pm 1\} \times \{\pm 1\} \cong C_2 \times C_2$, the Klein four group. We have the representation $(\mathbb{C}^2, \rho)$ given by

$$\rho(a, b) = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}.$$

This is the direct sum of the irreducible subrepresentations

$$\begin{aligned} \mathbb{C}e_1 &\cong (\mathbb{C}, \rho_1), & \rho_1(a, b) &= \begin{pmatrix} a \end{pmatrix} \\ \mathbb{C}e_2 &\cong (\mathbb{C}, \rho_2), & \rho_2(a, b) &= \begin{pmatrix} b \end{pmatrix} \end{aligned}$$

Example 1.9.10

We have the representation of $\mathbb{C}^\times$ on $(\mathbb{C}^2, \rho)$ given by

$$\rho(a + bi) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

We check this:

$$\begin{aligned} \rho(a + bi)\rho(c + di) &= \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} c & -d \\ d & c \end{pmatrix} = \begin{pmatrix} ac - bd & -ad - bc \\ cb + ad & -bd + ac \end{pmatrix} \\ &= \rho((ac + bd) + i(bc + ad)) = \rho((a + bi)(c + di)) \end{aligned}$$

We note that for $P = \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}$,

$$P^{-1} \begin{pmatrix} a & -b \\ b & a \end{pmatrix} P = \begin{pmatrix} a + bi & 0 \\ 0 & a - bi \end{pmatrix}.$$

Therefore, $(\mathbb{C}^2, \rho)$ is equivalent to the representation $(\mathbb{C}^2, \rho')$ defined by

$$\rho(z = a + bi) = \begin{pmatrix} z & 0 \\ 0 & \bar{z} \end{pmatrix}.$$

This is the direct sum of

$$\begin{aligned} \mathbb{C}e_1 &\cong (\mathbb{C}, \rho_1), & \rho_1(z) &= \begin{pmatrix} z \end{pmatrix} \\ \mathbb{C}e_2 &\cong (\mathbb{C}, \rho_2), & \rho_2(z) &= \begin{pmatrix} \bar{z} \end{pmatrix} \end{aligned}$$

Example 1.9.11

We've shown (in problem sheets) that the representation $(\mathbb{C}^2, \rho)$ of C_n ,

$$\rho(g) = \begin{pmatrix} \cos(\frac{2\pi}{n}) & -\sin(\frac{2\pi}{n}) \\ \sin(\frac{2\pi}{n}) & \cos(\frac{2\pi}{n}) \end{pmatrix}$$

is equivalent to

$$\rho'(G) = \begin{pmatrix} \zeta & 0 \\ 0 & \zeta^{-1} \end{pmatrix}, \quad \zeta = e^{\frac{2\pi i}{n}}$$

by conjugating by a matrix P . Thus, this representation of C_n is isomorphic to a direct sum of one-dimensional representations. In fact, the second example is obtained from the first by composing with the homomorphism of group $C_n \hookrightarrow \mathbb{C}^\times, g \mapsto \cos(\frac{2\pi}{n}) + i \sin(\frac{2\pi}{n})$.

Exercise 1.9.12. Given a group G and a representation $\rho: G \rightarrow GL_n(\mathbb{C})$, then $\{\rho(g) \mid g \in G\}$ are simultaneously diagonalizable if and only if $(\mathbb{C}^n, \rho)$ decomposes as a direct sum of one-dimensional representations.

Remark 1.9.13 — If G is finite and abelian, this implies $\{\rho(g) \mid g \in G\}$ are commuting and diagonalizable. Therefore (by linear algebra), they are simultaneously diagonalizable. By the exercise above, we get an alternative proof of corollary 1.9.6.

§1.10 One Dimensional Representations

We start by looking at nonabelian groups for motivation.

Example 1.10.1 (One Dimensional Representations of S_n)

A one dimensional matrix representation $(\mathbb{C}, \rho)$ of S_n has $\rho(12) = \pm 1$, since

$$1 = \rho(e) = \rho((12)^2) = \rho((12))^2.$$

Let $\rho((12)) := \epsilon$, so $\epsilon = \pm 1$. For any pair $(ij) \in S_n$, consider

$$(1i)(2j)(12) \underbrace{(2j)(1i)}_{[(1i)(2j)]^{-1}} = (ij).$$

This shows that (12) is conjugate to any other pair (ij) . Thus,

$$\rho((ij)) = \rho(g(12)g^{-1}) = \rho(g)\rho((12))\rho(g)^{-1} = \rho((12)) = \epsilon,$$

since $GL_1(\mathbb{C})$ is abelian. Pairs generate S_n , so a one-dimensional representation is completely determined by $\rho((12)) = \epsilon \in \{-1, 1\}$.

We note that both choices are possible:

- $\epsilon = 1$: trivial representation, $\forall g, \rho(g) = 1$.
- $\epsilon = -1$: sign representation, $\forall g, \rho(g) = \text{sign}(g)$.

Example 1.10.2 (One Dimensional Representations of D_n)

Let $G = D_n = \langle x, y \mid x^n = 1 = y^2, yxy^{-1} = x^{-1} \rangle$. Given a one dimensional representation of D_n , we must have that

$$\rho(x)^n = 1, \quad \rho(y)^2 = 1, \quad \rho(x) = \rho(y)\rho(x)\rho(y)^{-1} = \rho(x)^{-1}.$$

This gives $\rho(x) = \rho(x)^{-1}$, so $\rho(x)^2 = 1$. Therefore,

$$\rho(x)^{\gcd(n,2)} = 1.$$

There are two cases then:

- n is odd: We get $\rho(x) = 1$, so there are at most two one-dimensional representations (as $\rho(y) = \pm 1$).
- n is even: $\rho(x)^2 = 1$, so there are at most four one-dimensional representations (as $\rho(y) = \pm 1$).

To see that all these potential one-dimensional representations of D_n exist, consider the following:

- n odd: $\langle x, y \mid x = 1, y^2 = 1 \rangle \cong C_2$
- n even: $\langle x, y \mid x^2 = y^2 = 1 \rangle \cong C_2 \times C_2$

Check the the obvious homomorphism from D_n to the appropriate one of these groups is well-defined ($x \mapsto x, y \mapsto y$), then pull back representations of C_2 and $C_2 \times C_2$ under this homomorphism.

§1.10.i Abelianization

Definition 1.10.3. Let G be a group. For any $g, h \in G$, we define the **commutator** of g and h , denoted $[g, h]$, as

$$[g, h] = ghg^{-1}h^{-1}.$$

Definition 1.10.4. The **commutator subgroup** of G is $[G, G]$ is the subgroup generated by $\{[g, h] \mid g, h \in G\}$.

Lemma 1.10.5

$[G, G]$ is normal.

Proof. Note that $[g, h][h, g] = e$, so $[g, h]^{-1} = [h, g]$. Therefore, the elements of $[G, G]$ are all products of commutators. We have $[kgk^{-1}, khk^{-1}] = k[g, h]k^{-1}$. Given $\prod_{i=1}^n [g_i, h_i] \in [G, G]$, we thus have

$$k \left(\prod_{i=1}^n [g_i, h_i] \right) k^{-1} = \prod_{i=1}^n k[g_i, h_i]k^{-1} = \prod_{i=1}^n [kg_i k^{-1}, kh_i k^{-1}] \in [G, G],$$

so we're done. $\square$

Proposition 1.10.6

We have that:

1. $G/[G, G]$ is abelian.
2. For any normal subgroup N of G , G/N is abelian if and only if $[G, G] \subseteq N$.

Proof. It suffices to prove the second statement, as the first statement follows from it. We have

$$gh[h^{-1}, g^{-1}] = hg, \forall g, h \in G.$$

Thus, $ghN = hgN$ if and only if $[h^{-1}, g^{-1}] \in N$. Therefore, for all $g, h \in G$, $ghN = hgN$ if and only if for all $g, h \in G$, $[h^{-1}, g^{-1}] \in N$ if and only if $[G, G] \subseteq N$. $\square$

Remark 1.10.7 — If G is simple, then the only normal subgroups are $\{e\}$ and G itself, and so $[G, G]$ must be trivial.

Definition 1.10.8. $G_{ab} := G/[G, G]$ is the **abelianization** of G . (The maximal abelian quotient of G).

Notation 1.10.9. $q_{ab}: G \rightarrow G_{ab}$ the quotient map.

Corollary 1.10.10

Let $\phi: G \rightarrow H$ be a homomorphism of groups. Then, the following are equivalent:

1. $\text{Im}(\phi)$ is abelian
2. $[G, G] \subseteq \ker(\phi)$
3. ϕ factors through q_{ab} .

$$\begin{array}{ccc} G & \xrightarrow{\phi} & H \\ \downarrow q_{ab} & \searrow \bar{\phi} & \\ G/[G, G] & & \end{array}$$

Proof. (1 $\iff$ 2): This is by the first isomorphism theorem and proposition 1.10.6.

(2 $\iff$ 3): This holds because in general, $\phi: G \rightarrow H$ factors through $\Pi: G \rightarrow G/N$ if and only if $N \triangleleft \ker(\phi)$. $\square$

Corollary 1.10.11

Let G be a group, A an abelian group. Then, the map

$$\text{Hom}(G_{ab}, A) \rightarrow \text{Hom}(G, A), \quad \phi \mapsto \phi \circ q_{ab}$$

is a bijection.

Proof. The map is obviously injective, since q_{ab} is surjective. By corollary 1.10.10, every homomorphism $\phi \in \text{Hom}(G, A)$ is of the form $\phi \circ q_{ab}$ for some ϕ . $\square$

Since $GL_1(\mathbb{C}) = \mathbb{C}^\times$ is abelian, we obtain from corollary 1.10.11 the following:

Corollary 1.10.12

The one dimensional matrix representations of G are the same as those of G_{ab} .

We can apply exercise 1.9.8 to get:

Corollary 1.10.13

If G_{ab} is finite (e.g., if G is finite), then the number of one-dimensional matrix representations of G is equal to $|G_{ab}|$.

Remark 1.10.14 — This also tells us that the number of one-dimensional representations divide $|G|$.

Remark 1.10.15 — For S_n , we have $[S_n, S_n] = A_n$, and $(S_n)_{ab} = S_n/A_n \cong C_2$.

Remark 1.10.16 — For D_{2n+1} , we have $[D_{2n+1}, D_{2n+1}] \cong C_{2n+1}$ and $(D_{2n+1})_{ab} = D_{2n+1}/C_{2n+1} \cong C_2$.

For D_{2n} , we have $[D_{2n}, D_{2n}] \cong C_n$, and $(D_{2n})_{ab} = D_{2n}/C_n \cong C_2 \times C_2$.

Remark 1.10.17 — Note that if a representation contains a 1-dimensional subrepresentation, then there exists a common nonzero eigenvector for all of $\rho_V(g)$.

§1.11 Homomorphisms of Representations and Representations of Homomorphisms

Observe that if $S, T: (V, \rho_V) \rightarrow (W, \rho_W)$ are homomorphisms of representations, so is $aS + bT$ for any $a, b \in \mathbb{C}$. Thus, $\text{Hom}_G(V, W)$ forms a vector space.

Remark 1.11.1 — Note that Schur's lemma says that if V and W are finite dimensional irreducible representations, then

$$\dim \text{Hom}_G(V, W) = \begin{cases} 1 & V \cong W \\ 0 & V \not\cong W \end{cases}$$

The goal now is to find all irreducible representations, and prove that there is some uniqueness property. The way we do this is by considering the regular representation $\mathbb{C}[G]$.

Definition 1.11.2. Given representations (V, ρ_V) and (W, ρ_W) , define a linear action $G \times \text{Hom}(V, W) \rightarrow \text{Hom}(V, W)$, i.e., a homomorphism $\rho_{\text{Hom}(V, W)}: G \rightarrow GL(\text{Hom}(V, W))$, by

$$\rho_{\text{Hom}(V, W)}(g)(T) := g \cdot T = \rho_W(g) \circ (T) \circ \rho_V(g)^{-1}$$

Remark 1.11.3 — This inverse is necessary because we want associativity to hold:

$$\rho_{\text{Hom}(V, W)}(h) \circ \rho_{\text{Hom}(V, W)}(g)(T) = \rho_{\text{Hom}(V, W)}(gh)(T).$$

Remark 1.11.4 — Analogous to the three actions on $\mathbb{C}[G]$, we can define the three actions:

- Use both $\rho_W(g)$ and $\rho_V(g)^{-1}$
- Use only $\rho_W(g)$
- Use only $\rho_V(g)^{-1}$

This tells us the more general fact that $G \times G$ acts on $\text{Hom}(V, W)$, and, even more generally, $G \times H$ acts on $\text{Hom}(V, W)$ if V is a G -representation and W is a H -representation. However, here we're only interested in the first case.

Definition 1.11.5. If V is a G -representation, we can define

$$V^G := \{v \in V \mid \rho_V(g)(v) = v, \forall g \in G\}.$$

We call this the G -invariant subspace.

Remark 1.11.6 — We have V^G to be the sum of all subrepresentations where G acts trivially.

Remark 1.11.7 — We note that V^G is indeed a linear subspace of V .

Question: What is $\text{Hom}(V, W)^G$? That is, when do we have $\rho_{\text{Hom}(V, W)}(g)(T) = T$, for all g ? This is equivalent to the fact that T is a G -linear map (a homomorphism of representations). This is because

$$\rho_W(g) \circ (T) \circ \rho_V(g)^{-1} = T \iff \rho_W(g) \circ (T) = T \circ \rho_V(g).$$

This tells us that $\text{Hom}(V, W)^G = \text{Hom}_G(V, W)$, and allows us to define

$$\text{Hom}_G(V, W) := \{T: V \rightarrow W \text{ such that } T \text{ is } G\text{-linear}\} \subseteq \text{Hom}(V, W).$$

Remark 1.11.8 — From the above, we also have that:

$$\dim \text{Hom}(V, W)^G = \begin{cases} 1 & V \cong W \\ 0 & V \not\cong W \end{cases}$$

We know that $V^G \subseteq V$, but we now want to ask whether there is a map from V to V^G . The construction goes by averaging an element v over G (which requires G to be finite).

Proposition 1.11.9

For finite G , given any representation (V, ρ_V) , there is a G -linear projection $S: V \rightarrow V^G$ given by the formula

$$\text{avg}(v) := \frac{1}{|G|} \sum_{g \in G} \rho_V(g)(v).$$

Proof. As in the proof of Maschke's theorem, note that $\text{avg}(v) = v$ for all $v \in V$. Moreover, $\rho_V(g)(\text{avg}(v)) = \text{avg}(v)$, since

$$\{\rho_V(hg) \mid g \in G\} = \{\rho_V(g) \mid g \in G\}.$$

This implies that

$$\rho_V(h) \sum_{g \in G} \rho_V(g)(v) = \sum_{g \in G} \rho_V(g)(v) \implies \rho_V(h)(\text{avg}(v)) = \text{avg}(v).$$

Thus, the image of avg is exactly V^G and avg is the identity restricted to V^G . By lemma 1.5.18, we get that avg is a projection map.

To see that avg is G -linear, we have that

$$\text{avg}(\rho_V(g)(v)) = \text{avg}(v).$$

Therefore,

$$\rho_V(g)(\text{avg}(v)) = \text{avg}(\rho_V(g)(v)) = \text{avg}(v).$$

□

Remark 1.11.10 — Note as in Maschke's theorem the necessity of being able to invert $|G|$ here (which we get by working over $\mathbb{C}$).

In the proof of Maschke's theorem we applied this proposition replacing V by $\text{Hom}(V, V)$, v by a transformation $T: V \rightarrow V$, and $\text{avg}(v)$ by $\tilde{T}$.

§1.12 Decomposition of a Representations

Given V , which is a finite dimensional representation of a finite group G , we write

$$V \cong V_1^{r_1} \oplus \cdots \oplus V_m^{r_m},$$

where V_i are irreducible representations, and pairwise non-isomorphic ($V_i \not\cong V_j$, for all $i \neq j$).

Proposition 1.12.1

We have $r_i = \dim \text{Hom}_G(V_i, V) = \dim \text{Hom}_G(V, V_i)$.

Proof. This follows from Schur's lemma together with the following basic lemma. □

Lemma 1.12.2

We have

$$\mathrm{Hom}(V, U \oplus W) \cong \mathrm{Hom}(V, U) \oplus \mathrm{Hom}(V, W)$$

and

$$\mathrm{Hom}(V \oplus U, W) \cong \mathrm{Hom}(V, W) \oplus \mathrm{Hom}(U, W).$$

Proof. This follows by linear algebra. In the first, we can send $T \in \mathrm{Hom}(V, U \oplus W)$ to (T_1, T_2) such that $T(v) = T_1(v) + T_2(v)$. For the second, we can say $T \in \mathrm{Hom}(V \oplus U, W)$, we can send it to $(T|_V, T|_W)$. $\square$

Remark 1.12.3 — Note that the same holds for Hom_G .

Proposition 1.12.4

There exists a unique decomposition of a finite dimensional representation V of a finite group G into irreducibles, up to isomorphism and reordering.

Proof. Let

$$V \cong V_1^{r_1} \oplus \cdots \oplus V_m^{r_m}.$$

This follows because, up to isomorphism, each irreducible representation V' must occur exactly $r' = \dim \mathrm{Hom}_G(V', V)$ times. If $r' > 0$, since the V_i are all assumed to be (pairwise) nonisomorphic, there is exactly one i such that $V' \cong V_i$ and $r_i = r'$. For more detail, see the below.

Let W be an irreducible representation of G , and

$$\mathrm{Hom}_G(W, V) \cong \mathrm{Hom}_G(W, V_1^{r_1} \oplus \cdots \oplus V_m^{r_m}).$$

The lemma above then implies that:

$$\mathrm{Hom}_G(W, V_1^{r_1} \oplus \cdots \oplus V_m^{r_m}) \cong \bigoplus_{i=1}^m \mathrm{Hom}_G(W, V_i)^{r_i}$$

Schur's lemma then tells us that

$$\bigoplus_{i=1}^m \mathrm{Hom}_G(W, V_i)^{r_i} \cong \begin{cases} \mathbb{C}^{r_i} & W \cong V_i \\ 0 & W \not\cong V_i, \forall i \end{cases}.$$

Now define the **multiplicity** of W in V as $\dim \mathrm{Hom}_G(W, V) =: \mathrm{mult}(W, V)$. We now get that

$$V \cong \bigoplus_{\substack{\text{all irreducible representation} \\ W/\cong}} W^{\mathrm{mult}(W, V)}.$$

If W is irreducible, then

$$\mathrm{mult} = \begin{cases} r_i & W \cong V_i, \text{ some } i \\ 0 & \text{otherwise} \end{cases}.$$

This then tells us that the decomposition we had is unique, up to isomorphism and reordering $V_1, \dots, V_m$. $\square$

Corollary 1.12.5

We have that

$$\dim \operatorname{End}_G(V) = \dim \bigoplus_{i,j} \operatorname{Hom}(V_i, V_j)^{r_i r_j} = \sum_{i=1}^m r_i^2.$$

Proof. Decompose V as $V \cong V_1^{r_1} \oplus \cdots \oplus V_m^{r_m}$. Applying lemma 1.12.2, we get

$$\operatorname{End}_G(V) \cong \operatorname{Hom}_G(V_1^{r_1} \oplus \cdots \oplus V_m^{r_m}, V) \cong \bigoplus_{i=1}^m \operatorname{Hom}_G(V_i, V)^{r_i}.$$

Taking dimensions then proves the result. $\square$

Example 1.12.6

If $\dim \operatorname{End}_G(V) = 3$ implies that $V \cong V_1 \oplus V_2 \oplus V_3$.

Example 1.12.7

If $\dim \operatorname{End}_G(V) = 4$, then there are two cases: $V \cong V_1^2 = V_1 \oplus V_1$ or $V = V_1 \oplus V_2 \oplus V_3 \oplus V_4$.

Remark 1.12.8 — The same argument shows that as a $\mathbb{C}$ -algebra,

$$\operatorname{End}_G(V) \cong \bigoplus M_{r_i}(\mathbb{C}).$$

§1.13 Decomposition of $\mathbb{C}[G]$

Remark 1.13.1 — Note that we have

$$\dim \operatorname{Hom}_G(\mathbb{C}[G], V) = \dim V,$$

for every representation (V, ρ) by the isomorphism $\operatorname{Hom}_G(\mathbb{C}[G], V) \cong V$ from exercise 1.4.6. We can apply this result to irreducible representations to get the following:

Corollary 1.13.2

There are finitely many nonisomorphic irreducible representations of G . Calling them $(V_1, \rho_1), \dots, (V_m, \rho_m)$ up to isomorphism, there exists a G -linear isomorphism

$$\mathbb{C}[G] \cong \bigoplus_{i=1}^m V_i^{\dim(V_i)}$$

Proof. By Maschke's theorem, we can write $\mathbb{C}[G] \cong V_1^{r_1} \oplus \cdots \oplus V_m^{r_m}$ for some nonisomorphic irreducible representations V_i . Now if V is any irreducible representation, remark 1.13.1 implies that $V \cong V_i$ for some i with $r_i = \dim V$. Thus, every irreducible

representation is isomorphic to one of the V_i . Moreover, $r_i = \dim V = \dim V_i$, so we've proven true. $\square$

Taking dimensions, we obtain the following important identity.

Corollary 1.13.3

We have

$$\sum_i (\dim V_i)^2 = |G|.$$

Proof. This is because $\dim(V \oplus W) = \dim V + \dim W$ and $\dim V^r = r \cdot \dim V$. $\square$

Corollary 1.13.4

Unless G is trivial, the dimension of every irreducible representation is less than $\sqrt{|G|}$.

Proof. For every irreducible representation (V, ρ) other than the trivial representation,

$$|G| = \dim \mathbb{C}[G] = \dim \left(\mathbb{C} \oplus_i V_i^{\dim V_i} \right) \geq 1 + (\dim V)^2$$

$\square$

Remark 1.13.5 — Note that we get $\dim V = \sqrt{|G|}$ if and only if there exists a unique irreducible representation $|G| = (\dim V)^2$, which implies that V is trivial $G = \{e\}$ (this can be proven by observing that the trivial representation is always present).

§1.14 Examples: S_3 , S_4 , and Dihedral Groups

§1.14.i S_3 and S_4

Here we classify irreducible representations of S_3 , the Dihedral Groups, and of S_4 up to isomorphism. The following result will be important. Recall that the reflection representation of $G = S_n$ is the representation (V, ρ) with

$$V = \{(a_1, \dots, a_n) \mid a_1 + \dots + a_n = 0\} \subseteq \mathbb{C}^n$$

and ρ is obtained by restricting the permutation representation. Explicitly,

$$\rho(\sigma)(a_1, \dots, a_n) = (a_{\sigma^{-1}(1)}, \dots, a_{\sigma^{-1}(n)}).$$

Proposition 1.14.1

For $n \geq 2$, the reflection representation (V, ρ) is irreducible.

Proof. Let $W \subseteq V$ be a nonzero subrepresentation, and let $0 \neq w \in W$. Say $w = (a_1, \dots, a_n)$ with $\sum a_i = 0$. Then, we know there exists i, j such that $a_i \neq a_j$. Then, note that $w - \rho((ij))w \in W$. We have

$$w - \rho((ij))w = (a_i e_i + a_j e_j) - (a_j e_i + a_i e_j) = (a_i - a_j)(e_i - e_j).$$

As $a_i \neq a_j$, we have that $e_i - e_j$ is in W . Thus, for any $i', j' \in \{1, \dots, n\}$ with $i' \neq j'$, by choosing $\sigma \in S_n$ such that $\sigma(i') = i$ and $\sigma(j') = j$, we have

$$e_{i'} - e_{j'} = e_{\sigma^{-1}(i)} - e_{\sigma^{-1}(j)} = \rho(\sigma)(e_i - e_j) \in W.$$

Therefore, $\{e_i - e_j \mid i, j \in \{1, \dots, n\}, i \neq j\}$ spans V , and so $W = V$. □

Remark 1.14.2 — The reflection representation has dimension $n - 1$.

Remark 1.14.3 — In characteristic 2, $(1, 1) \neq 0$, but $1 + 1 = 0$, even though $1 \neq 1$. This shows that the argument above doesn’t hold for arbitrary fields. In order for the argument in the proof to work, we need the condition that $\forall a \in F$ (for an arbitrary field F),

$$na = \sum_{i \in \{1, \dots, n\}} a = 0 \implies a = 0.$$

This works as long as $\text{char}(F) \nmid n$.

Example 1.14.4 (Irreducible representations of S_3)

We know that there are exactly two one-dimensional representations: the trivial and the sign representation. We also have, by proposition 1.14.1, the two-dimensional reflection representation, which is also irreducible. Since $1^2 + 1^2 + 2^2 = 6 = |S_3|$, these are all of the representations.

Example 1.14.5 (Irreducible Representations of S_4)

We next classify the irreducible representations of S_4 . Again, there are two one-dimensional representations. If $d_1 = 1, d_2 = 1, d_3, \dots, d_r$ are the dimensions of the irreducible representations, then

$$1^2 + 1^2 + d_3^2 + \dots + d_r^2 = 24.$$

Thus $d_3^2 + \dots + d_r^2 = 22$, with all squares appearing being 4, 9, or 16. We can’t have 16 appear since 6 is not a sum of these squares. With 4 and 9 we can’t have an odd number of 9’s appearing, and also clearly can’t have only 4’s appearing, so the only possibility is $4 + 9 + 9$. Thus, up to reordering, $d_3 = 2, d_4 = 3, d_5 = 3$.

The above example motivates the following constructions in the exercise.

- Exercise 1.14.6.**
1. Let (V, ρ) be a representation of a group G and $\theta: G \rightarrow \mathbb{C}^\times$ a one-dimensional representation. Show that (V, ρ') given by $\rho'(g) = \theta(g)\rho(g)$ is also a representation, and that it is irreducible if and only if (V, ρ) is irreducible.
 2. Let (V, ρ) be a representation of G and $\phi: H \rightarrow G$ a homomorphism of groups. Show that $(V, \rho \circ \phi)$ is a representation of H . In the case that (V, ρ) is irreducible and ϕ is

surjective, show that $(V, \rho \circ \phi)$ is also irreducible.

Example 1.14.7

We've already seen an example of the second statement: $q_{ab}: H \rightarrow H_{ab}$, the homomorphism from a group to its abelianization.

Example 1.14.8

Continuing from the example with S_4 , we can realize these representations. One of the three-dimensional representations is the reflection representation, call it (V, ρ) , with $V = \{(a_1, \dots, a_4) \mid a_1 + \dots + a_4 = 0\} \subseteq \mathbb{C}^4$. Another one can be given by multiplying by the sign: (V, ρ') with $\rho'(\sigma) = \text{sign}(\sigma)\rho(\sigma)$ (it is either a permutation matrix or just like a permutation matrix but with -1 's replacing 1 throughout). Finally, we need a two-dimensional irreducible representation. We can get it from the reflection representation of S_3 , since there is a surjective homomorphism $q: S_4 \rightarrow S_3$, given by the action of S_4 by permuting the elements of the conjugacy class $\{(14)(23), (13)(24), (12)(34)\}$ (or we can explicitly write the formula, $q((12)) = (12)$, $q((23)) = (23)$, $q((34)) = (12)$ and the elements $(12), (23), (34)$ generate S_4). Let (W, ρ_W) be the reflection representation of S_3 , $\rho_W: S_3 \rightarrow GL(W)$ with $\dim W = 2$. Then we can take $(W, \rho_W \circ q)$, and since (W, ρ_W) is irreducible, so is this one by the next exercise.

Example 1.14.9

Let $G = C_4 = \{1, g, g^2, g^3\}$ and $\rho(g) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ the rotation matrix. Then for the first part of the exercise above, we can let $\theta(g) = i \in \mathbb{C}$, and then $\rho'(g) = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$. For the second part, we can consider the automorphism $\phi: G \rightarrow G$ given by $\phi(g) = g^{-1} = g^3$, and then $\rho \circ \phi = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

§1.14.ii Dihedral Group

Example 1.14.10

We can classify the representations of D_{2n} for low n . For $n = 3$ we have $D_6 = S_3$ so that is already done. Let's try D_8 . By example 1.10.2, there are four one-dimensional representations, so

$$8 = 1^2 + 1^2 + 1^2 + 1^2 + 2^2$$

shows that there can be exactly one two-dimensional irreducible representation, which is then the one of example 1.1.6.

Recall that we have a family $(V_i)_{1 \leq i < \frac{n}{2}}$ of irreducible 2 dimensional representations of D_n . For example, for $n = 3$, we have just one i ($i = 1$). Setting $\theta = \frac{2\pi}{3}$, we have

$$\rho(x) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad \rho(y) = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$$

In general, if we set $\theta = \frac{2\pi}{n}$ and write $D_n = \langle x, y \mid x^n = y^2 = 1, yxy^{-1} = x^{-1} \rangle$, the two dimensional representations of D_n , say $(V_j, \rho_j)_{1 \leq j < \frac{n}{2}}$, are given by

$$\rho_j(x) = \begin{pmatrix} \cos j\theta & -\sin j\theta \\ \sin j\theta & \cos j\theta \end{pmatrix}, \quad \rho_j(y) = \begin{pmatrix} \cos 2j\theta & \sin 2j\theta \\ \sin 2j\theta & -\cos 2j\theta \end{pmatrix}$$

Given this, we have four 1 dimensional subrepresentations when n is even and two 1-dimensional representations when n is odd. In general, for odd n ($=: 2k+1$), we have a sum-of-squares of dimensions:

$$2 \cdot 1^2 + k \cdot 2^2 = 2(2k+1) = |D_{2k+1}|.$$

Similarly, for even n ($=: 2k$), we have a sum-of-squares of dimensions as well:

$$4 \cdot 1^2 + (k-1) \cdot 2^2 = 4(1 + (k-1)) = 2(2k) = |D_{2k}|.$$

Note that by corollary 1.13.3, there are no other irreducible representations of D_n , as the sum is exhausted.

Exercise 1.14.11. Find all irreducible representations of D_n . Other than the one-dimensional representations (see example 1.10.2), show that they are all obtained from the one of example 1.1.6, by the construction of exercise 1.14.6. In more detail, consider $(\mathbb{C}^2, \rho \circ \phi_j)$ for the endomorphisms $\phi_j: D_n \rightarrow D_n$, $\phi_j(x^a y^b) = x^{ja} y^b$ (note that these are not automorphisms). Show that they are irreducible for $1 \leq j < \frac{n}{2}$. Moreover, show that they are nonisomorphic for these values of j , by showing that $\text{Tr}(\rho(x^j)) \neq \text{Tr}(\rho(x^k))$ for $1 \leq j < k < \frac{n}{2}$.

Remark 1.14.12 — Here is a useful observation: Since the number of one-dimensional representations of a finite group G equals the size of G_{ab} (corollary 1.10.13), it is in particular a factor of $|G|$. If G is not itself abelian, this is a proper factor.

Exercise 1.14.13. Classify the irreducible representations of A_4 . First, using the remark above, show that the possibilities are one of the following:

1. $1^2 + 1^2 + 1^2 + 3^2$
2. $1^2 + 1^2 + 1^2 + 1^2 + 2^2 + 2^2$

Then, use the surjective homomorphism $q: S_4 \rightarrow S_3$, restricted to A_4 , to get $q|_{A_4}: A_4 \rightarrow A_3$, a surjective homomorphism with abelian image. Using corollary 1.10.10, now show that we are in the first case. Then, find the one-dimensional representations. Finally, prove that the restriction of the three-dimensional reflection representation of S_4 to A_4 is irreducible, completing the classification.

§1.15 The Number of Irreducible Representations

Let $(V_1, \rho_1), \dots, (V_m, \rho_m)$ be representatives of the isomorphism classes of irreducible representations of a finite group G . Then, recall that corollary 1.13.3 says that

$$|G| = \sum_{i=1}^m (\dim V_i)^2.$$

This means that $\mathbb{C}[G]$ and $\bigoplus_{i=1}^m \text{End}(V_i)$ have the same dimension, so there is some isomorphism between them.

Remark 1.15.1 — Note that $\text{End}(V_i)$ has dimension $(\dim V_i)^2$.

A remarkable fact is there is a *canonical* isomorphism between these vector spaces. We have that $\rho_i: G \rightarrow GL(V_i) \subseteq \text{End}(V_i)$. By abuse of notation, also write $\rho_i: \mathbb{C}[G] \rightarrow \text{End}(V_i)$, given by

$$\rho_i \left(\sum_{g \in G} a_g g \right) := \sum_{g \in G} a_g \rho_i(g) \in \text{End}(V_i).$$

That is, we extend from a standard basis. Each of these maps $\rho_i: G \rightarrow \text{End}(V_i)$ is clearly linear. Let $\Phi = (\rho_1, \dots, \rho_m): \mathbb{C}[G] \rightarrow \bigoplus_{i=1}^m \text{End}(V_i)$. This is also clearly linear.

Theorem 1.15.2

The map Φ is a linear isomorphism.

Proof. Since the two spaces have the same dimension, it suffices to prove Φ is either injective or surjective. We will show here that Φ is injective. Let $f = \sum_{g \in G} a_g g \in \mathbb{C}[G]$ and suppose that

$$\Phi(f) = 0.$$

Then, for each i , we must have that

$$0 = \rho_i(f) = \rho_i \left(\sum_{g \in G} a_g g \right) = \sum_{g \in G} a_g \rho_i(g).$$

Since the (V_i, ρ_i) cover representatives for all isomorphism classes of irreducible representations of G , this shows that for any irreducible representation (W, ρ_W) of G ,

$$0 = \sum_{g \in G} a_g \rho_W(g).$$

Since every (finite dimensional) representation of G can be decomposed as a direct sum of irreducible representation, in fact for any (W, ρ_W) that is a finite dimensional (not necessarily irreducible) representation of G ,

$$0 = \sum_{g \in G} a_g \rho_W(g).$$

In particular, applying this fact for $(\mathbb{C}[G], \rho)$, the regular representation,

$$0 = (0)(e) = \left(\sum_{g \in G} a_g \rho(g) \right)(e) = \sum_{g \in G} a_g (ge) = \sum_{g \in G} a_g g = f.$$

Therefore, if $\Phi(f) = 0$, $f = 0$, so we've shown injectivity. $\square$

Remark 1.15.3 — This tells us that $\mathbb{C}[G] \cong \bigoplus_{i=1}^m \text{End}(V_i)$.

Remark 1.15.4 — We have constructed this Φ , but it is *mysterious*.

Remark 1.15.5 — We have also not yet shown that Φ is G -linear.

Example 1.15.6

For instance, the irreducible representations of S_3 are as follows:

- $\rho_+ : S_3 \rightarrow \mathbb{C}^\times$ (trivial)
- $\rho_- : S_3 \rightarrow \mathbb{C}^\times$ (sign)
- $\rho : GL_2(\mathbb{C}) \rightarrow \mathbb{C}^\times$ (reflection)

Theorem 1.15.2 tells us that there exists a unique six-tuple $(a_\sigma)_{\sigma \in S_3}$ such that

$$\begin{aligned} \sum_{\sigma} a_{\sigma} &= \sum_{\sigma} a_{\sigma} \rho_+(\sigma) = 0 \in \mathbb{C} \\ \sum_{\sigma} a_{\sigma} \rho(\sigma) &= 0 \in M_2(\mathbb{C}) \\ \sum_{\sigma} a_{\sigma} \rho_-(\sigma) &= 1 \in \mathbb{C} \end{aligned}$$

These have 1, $2^2 = 4$, and 1 constraint(s) respectively, so this makes sense. However, there are no obvious formula for (a_σ) .

Recall that each $\text{End}(V)$ (for (V, ρ) a representation of G) is naturally a representation of G : for $f \in \text{End}(V)$,

$$\rho_{\text{End}(V)}(g) = \rho(g) \circ f \circ \rho(g)^{-1}.$$

Therefore, $\bigoplus_{i=1}^m \text{End}(V_i)$ is also naturally a representation of G . We now ask whether $\Phi : \mathbb{C}[G] \rightarrow \bigoplus_{i=1}^m \text{End}(V_i)$ G -linear? The answer is yes, but not the way we think it is.

Lemma 1.15.7

For $g, h \in G$, $\rho_{\bigoplus_i \text{End}(V_i)}(g)(\Phi(h)) = \Phi(ghg^{-1})$.

Proof. Let $i \in \{1, \dots, m\}$, and compare the $\text{End}(V_i)$ components of the LHS and RHS. The LHS gives

$$\rho_{\text{End}(V_i)}(g)(\rho_i(h)) = \rho_i(g)\rho_i(h)\rho_i(g)^{-1} = \rho_i(ghg^{-1}).$$

The RHS gives

$$\rho_i(ghg^{-1}),$$

so the two agree in each component; therefore, are the same. $\square$

Definition 1.15.8. The **adjoint** representation $(\mathbb{C}[G], \rho_{\text{ad}})$ is given by

$$\rho_{\text{ad}}(g) \left(\sum_{h \in G} a_h h \right) = \sum_{h \in G} a_h ghg^{-1}.$$

Exercise 1.15.9. Show that this formula gives a well defined representation.

Corollary 1.15.10

Let G be finite and $(V_1, \rho_1), \dots, (V_m, \rho_m)$ are the representatives of the isomorphism classes of irreducible representations of G . Then, $\Phi: (\mathbb{C}[G], \rho_{\text{ad}}) \xrightarrow{\sim} \bigoplus_{i=1}^m \text{End}(V_i)$ defined by $\Phi = (\rho_1, \dots, \rho_m)$, where each $\rho_i: \mathbb{C}[G] \rightarrow \text{End}(V_i)$ linearly extends to $\rho_i: G \rightarrow GL(V_i)$, is an isomorphism of representations.

Remark 1.15.11 — Note that we have a finite m , since G is finite (sum of squares of dimension is equal to $|G|$).

Remark 1.15.12 — We focus on $\left(\bigoplus_{i=1}^m \text{End}(V_i)\right)^G$. We have

$$\left(\bigoplus_{i=1}^m \text{End}(V_i)\right)^G = \bigoplus_{i=1}^m \text{End}(V_i)^G \cong \bigoplus_{i=1}^m \mathbb{C} = \mathbb{C}^m,$$

by Schur's lemma. Therefore, we have a (canonical) isomorphism

$$(\mathbb{C}[G], \rho_{\text{ad}})^G \xrightarrow{\sim} \mathbb{C}^m.$$

This is contained in the following corollary.

Corollary 1.15.13

We have

$$\Phi: (\mathbb{C}[G], \rho_{\text{ad}})^G \xrightarrow{\sim} \left(\bigoplus_{i=1}^m \text{End}(V_i)\right)^G = \bigoplus_{i=1}^m \text{End}(V_i)^G \cong \mathbb{C}^m$$

Proof. First the LHS: We study $(\mathbb{C}[G], \rho_{\text{ad}})^G$. Let $\sum_{h \in G} a_h h \in (\mathbb{C}[G], \rho_{\text{ad}})^G$. Let $g \in G$. Then,

$$\sum_{h \in G} a_h h = \rho_{\text{ad}}(g) \left(\sum_{h \in G} a_h h \right) = \sum_{h \in G} g h g^{-1} = \sum_{h' \in G} a_{g^{-1}h'g} h'.$$

Note that the last equality follows from sends $h \mapsto g h g^{-1}$, which is a bijection. Matching coefficients, then $a_h = a_{g^{-1}h'g}$. Therefore, $(\mathbb{C}[G], \rho_{\text{ad}})^G$ consists of

$$\left\{ \sum_{h \in G} a_h h \mid \forall g, h \in G, a_h = a_{g^{-1}h'g} \right\}.$$

The condition $\forall g, h \in G, a_h = a_{g^{-1}h'g}$, means that the function $g \mapsto a_g$ is constant on conjugacy classes. Therefore, $\dim(\mathbb{C}[G], \rho_{\text{ad}})^G$ is equal to the number of conjugacy classes of G .

Now the RHS: We have

$$\left(\bigoplus_{i=1}^m \text{End}(V_i) \right)^G = \bigoplus_{i=1}^m \text{End}(V_i)^G = \bigoplus_{i=1}^m \text{End}_G(V_i) = \bigoplus_{i=1}^m \mathbb{C} \cong \mathbb{C}^m.$$

Note that $\text{End}(V_i)^G = \text{End}_G(V_i)$. Therefore, taking dimensions, we have that

$$\dim \left(\bigoplus_{i=1}^m \text{End}(V_i) \right) = m,$$

which is the number of isomorphism classes of irreducible representations of G . $\square$

Remark 1.15.14 — The dimensions of the two spaces above are equal, but there is no combinatorial argument where we identify one item with another—there is no *clear* bijection.

Theorem 1.15.15

Let G be a finite group and $(V_1, \rho_1), \dots, (V_m, \rho_m)$ a full set of irreducible representations. Then m equals the number of conjugacy classes of G .

We now just go through some examples to illustrate this theorem.

Example 1.15.16 (Abelian Groups)

We've seen that the number of representations is equal to $|G|$, and every element is its own conjugacy class (so is equal).

Example 1.15.17 (Even Dihedral)

D_{2n} had four 1-dimensional irreducible representations and $n - 1$ 2-dimensional irreducible representations. Thus, there are $n + 3$ irreducible representations in total.

There are $n + 1$ rotation conjugacy classes: $1, x^{\pm 1}, x^{\pm 2}, \dots, x^n$. There are 2 reflection conjugacy classes:

$$\{y, x^2y, x^4y, \dots, x^{2n-2}y\}, \quad \{xy, x^3y, \dots, x^{2n-1}y\}.$$

Thus, there are $n + 3$ conjugacy classes, so the cardinalities match.

Example 1.15.18 (Odd Dihedral)

D_{2n+1} had two 1-dimensional irreducible representations and n 2-dimensional irreducible representations. Thus, there are $n + 2$ irreducible representations in total.

There are $n + 1$ rotation conjugacy classes: $1, x^{\pm 1}, x^{\pm 2}, \dots, x^{\pm n}$. There is only one reflection conjugacy class. Thus, there are $n + 2$ conjugacy classes, so the cardinalities match.

Example 1.15.19 (S_3)

S_3 had three irreducible representations: the trivial, sign, and reflection representations. There are three conjugacy classes: $()$, (12) , (123) .

Example 1.15.20 (S_4)

S_3 had five irreducible representations: the trivial, sign, and reflection, reflection $\times$ sign, pullback of reflection representation of S_3 representations. There are five conjugacy classes: $()$, (12) , (123) , $(12)(34)$, (1234) .

§1.16 Duals and Tensor Products

Recall that, given a vector space V over $\mathbb{C}$, the dual vector space is

$$V^* := \text{Hom}((V, \rho_V), (\mathbb{C}, \rho_{\text{triv}})).$$

When (V, ρ_V) is a representation of G , we also obtain a representation (V^*, ρ_{V^*}) . Let us write explicitly the formula: for $f \in V^*$, $g \in G$ and $v \in V$:

$$(\rho_{V^*}(g)(f))(v) = \rho_{\text{triv}}(g)f(\rho_V(g)^{-1}(v)) = f(\rho_V(g)^{-1}(v)).$$

That is, $\rho_{V^*}(g)f = f \circ \rho(g)^{-1}$. This is an equality of maps from $V \rightarrow \mathbb{C}$.

Example 1.16.1

Let $\rho: G \rightarrow \mathbb{C}^\times$ be a one-dimensional representation. Let us compute its dual. Let $V = \mathbb{C}$. We obtain (V, ρ_V) where $\rho_V(g)(v) = \rho(g)v$ for all $g \in G$ and $v \in V$. Then the dual (V^*, ρ_{V^*}) satisfies $\rho_{V^*}(g)(f) = f \circ \rho_V(g)^{-1} = \rho(g)^{-1}f$, for $f: \mathbb{C} \rightarrow \mathbb{C}$ linear. That is, dualizing one-dimensional representations inverts $\rho(g)$ for all g .

Recall that given a vector space V , we have a canonical inclusion (injective linear map) from $V \hookrightarrow (V^*)^*$, by $v \mapsto \phi_v$. This inclusion goes as follows. $\phi_v \in (V^*)^*$, so $\phi_v: V^* \rightarrow \mathbb{C}$ has, for all $f \in V^*$, $\phi_v(f) = f(v)$. Check that this is linear in f , and we show injectivity. Suppose that $\phi_v = 0$. Then, for every $f \in V^*$,

$$f(v) = \phi_v(f) = 0.$$

We note that if for all $f: V \rightarrow \mathbb{C}$, $f(v) = 0$, then $v = 0$ (prove this!). It then follows by this fact that $v = 0$, and so we get injectivity.

Remark 1.16.2 — If V is finite dimensional, we have that $V \xrightarrow{\sim} (V^*)^*$.

Remark 1.16.3 — For V a representation of G , $V \hookrightarrow (V^*)^*$ is a homomorphism of representations.

Proposition 1.16.4

We have a canonical (and G -linear) inclusion $V \hookrightarrow (V^*)^*$ of vector spaces, $v \mapsto \phi_v$, $\phi_v(f) := f(v)$. This is an isomorphism if V is finite-dimensional.

Proof. The given map is obviously linear. To see it is injective, suppose that $v \neq 0$. Then we can extend v to a basis of V , and therefore define a linear map $f \in V^*$ such that $f(v) = 1$ (and say $f(v') = 0$ for other elements of the basis). Then $\phi_v(f) = f(v) \neq 0$, so $\phi_v \neq 0$. To check G -linearity, we trace through the definitions:

$$\rho_{(V^*)^*}(g)(\phi_v)(f) = \phi_v(\rho_{V^*}(g)^{-1}(f)) = \phi_v(f \circ \rho_V(g)) = f(\rho_V(g)(v)) = \phi_{\rho_V(g)(v)}(f).$$

□

Remark 1.16.5 — Let us work out the formula in terms of matrices for $\rho_{V^*}(g)$. Suppose $\mathcal{B} = \{v_1, \dots, v_n\}$ is a basis of V , and let $\mathcal{C} = \{f_1, \dots, f_n\}$ be the dual basis, given by

$$f_i(v_j) = \delta_{ij},$$

where δ_{ij} denotes the Kronecker delta function. We now compute

$$\rho_{V^*}(g)(f_i)(v_j) = f_i(\rho_V(g^{-1})(v_j)).$$

The LHS is the (j, i) -th entry of $[\rho_{V^*}(g)]_{\mathcal{C}}$. The RHS, on the other hand, is the (i, j) -th entry of $[\rho_V(g)]_{\mathcal{B}} = ([\rho_V(g)]_{\mathcal{B}})^{-1}$. Putting this together, we get that

$$[\rho_{V^*}(g)]_{\mathcal{C}} = (([\rho_V(g)]_{\mathcal{B}})^{-1})^T,$$

where the superscript of T denotes the transpose.

§1.16.i Definition of the Tensor Product of Vector Spaces

Next, given representations (V, ρ_V) and (W, ρ_W) of G , we want to define the representation $(V \otimes W, \rho_{V \otimes W})$. To do this, we first must define the tensor product. The best definition is by a *universal property*. We seek a vector space $V \otimes W$ and a bilinear map $V \times W \rightarrow V \otimes W$ which is *universal* amongst bilinear maps with source $V \times W$. Here, recall that bilinear map $F: V \times W \rightarrow U$ is one that satisfies the following:

- $F(av, w) = aF(v, w) = F(v, aw)$
- $F(v + v', w) = F(v, w) + F(v', w)$
- $F(v, w + w') = F(v, w) + F(v, w')$

A bilinear map $\Psi: V \times W \rightarrow V \otimes W$ is **universal** if, for any other bilinear map $F: V \times W \rightarrow U$, there exists a unique $\tilde{F}: V \otimes W \rightarrow U$ such that $F = \tilde{F} \circ \Psi$, i.e., the following diagram is commutative:

$$\begin{array}{ccc} V \times W & \xrightarrow{\Psi} & V \otimes W \\ \downarrow F & \swarrow \exists! \tilde{F} & \\ V \otimes W & & \end{array}$$

Given a universal Ψ , we denote $v \otimes w := \Psi(v, w) \in V \otimes W$, for all $v, w \in w$.

Example 1.16.6

Let us show that if $W = \mathbb{C}$, then the universal $V \otimes \mathbb{C}$ is nothing but V itself. Namely, we show that $\Psi: V \times \mathbb{C} \rightarrow V$, $(v, a) \mapsto av$, is universal. Given any bilinear map $F: V \times \mathbb{C} \rightarrow U$, we have $F(v, a) = aF(v, 1)$. Therefore, we have $F = \Psi \circ \tilde{F}$, for $\tilde{F}: V \rightarrow U$ the map $\tilde{F}(v) = F(v, 1)$. This is clearly linear and unique.

For theoretical purpose, i.e., proving anything about the tensor product, it is cleanest and most elegant to let $V \otimes W$ be any universal $\Psi: V \times W \rightarrow V \otimes W$. However, we must prove that this exists and is unique (in an appropriate sense).

Definition 1.16.7. The **tensor product** $V \otimes W := \mathbb{C}[V \times W]/R$ of vector space V and W is the quotient of the vector space

$$\mathbb{C}[V \times W] := \left\{ \sum_{i=1}^n a_i(v_i, w_i) \mid a_i \in \mathbb{C}, v_i \in V, w_i \in W \right\},$$

where the pairs (v_i, w_i) are distinct, with basis $V \times W$, by the subspace R spanned by the elements (called **relations**):

- $a(v, w) - (av, w)$
- $a(v, w) - (v, aw)$
- $(v + v', w) - (v, w) - (v', w)$
- $(v, w + w') - (v, w) - (v, w')$

The map $\Phi: V \times W \rightarrow V \otimes W$ is given by $\Phi(v, w) := (v, w) + R$, which we denote $v \otimes w \in V \otimes W$.

Remark 1.16.8 — Φ is bilinear, since for $a \in \mathbb{C}, v \in V, w \in W$, we have as follows:

$$\begin{aligned} \Phi(av, w) &= (av, w) + R \\ &= (av, w) + (a(v, w) - (av, w)) + R \\ &= a(v, w) + R \\ &= a\Phi(v, w) \end{aligned}$$

We can perform a similar argument for the remaining conditions.

Remark 1.16.9 — Note that the elements $v \otimes w$ span $V \otimes W$, since $a(v \otimes w) = (av) \otimes w$.

Example 1.16.10 (Example 1.16.6 Revisited)

We revisit example 1.16.6 with the concrete definition above. Let $W = \mathbb{C}$. Then, we have an isomorphism $\phi: V \otimes \mathbb{C} \rightarrow V$ given by $(v \otimes a) \mapsto av$. First, let us check that this is well-defined. Obviously, we have a map $\tilde{\phi}: \mathbb{C}[V \times \mathbb{C}] \rightarrow V$ given by $(v, a) \mapsto av$, and we just need to observe the relations are satisfied. This is true because $(v, a) \mapsto av$ is bilinear; for instance,

$$\tilde{\phi}((a(v, b) + a'(v', b))) = abv + a'bv' = \tilde{\phi}(av + a'v, b).$$

To see that ϕ is an isomorphism, note first that it is obviously surjective and linear. Let us show injectivity. Suppose some element $x = \sum_i a_i(v_i \otimes v_i)$ is in the kernel. Applying ϕ , we get that $\sum_i a_i b_i v_i = 0$. However, also $x = \sum_i (a_i b_i b_i) \otimes 1$, applying linearity. Therefore, $x = 0 \otimes 1 = 0$.

Proposition 1.16.11 (Universal Property of the Tensor Product)

The map $\Phi: V \times W \rightarrow V \otimes W$ is universal amongst bilinear maps with source $V \times W$. The composition with Φ induces an isomorphism $\text{Hom}(V \otimes W, U) \cong \text{Hom}_{\text{bilinear}}(V \times W, U)$ for every vector space U .

Proof. First note that Φ is bilinear since R contains all bilinear relations. Construct the linear map $F': \mathbb{C}[V \times W] \rightarrow U$ by linear extension of F . Because F is bilinear, the kernel includes the rotations, as in the preceding proof. Therefore, we get a well-defined map $\tilde{F}: V \otimes W \rightarrow U$ (as R is contained in the span of the rotational elements). This map is unique since $V \otimes W$ is spanned by $V \times W$. Finally, for every linear map $V \otimes W \rightarrow U$, the composition with Φ is bilinear by construction. So we get the desired isomorphism. $\square$

The definition and universal property are rather abstract, but we will now use them to deduce all the nice concrete properties of tensor product in an elegant fashion, that you may prefer to use to think about the tensor product when you don't need to prove something about it. (In practice, the universal property is only usually needed to prove that a map from $V \otimes W$ is well-defined.)

Proposition 1.16.12

Suppose that $\mathcal{B}$ is a basis of V and $\mathcal{C}$ is a basis of W . Then, $\Phi(\mathcal{B} \times \mathcal{C}) = \{v \otimes w \mid v \in \mathcal{B}, w \in \mathcal{C}\}$ is a basis of $V \otimes W$. In particular, we have

$$\dim(V \otimes W) = (\dim V)(\dim W).$$

Proof. For every vector space U , we have $\text{Hom}(V \otimes W, U) \cong \text{Hom}_{\text{bilinear}}(V \times W, U)$, with the map given by restriction to $V \times W$. Now, bilinear maps from $V \times W$ to U are uniquely determined by their values on the set $\mathcal{B} \times \mathcal{C}$ for bases $\mathcal{B}, \mathcal{C}$ of V, W . Now, this means that linear maps $V \otimes W \rightarrow U$ are uniquely determined on the subset $\Phi(\mathcal{B} \times \mathcal{C}) = \{v \otimes w \mid v \in \mathcal{B}, w \in \mathcal{C}\}$ indicated. This property characterizes a basis. $\square$

Proposition 1.16.13

There is an injective homomorphism

$$\iota: V^* \otimes W \rightarrow \text{Hom}(V, W), \quad \iota(f \otimes w) := T_{f,w}, \quad T_{f,w}(v) = f(v)w.$$

The image of ι is the set of finite rank homomorphisms from V to W . In particular, if V or W is finite dimensional, then ι is an isomorphism.

Proof. To check that this map ι is linear and well-defined, by the universal property we only need to observe that the map $V^* \otimes W \rightarrow \text{Hom}(V, W)$ is bilinear. To see this, note first that

$$T_{f+g,w}(v) = (f(v) + g(v))w = f(v)w + g(v)w = T_{f,w} + T_{g,w},$$

and also that

$$T_{cf,w}(v) = cf(v)w.$$

Next, we have that

$$T_{f,w_1+w_2}(v) = f(v)(w_1 + w_2) = f(v)w_1 + f(v)w_2,$$

and that

$$T_{f,cw} = f(v)(cw) = cf(v)w.$$

We now show that this map is injective. Suppose that we have $\alpha = \sum_{i=1}^n f_i \otimes w_i$ to be in the kernel. Then by bilinearity, we can assume that the w_i are all linearly independent. We can also assume that $f_i \neq 0$, otherwise we can remove the summand. If the element is nonzero, then $n \geq 1$, and we can find v such that $f_1(v) \neq 0$. Then, $\iota(\alpha)(v) = f_1(v)w_1 + \sum_{i>1} c_i w_i$, for some $c_i \in \mathbb{C}$. Since $f_1(v) \neq 0$, $\sum_{i>1} c_i w_i$ is nonzero, so $\iota(\alpha) \neq 0$.

To see that the image of ι is the set of finite rank homomorphisms, first note that $T_{f,w}$ has rank one, and $\sum_{i=1}^n T_{f_i,w_i}$ has rank at most n (the image is in the span of the w_i). Conversely, if $\phi: V \rightarrow W$ is a finite rank homomorphism, then by taking a basis w_i of the image of ϕ , we can write it as $\sum_{i=1}^n T_{f_i,w_i}$, for some $f_i \in V^*$. Note finally that if V or W is finite-dimensional, then all homomorphisms are finite rank, so the isomorphism follows. $\square$

The main thing is that we need to think of $V \otimes W$ as a vector space whose elements are linear combinations (or sums) of elements of the form $v \otimes w$, $v \in V, w \in W$ subject to the defining relations (of the rotation elements) which in terms of tensor products can be written as the following equalities for all $v, v' \in V$, $w, w' \in W$, and $a, a' \in \mathbb{C}$:

- $(av + a'v') \otimes w = a(v \otimes w) + a'(v' \otimes w)$
- $v \otimes (aw + a'w') = a(v \otimes w) + a'(v \otimes w')$,

and a basis of $V \otimes W$ can be obtained by taking the product of any basis of V with any basis of W . In particular, the isomorphism $V \otimes \mathbb{C} \cong V \cong \mathbb{C} \otimes V$ is particularly useful. There will be more useful isomorphisms presented in the next subsection.

§1.16.ii Natural Isomorphisms Involving Tensor Products

Proposition 1.16.14

We have the following:

1. There are the following explicit isomorphisms, for U, V , and W arbitrary vector spaces:

$$\begin{aligned} V \otimes W &\xrightarrow{\sim} W \otimes V, & (v \otimes w) &\mapsto (w \otimes v) \\ (U \otimes V) \otimes W &\xrightarrow{\sim} U \otimes (V \otimes W), & (u \otimes v) \otimes w &\mapsto u \otimes (v \otimes w) \\ U \otimes (V \oplus W) &\xrightarrow{\sim} (U \otimes V) \oplus (U \otimes W), & u \otimes (v, w) &\mapsto (u \otimes v, u \otimes w) \\ \Phi: \text{Hom}(U \otimes V, W) &\xrightarrow{\sim} \text{Hom}(U, \text{Hom}(V, W)), & \Phi(f)(u)(v) &= f(u \otimes v) \end{aligned}$$

2. For arbitrary V and W , there is an explicit linear injection,

$$\Phi: V^* \otimes W^* \rightarrow (V \otimes W)^*, \quad \Phi(f \otimes g)(v \otimes w) = f(v)g(w).$$

This is an isomorphism if V is finite-dimensional.

Proof. (1): The maps are obviously linear. It is easy to construct an explicit formula for their inverses. This is left as an exercise. For example, the inverse of the third one is given by the sum of the injective map $U \otimes V \cong U \otimes (V \oplus \{0\}) \hookrightarrow U \otimes (V \oplus W)$ and the same map with V replaced by W .

(2): This map is obviously linear. To see it is injective, by proposition 1.16.13, we have an injection $V^* \otimes W^* \rightarrow \text{Hom}(V, W^*)$, and by the first part, we can compose this with $\text{Hom}(V, W^*) = \text{Hom}(V, \text{Hom}(W, \mathbb{C})) \cong \text{Hom}(V \otimes W, \mathbb{C}) = (V \otimes W)^*$ to get a linear injection. It is easy to check that the result is the same as the given map Φ . Finally, if V is finite-dimensional, by proposition 1.16.13 again, the injection is an isomorphism. $\square$

If we take dimensions of vector spaces, the first three isomorphisms in the first statement of the proposition above become the commutativity, associativity, and distributivity rules on $\mathbb{N}$. Namely, for $a = \dim U$, $b = \dim V$, $c = \dim W$, we get:

$$b \cdot c = c \cdot b, \quad (a \cdot b) \cdot c = a \cdot (b \cdot c), \quad a \cdot (b + c) = a \cdot b + a \cdot c.$$

§1.16.iii Tensor Products of Representations

Definition 1.16.15. The **tensor product representation** $(V \otimes W, \rho_{V \otimes W})$ is given by

$$\rho_{V \otimes W}: G \rightarrow GL(V \otimes W), \quad \rho_{V \otimes W}(g)(v \otimes w) = \rho_V(g)(v) \otimes \rho_W(g)(w).$$

Remark 1.16.16 — To show that this is well-defined, we need to argue that the formula above is bilinear in v and w .

Example 1.16.17

Let $(V, \rho_V), (V', \rho_{V'})$ be one-dimensional representations. Let $\rho, \rho': G \rightarrow \mathbb{C}^\times$ be defined by the properties $\rho_V(g)(v) = \rho(g)v$ and $\rho_{V'}(g)(v') = \rho'(g)v'$ for all $g \in G, v \in V$, and $v' \in V'$. Then $(V \otimes V', \rho_{V \otimes V'})$ has the property $\rho_{V \otimes V'}(v \otimes v') = \rho(g)\rho'(g)(v \otimes v')$. That is, the tensor product of the one-dimensional representations $\rho, \rho': G \rightarrow \mathbb{C}^\times$ in terms of matrices is the product,

$$(\rho \otimes \rho')(g) = \rho(g)\rho'(g).$$

Example 1.16.18

For (W, ρ_W) a 1-dimensional representation $\theta: G \rightarrow \mathbb{C}$, this covers the *product of a representation and a 1-dimensional representation* construction.

Remark 1.16.19 — Suppose we have bases $\mathcal{B}$ for V and $\mathcal{C}$ for W , and matrices $M := \rho_V^{\mathcal{B}}(g) \in M_m(\mathbb{C})$ and $N := \rho_W^{\mathcal{C}}(g) \in M_n(\mathbb{C})$, where $|\mathcal{B}| = m, |\mathcal{C}| = n$. We should then get that $\rho_{V \otimes W}^{\mathcal{B} \times \mathcal{C}}(g) \in M_{mn}(\mathbb{C})$. In coordinates, if $\mathcal{B} = (v_1, \dots, v_m), \mathcal{C} = (w_1, \dots, w_n)$, then

$$\rho_{V \otimes W}(g)(v_k \otimes w_l) = \rho_V(g)(v_k) \otimes \rho_W(g)(w_l).$$

Then, we note that

$$\rho_V(g)(v_k) = \sum_i M_{i,k} v_i, \quad \rho_W(g)(w_l) = \sum_j M_{j,l} w_l.$$

Therefore, we have that

$$\begin{aligned} \rho_{V \otimes W}(g)(v_k \oplus w_l) &= \left(\sum_i M_{i,k} v_i \right) \otimes \left(\sum_j M_{j,l} w_l \right) \\ &= \sum_{i,j} (M_{i,k} N_{j,l}) v_i \otimes w_l. \end{aligned}$$

This is sometimes called the Kronecker product of M and N .

Exercise 1.16.20. Show that under the isomorphism $\iota: V \otimes W \rightarrow \text{Hom}(V^*, W)$, that the definition for the tensor product representation $(V \otimes W, \rho_{V \otimes W})$ agrees with the definition for the ‘Hom’-representation $(\text{Hom}(V, W), \rho_{\text{Hom}(V, W)})$.

Now instead of using the formula for the representation $\text{Hom}(V, W)$ we can feel free to begin with V^* and once we understood its formula, to use $V^* \otimes W$. Here is an example which shows why this might be useful:

Example 1.16.21

Let $(\mathbb{C}_\theta, \theta)$ be a one-dimensional representation, given by $\mathbb{C}_\theta = \mathbb{C}$ and $\theta: G \rightarrow \mathbb{C}^\times = GL_1(\mathbb{C})$. Let (W, ρ_W) be an arbitrary representation. To compute the representation on $\mathbb{C}_\theta \otimes W$, first note that $\mathbb{C}_\theta \otimes W \cong W$ as vector spaces by example 1.16.10. In terms of this, we have

$$\rho_{\mathbb{C}_\theta \otimes W}(g)(w) = \rho_{\mathbb{C}_\theta \otimes W}(1 \otimes w) = \theta(g)(1) \otimes \rho_W(g)(w) = \theta(g) \rho_W(g)(w).$$

We now just multiply ρ_W by θ , and we see it’s just the construction of exercise 1.14.6.

On the other hand, if we want to compute the representation $\text{Hom}(\mathbb{C}_\theta, W)$, this vector space is also isomorphic to W , but the correct representation is given by $\text{Hom}(\mathbb{C}_\theta, W) \cong \mathbb{C}_{\theta^*} \otimes W$ so that we have to invert the $\theta(g)$. We get that

$$\rho_{\text{Hom}(\mathbb{C}_\theta, W)}(g)(w) = \theta(g)^{-1} \rho_W(g)(w).$$

This (personally) is easier to think about than computing directly from the definition introduced previously for $\text{Hom}(\mathbb{C}_\theta, W)$ in terms of linear maps $\mathbb{C}_\theta \rightarrow W$.

§1.16.iv External Tensor Product

Definition 1.16.22. Let G, H be groups with representations (V, ρ_V) , (W, ρ_W) of G, H , respectively, the **external tensor product** $V \boxtimes W$ of these representations (on the vector space $V \otimes W$) is the representation of $G \times H$ on $V \otimes W$ defined by

$$\rho_{V \boxtimes W}: G \times H \rightarrow GL(V \otimes W), \quad \rho_{V \boxtimes W}(g, h) = \rho_V(g) \otimes \rho_W(h).$$

Exercise 1.16.23. Show that for $H = G$, $\rho_{V \otimes W}(g) = \rho_{V \boxtimes W}(g, g)$. Note that here, (V, ρ_V) and (W, ρ_W) are both representations of G .

Exercise 1.16.24. More abstractly, consider the diagonal homomorphism $\Delta: G \rightarrow G \times G$, $\Delta(g) = (g, g)$. Then, $\rho_{V \otimes W} = \rho_{V \boxtimes W} \circ \Delta$.

To summarize the two exercises above, we have the following diagram:

$$\begin{array}{ccc} G & \xrightarrow{\Delta} & G \times G \\ & \searrow \rho_{V \otimes W} & \downarrow \rho_{V \boxtimes W} \\ & & GL(V \otimes W) \end{array}$$

Proposition 1.16.25

Let G and H be finite groups. Then,

1. Every irreducible representation of $G \times H$ is isomorphic to $V \boxtimes W$, where $(V, \rho_V), (W, \rho_W)$ are some irreducible representations of G, H , respectively.
2. Given V, V' irreducible representations of G and W, W' irreducible representations of H , we have $V' \boxtimes W' \cong V \boxtimes W$ if and only if $V' \cong V$ and $W' \cong W$.

We will not prove this here, but come back to it later.

2 Character Theory

The main idea about character theory is to use the trace of a matrix in order to distinguish distinct representations.

§2.1 Traces and Definitions

Recall from linear algebra: given a square matrix A , the **trace** of A is the sum of the diagonal entries of A .

Example 2.1.1

$$\text{Tr} \begin{pmatrix} 3 & 0 \\ 1 & 4 \end{pmatrix} = 7.$$

Properties 2.1.2 (Properties of Trace from Linear Algebra)

Here are some properties of the trace from linear algebra. Let A, B be square matrices of the same size. Then:

1. $\text{Tr}(AB) = \text{Tr}(BA)$
2. If A and B are conjugate matrices, i.e., $B = PAP^{-1}$, we have that $\text{Tr}(A) = \text{Tr}(B)$.

Definition 2.1.3. Given $T: V \rightarrow V$ a linear endomorphism of a finitely dimensional vector space V ,

$$\text{Tr}(T) := \text{Tr}([T]_{\mathcal{B}}),$$

where $\mathcal{B}$ is a basis of V .

We make sure that this is well-defined (doesn't depend on choice of basis). If $\mathcal{B}'$ is another basis of V , let $P = [I]_{\mathcal{B}', \mathcal{B}}$. Then,

$$\text{Tr}([T]_{\mathcal{B}'}) = \text{Tr}(P^{-1}[T]_{\mathcal{B}}P) = \text{Tr}([T]_{\mathcal{B}}).$$

Remark 2.1.4 (Non-Examinable) — This definition relies on a basis, so we now try to define the trace without using a basis.

$\text{End}(V) = \text{Hom}(V, V) \cong V^* \otimes V$, for finite dimensional V . We have a bilinear map from $V^* \times V$ to $\mathbb{C}$ by $(f, v) \mapsto f(v)$. By the universal property of $\otimes$, we have a linear map $V^* \times V \rightarrow \mathbb{C}$ such that for all f, v $f \otimes v \mapsto f(v)$. Identifying $V^* \otimes V$ with $\text{End}(V)$, this linear map is $\text{Tr}: \text{End}(V) \rightarrow \mathbb{C}$.

Remark 2.1.5 — For square matrices A, B we had that $\text{Tr}(AB) = \text{Tr}(BA)$. For linear maps $T: V \rightarrow W, S: W \rightarrow V$, we have

$$\text{Tr}(T \circ S) = \text{Tr}(S \circ T).$$

Remark 2.1.6 — Recall that if $T: V \rightarrow V$ has eigenvalues (with multiplicity) $\lambda_1, \dots, \lambda_n$ ($n = \dim(V)$), then

$$\text{Tr}(T) = \lambda_1 + \dots + \lambda_n.$$

This is by the Jordan normal form.

Definition 2.1.7. The **character** of a matrix representation $\rho: G \rightarrow GL_n(\mathbb{C})$ of a group G is $\chi_\rho: G \rightarrow \mathbb{C}$ defined by,

$$\chi_\rho(g) = \text{Tr}(\rho(g)).$$

Definition 2.1.8. The **character** of a representation (V, ρ_V) of a group G (with $\rho_V: G \rightarrow GL(V)$) is $\chi_V: G \rightarrow \mathbb{C}$ defined by

$$\chi_V(g) = \text{Tr}(\rho_V(g)).$$

Proposition 2.1.9

Isomorphic representations have the same character.

Proof. Let $T: (V, \rho_V) \rightarrow (W, \rho_W)$ be an isomorphism between representations of a group G . Letting $g \in G$,

$$\begin{aligned} \text{Tr}(\rho_W(g)) &= \text{Tr}(T \circ \rho_V(g) \circ T^{-1}) \\ &= \text{Tr}(T \circ T^{-1} \circ \rho_V(g)) \\ &= \text{Tr}(\rho_V(g)). \end{aligned}$$

□

We state the converse of the proposition above, but will prove this later.

Exercise 2.1.10. Let V be a finite dimensional representation of G . If there is a subrepresentation $W \subseteq V$ of G , show that

$$\chi_V = \chi_W + \chi_{V/W}.$$

Solution. Let $k = \dim(W)$ and take k vectors of V that span W . Then, every representation $\rho_V(g)$ for $g \in G$ can be written as

$$\begin{pmatrix} A & B \\ 0 & C \end{pmatrix}.$$

Here, A and C are square matrices that are just $\rho_V(g)|_W$ and $\rho_V(g)|_{V/W}$, respectively. B is a projector to W . Therefore, taking traces, we get that

$$\chi_V = A + C = \chi_W(g) + \chi_{V/W}(g).$$

□

Theorem 2.1.11

Let G be finite, and $(V, \rho_V), (W, \rho_W)$ be finite dimensional representations of G . Then, $\chi_V = \chi_W$ if and only if (V, ρ_V) and (W, ρ_W) are isomorphic.

Example 2.1.12

If $(\mathbb{C}, \rho_{\text{triv}})$ is the trivial representation of G , then $\forall g \in G, \chi_{\rho_{\text{triv}}}(g) = 1$.

Example 2.1.13

Let $\rho: G \rightarrow \mathbb{C}^\times$ be a one-dimensional representation of G . Then, for all $g \in G$, $\chi_\rho(g) = \rho(g)$. In particular, the character of a 1-dimensional representation is multiplicative:

$$\chi(g)\chi(h) = \chi(gh).$$

Example 2.1.14

In general, for a finite dimensional (V, ρ) of a group G ,

$$\chi_V(e) = \text{Tr}(\rho(e)) = \text{Tr}(\text{id}_V) = \dim(V).$$

In particular, this tells us that if χ_V is multiplicative, then

$$(\dim V)^2 = \chi_V(e)^2 = \chi_V(e^2) = \chi_V(e) = \dim(V),$$

and so $\dim(V)$ is either 0 or 1.

Example 2.1.15

Let $G = \{\pm 1\} \times \{\pm 1\} \cong C_2 \times C_2$ and

$$\rho(a, b) = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}.$$

Then, $\chi_\rho(a, b) = a + b$. Indeed, we have that $\chi_\rho(1, 1) = 2$, which is the dimension of the representation as expected. We also get

$$\chi_\rho(1, -1) = \chi_\rho(-1, 1) = 0,$$

and

$$\chi_\rho(-1, -1) = -2.$$

Example 2.1.16

Let G act on a finite set X and take the associated representation $(\mathbb{C}[X], \rho)$. Using the basis X of $\mathbb{C}[X]$ (put in any ordering), $[\rho(g)]_X$ is a permutation matrix. The trace of a permutation matrix P_σ is the number of diagonal entries, i.e., the number of basis vectors e_i such that $P_\sigma e_i = e_i$, or equivalently the number of i such that $\sigma(i) = i$. Thus here we have $\chi_\rho(g) = \text{Tr}(\rho(g)) = |\{x \in X \mid g \cdot x = x\}|$. This can also be written as $|X^g|$, by the following definition.

Definition 2.1.17. For $g: X \rightarrow X$, the **fixed point set** X^g is defined as $X^g := \{x \in X \mid g(x) = x\}$. If $G \times X \rightarrow X$ is an action then we use the same notation for $g \in G$, via $g(x) := g \cdot x$.

Remark 2.1.18 (Why Focus on the Trace?) — There are quite a lot of conjugation-invariant functions on $\text{End}(V)$ (or $M_n(\mathbb{C})$). For instance, the trace and determinant are examples of these, so we ask why our particular focus is on the trace (and why not others?) We actually get nothing new by studying the others. The trace alone contains full information.

Here is now an example that illustrates this phenomena with the determinant.

Example 2.1.19

For example, for a 2-dimensional matrix A ,

$$\det(A) = \frac{\text{Tr}(A)^2 - \text{Tr}(A^2)}{2}.$$

For this, let’s focus on diagonalizable matrices (for others, those are niche cases).

Considering a diagonal matrix, $\begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}$, we see that

$$\lambda\mu = \frac{(\lambda + \mu)^2 - (\lambda^2 + \mu^2)}{2}.$$

Exercise 2.1.20. We attain a similar formula for the determinant of a 3×3 matrix.

Therefore, for a 2-dimensional representation (V, ρ) of a group G , $\forall g \in G$,

$$\det(\rho(g)) = \frac{\chi_V(g)^2 - \chi_V(g^2)}{2}.$$

§2.2 Characters, Class Functions, and Other Basic Properties

Definition 2.2.1. A function $f: G \rightarrow \mathbb{C}$ is a **class function** if $f(g) = f(h^{-1}gh)$ for all $h \in G$.

Notation 2.2.2. Let $\text{Fun}(G, \mathbb{C})$ denote the vector space of all complex-valued functions on G , and let $\text{Fun}_{\text{cl}}(G, \mathbb{C}) \subseteq \text{Fun}(G, \mathbb{C})$ denote the vector space of all class functions.

Exercise 2.2.3. Show that the dimension of $\text{Fun}_{\text{cl}}(G, \mathbb{C})$ is equal to the number of conjugacy classes.

Recall that $\mathbb{C}[G] = \{ \sum_{g \in G} a_g g \mid (a_g)_{g \in G} \text{ all in } \mathbb{C} \}$. In fact, a formal linear combination $\sum_{g \in G} a_g g$ is the same thing as an element of $\text{Fun}(G, \mathbb{C})$, namely $g \mapsto a_g$, a function from G to $\mathbb{C}$.

Further recall the adjoint representation on $\mathbb{C}[G]$: for $g \in G$,

$$\rho_{\text{ad}}(g) \left(\sum_{h \in G} a_h h \right) = \sum_{h \in G} a_h g h g^{-1}.$$

We re-index this so that we have

$$\rho_{\text{ad}}(g) \left(\sum_{h \in G} a_h h \right) = \sum_{h' \in G} a_{g^{-1}h'g} h'.$$

That is, $\rho_{\text{ad}}(g)$ sends the function $h \mapsto a_h$ to the function $h \mapsto a_{g^{-1}h'g}$.

Exercise 2.2.4. Show that $\text{Fun}(G, \mathbb{C})$ is a representation of G under the adjoint action defined by, for $f: G \rightarrow \mathbb{C}$ and $g, h \in G$

$$(\rho(g)f)(h) = f(g^{-1}hg).$$

Further, show that the class functions are the invariant subspaces of the representation:

$$\text{Fun}_{\text{cl}}(G, \mathbb{C}) = \text{Fun}(G, \mathbb{C})^G.$$

Proposition 2.2.5

Every character χ_V is a class function.

Proof. We have

$$\chi_V(h^{-1}gh) = \text{Tr}(\rho_V(h^{-1}gh)) = (\rho_V(h)^{-1} \rho_V(g) \rho_V(h)) = (\rho_V(g)) = \chi_V(g).$$

□

Lemma 2.2.6

Suppose that $T \in GL(V)$ has finite order: $T^m = I$ for some $m \geq 1$. Then T is diagonalizable with eigenvalues which are m -th roots of unity, i.e., in some basis $\mathcal{B}$ of V , we have $[T]_{\mathcal{B}}$ is diagonal whose entries are in $\{\zeta \in \mathbb{C} \mid \zeta^m = 1\}$.

Proof. This is linear algebra. T satisfies the polynomial $x^m - 1 = 0$, which has distinct roots (the m -th roots of unity). The minimal polynomial $p(x)$ of T is a factor of $x^m - 1$, so also has roots of multiplicity one, which are m -th roots of unity. Therefore T is diagonalizable to a diagonal matrix whose entries are m -th roots of unity. (In other words, the Jordan normal form of T has diagonal entries which are m -th roots of unity, and nothing above the diagonal since $p(T) = 0$ and p has distinct roots.) □

Alternative Proof via Representation Theory. We can define a representation $\rho: C_m = \{1, g, \dots, g^{m-1}\} \rightarrow GL(V)$ given by $\rho(g^k) = T^k$. Then Maschke’s Theorem together with corollary 1.9.7 show that $V = V_1 \oplus \dots \oplus V_n$ for some one-dimensional representations $V_1, \dots, V_n$. Picking a basis $\mathcal{B} := (v_1, \dots, v_n)$ such that $v_i \in V_i$ for all i , we get that $[\rho(g)]_{\mathcal{B}}$ is diagonal and the entries are obviously m -th roots of unity. $\square$

Exercise 2.2.7. *Unwrap* the proof of Maschke’s theorem to prove lemma 2.2.6 directly in an elementary way (no representation theory, no Jordan normal form). The key lemma to prove this is that given $v \in V$, which is an eigenvector of T , produce a complementary T -invariant subspace $U \subseteq V$, i.e., $\mathbb{C}v \oplus U = V$.

Proposition 2.2.8

Let (V, ρ_V) be a finite-dimensional representation of G and $g \in G$ an element of finite order. Then:

1. $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.
2. $|\chi_V(g)| \leq \dim V$, with equality holding if and only if g is a scalar matrix (a scalar multiple of the identity).
3. $\chi_V(1) = \dim(V)$.

Proof. (1): Let the eigenvalues of $\rho_V(g)$ be, with multiplicity, $\zeta_1, \dots, \zeta_n$. Then, $\chi_V(g) = \zeta_1 + \dots + \zeta_n$. Also, the eigenvalues of $\rho_V(g^{-1}) = \rho_V(g)^{-1}$ are $\zeta_1^{-1}, \dots, \zeta_n^{-1}$. Since each ζ_j is a root of unity, it follows that $\zeta_j^{-1} = \zeta_k$: after all, $\zeta_j \overline{\zeta_j} = |\zeta_j| = 1$. Thus,

$$\chi_V(g^{-1}) = \zeta_1^{-1} + \dots + \zeta_n^{-1} = \overline{\zeta_1} + \dots + \overline{\zeta_n} = \overline{\chi_V(g)}.$$

(2): We apply the triangle inequality. We have

$$|\chi_V(g)| = |\zeta_1 + \dots + \zeta_n| \leq |\zeta_1| + \dots + |\zeta_n| = n = \dim V,$$

with equality holding if and only if all of the ζ_j are positive multiples of each other. But to have this, since $|\zeta_j| = 1$, all the ζ_j have to be equal.

(3): We have seen this before. $\square$

Exercise 2.2.9. Find the functions $f: C_2 = \{1, g\} \rightarrow \mathbb{C}$ that satisfy the conditions of the above lemmas. Further, give an example to show that they cannot all be characters. Finally, compute all functions which actually are characters. (Hint: $T = \rho(g)$ will have to be diagonalizable with entries ± 1 , since $T^2 = I$ hence $(T - I)(T + I) = 0$.)

Exercise 2.2.10. Let G be a finite group. Show that $z \in Z(G)$ if and only if for all irreducible representations V of G , $|\chi_V(z)| = \dim(V)$.

§2.3 Characters of Direct Sums and Tensor Products

We explain now how characters turn direct sums and tensor products into ordinary sums and products, and we will also discuss duals and Hom spaces. As an introduction, recall

the formulas

$$\dim(V \oplus W) = \dim V + \dim W, \quad \dim(V \otimes W) = \dim V \cdot \dim W$$

Since $\chi_V(1) = \dim V$, we obtain that, for $g = 1$, with V and W finite-dimensional,

$$\chi_{V \oplus W}(g) = \chi_V(g) + \chi_W(g), \quad \chi_{V \otimes W}(g) = \chi_V(g)\chi_W(g).$$

Below we will explain that in fact this holds for all g .

Proposition 2.3.1

Let (V, ρ_V) and (W, ρ_W) be finite-dimensional representations of a group G . Then:

1. $\chi_{V \oplus W} = \chi_V + \chi_W$
2. $\chi_{V \otimes W} = \chi_V \cdot \chi_W$

Now suppose G is finite.

3. $\chi_{V^*} = \overline{\chi_V}$
4. $\chi_{\text{Hom}(V, W)} = \overline{\chi_V} \chi_W$

Proof. Throughout let $\mathcal{B} = \{v_1, \dots, v_m\}$ and $\mathcal{C} = \{w_1, \dots, w_n\}$ be bases of V and W , respectively. Fix $g \in G$ and let $M := [\rho_V(g)]_{\mathcal{B}}$ and $N := [\rho_W(g)]_{\mathcal{C}}$. Thus, $\chi_V(g) = \text{Tr}(M)$ and $\chi_W(g) = \text{Tr}(N)$. Then the computations reduce to finding the matrices for the representations on the LHS and computing their traces.

(1): Let $\mathcal{D} := \mathcal{B} \sqcup \mathcal{C}$ be a basis of $V \oplus W$ (to be precise, for $v \in \mathcal{B}$, then $(v, 0) \in (V \oplus W)$ is the corresponding basis element in $\mathcal{D}$). Then we get the block-diagonal matrix for $\rho_{V \oplus W}(g)$:

$$[\rho_{V \oplus W}(g)]_{\mathcal{D}} = \begin{pmatrix} M & 0 \\ 0 & N \end{pmatrix},$$

since $\rho_{V \oplus W}(g)$ sends elements of $\mathcal{B}$ to linear combinations of $\mathcal{B}$ according to $\rho_V(g)$, and similarly for $\mathcal{C}$ and W . Then,

$$\chi_{V \oplus W}(g) = \text{Tr}[\rho_{V \oplus W}(g)]_{\mathcal{D}} = \text{Tr}(M) + \text{Tr}(N) = \chi_V(g) + \chi_W(g).$$

(2): We take the basis $\mathcal{D} = \{v_i \otimes w_j\}$ of $V \otimes W$. By the formula in remark 1.16.19, we have $[\rho_{V \otimes W}(g)]_{\mathcal{D}} = M \otimes N$ where $(M \otimes N)_{ij,kl} = M_{ik}N_{jl}$ for $i, k \in \{1, \dots, m\}$ and $j, l \in \{1, \dots, n\}$. Thus,

$$\begin{aligned} \chi_{V \otimes W}(g) &= \text{Tr}(M \otimes N) = \sum_{i,j} (\rho_V(g) \otimes \rho_W(g))_{ij,ij} = \sum_{i,j} \rho_V(g)_{ii} \rho_W(g)_{jj} \\ &= \sum_i \rho_V(g)_{ii} \sum_j \rho_W(g)_{jj} = \text{Tr}(\rho_V(g)) \text{Tr}(\rho_W(g)) = \chi_V(g) \chi_W(g). \end{aligned}$$

(3): Here we take the dual basis $\mathcal{D} = \{f_1, \dots, f_m\}$ of V^* , given by $f_i(v_j) = \delta_{ij}$. By remark 1.16.5, we get

$$\chi_{V^*}(g) = \text{Tr}[\rho_{V^*}(g)]_{\mathcal{D}} = \text{Tr}([\rho_V(g)]_{\mathcal{B}}^{-1})^T = \text{Tr}([\rho_V(g)]_{\mathcal{B}}^{-1}) = \text{Tr}[\rho_V(g^{-1})]_{\mathcal{B}} = \chi_V(g^{-1}).$$

The result now follows from proposition 2.2.8.

(4): By proposition 1.16.13, $\text{Hom}(V, W) \cong V^* \otimes W$, so by proposition 2.1.9, the result follows from the second and third statement. $\square$

Remark 2.3.2 — There is a hidden calculation in the fourth statement that requires to check the following diagram commutes.

$$\begin{array}{ccc} \mathrm{Hom}(V, W) & \cong & V^* \otimes W \\ \downarrow g & & \downarrow g \\ \mathrm{Hom}(V, W) & \cong & V^* \otimes W \end{array}$$

Example 2.3.3

Here’s an example to show why the third and fourth statement require G to be finite. Let $G = \mathbb{Z}$ and $a \in \mathbb{C}^\times$. Then we can take the one-dimensional representation $\rho: G \rightarrow \mathbb{C}^\times, \rho(m) = a^m$. Then as in example 2.1.13, $\chi_\rho = \rho$, but $\chi_\rho(-1) = a^{-1} \neq \bar{a}$ in general. [Note that the third and fourth statement hold in this case if and only if $|a| = 1$, which is weaker than the condition that a is a root of unity.]

Example 2.3.4

Let’s take $G = S_n$ and let $(\mathbb{C}^n, \rho_{\mathrm{perm}})$ be the permutation representation and (V, ρ_V) be the reflection representation ($V \subset \mathbb{C}^n$). Let $(\mathbb{C}, \rho_{\mathrm{triv}})$ denote the trivial representation. Then $\mathbb{C}^n = V \oplus \mathbb{C} \cdot (1, \dots, 1)$, with $\mathbb{C} \cdot (1, \dots, 1) = \{(a, \dots, a) \mid a \in \mathbb{C}\}$ isomorphic to the trivial representation. Hence $\chi_{\mathbb{C}^n} = \chi_V + \chi_{\mathbb{C}}$. By examples 2.1.12 and 2.1.16, we get

$$\chi_V(\sigma) = |\{1, \dots, n\}^\sigma| - 1.$$

Example 2.3.5

Now let (V, ρ_V) be an arbitrary representation of a group, and $(\mathbb{C}_\theta, \theta)$ a one-dimensional representation, for $\theta: G \rightarrow \mathbb{C}^\times$ a homomorphism. Then taking the tensor product, we get $\chi_{\mathbb{C}_\theta \otimes V} = \theta \cdot \chi_V$. This equals χ_V if and only if $\chi_V(g) = 0$ whenever $\theta(g) \neq 1$, i.e., whenever $g \notin \ker \theta$. In the case that G is finite, by theorem 2.1.11 (still to be proved), this gives an explicit criterion when the construction of exercise 1.14.6 produces a nonisomorphic representation to the first one.

Example 2.3.6

As a special case of the preceding, by theorem 2.1.11, for $(\mathbb{C}_-, \rho_{\mathrm{sign}})$ the sign representation of S_n and (V, ρ_V) the reflection representation, we see that $\mathbb{C}_- \otimes V \cong V$ if and only if $\forall \sigma \in S_n, \chi_{\mathbb{C}_-} \neq 1$ implies that $\chi_V(\sigma) = 0$ (via example 2.3.5). This occurs if and only if $|\{1, \dots, n\}^\sigma| = 1$ whenever σ is an odd permutation. We see that this is true if and only if $n = 3$. (Even for $n = 2$ it is not true, and indeed in this case the reflection and sign representations are isomorphic, and their tensor product is the trivial representation which is not isomorphic to the reflection representation!) Thus for $n \geq 4$, the reflection representation and the reflection tensor sign are two nonisomorphic irreducible representations of dimension $n - 1$.

Example 2.3.7

Let $(\mathbb{C}[G], \rho_{\mathbb{C}[G]})$ be the left regular representation of a finite group G , with

$$\rho_{\mathbb{C}[G]}(g)\left(\sum_{h \in G} a_h h\right) = \sum_{h \in G} a_h gh.$$

This representation is induced by the group action $G \times G \rightarrow G$, so $\chi_{\mathbb{C}[G]}(g) = |G^g|$ where $G^g := \{h \in G \mid gh = h\}$. But applying cancellation, $gh = h$ if and only if $g = 1$. Therefore $G^g = \emptyset$ for $g \neq 1$ and $G^1 = G$. We get

$$\chi_{\mathbb{C}[G]}(g) = \begin{cases} |G|, & g = 1 \\ 0, & g \neq 1 \end{cases}.$$

Example 2.3.8

The same applies to the right regular representation, $g \cdot \left(\sum_{h \in G} a_h h\right) = \sum_{h \in G} a_h hg^{-1}$. So these two have the same character; so by theorem 2.3.4 we see that the left and right regular representations are isomorphic (we can also write an explicit isomorphism).

Example 2.3.9

On the other hand, the adjoint representation $\rho_{\text{ad}}(g)(h) = ghg^{-1}$ does not have the same character: $\chi_{\rho_{\text{ad}}}(g) = |Z_G(g)|$ where $Z_G(g) := \{h \in G \mid gh = hg\}$, which is a subgroup of G called the centralizer of g . Note that, since it is a subgroup, $|Z_G(g)| \geq 1$ (i.e., $1 \in Z_G(g)$ for all g), so $\chi_{\rho_{\text{ad}}} \neq \chi_{\mathbb{C}[G]}$ unless $G = \{1\}$. So the adjoint representation is not isomorphic to the left and right regular representations.

§2.4 Inner Product of Functions and Characters

Suppose $f_1, f_2: G \rightarrow \mathbb{C}$ are functions and G is a finite group. Then we consider the (complex) inner product

$$\langle f_1, f_2 \rangle := |G|^{-1} \sum_{g \in G} f_1(g) \overline{f_2(g)}.$$

Remark 2.4.1 — This makes sense if G is replaced by any finite set. There is also an analogue for functions on more general spaces, where replace the finite sum by an infinite sum or even by an integral. For example, there is the complex vector space $\mathcal{L}^2(\mathbb{R})$ of square-integrable functions on $\mathbb{R}$ (functions $f: \mathbb{R} \rightarrow \mathbb{C}$ such that $\int_{\mathbb{R}} |f(x)|^2 dx < \infty$), with the inner product $\langle f, g \rangle = \int_{\mathbb{R}} f(x) \overline{g(x)} dx$.

Definition 2.4.2. Let V be a complex vector space. A **(Hermitian) inner product** is a pairing $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{C}$ satisfying, for all $\lambda \in \mathbb{C}$ and vectors $u, v, w \in V$:

1. $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$ and $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$
2. $\langle \lambda u, v \rangle = \lambda \langle u, v \rangle = \langle u, \overline{\lambda} v \rangle$ (sequilinearity)

3. $\langle u, v \rangle = \overline{\langle v, u \rangle}$ (conjugate-symmetry)
4. $\langle v, v \rangle \geq 0$ with equality if and only if $v = 0$ (positive-definiteness)

Remark 2.4.3 — This means that $\|v\| = \sqrt{\langle v, v \rangle}$ is a norm on V , so we can give this space a topology.

Lemma 2.4.4

The pairing $\langle \cdot, \cdot \rangle$ on $\text{Fun}(G, \mathbb{C})$ is an inner product defined by

$$\langle f_1, f_2 \rangle = \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)}.$$

Proof. Note that the first three properties are immediate from the definition. For the fourth, observe that $\langle f, f \rangle = \frac{1}{|G|} \sum_{g \in G} |f(g)|^2$, which implies the statement. $\square$

Definition 2.4.5. Let V be a complex vector space and $\langle \cdot, \cdot \rangle$ a Hermitian inner product on V . Then, $v, w \in V$ are **orthogonal**, written $v \perp w$, if $\langle v, w \rangle = 0$. A set $v_1, \dots, v_n$ is **orthonormal** if

$$\langle v_i, v_j \rangle = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}.$$

Lemma 2.4.6

If vectors $(v_1, \dots, v_n)$ in a Hermitian inner product space are orthonormal, then they are linearly independent and for any $v = a_1 v_1 + \dots + a_n v_n$, we have $\langle v, v_i \rangle = a_i$.

Proof. The second part is quite basic from definitions. Then, for the first suppose that $a_1 v_1 + \dots + a_n v_n = 0$. Then by the second part, we have that for all i , $a_i = \langle v, v_i \rangle = 0$, so we’re done. $\square$

§2.5 Dimension of Homomorphisms via Characters

Our next target is the following theorem, whose proof we will postpone until we introduce some lemmas.

Theorem 2.5.1

Let (V, ρ_V) and (W, ρ_W) be finite-dimensional representations of a finite group. Then,

$$\langle \chi_V, \chi_W \rangle = \dim \text{Hom}_G(V, W)$$

The main idea of the proof is to count the dimension of $\text{Hom}_G(V, W) = \text{Hom}(V, W)^G$ via the projection $\text{Hom}(V, W) \rightarrow \text{Hom}(V, W)^G$, onto G -invariants. The count is done using the following linear algebra result:

Lemma 2.5.2

Let $S: V \rightarrow V$ be a linear projection onto $W := \text{Im}(S)$. Then, $\dim(W) = \text{Tr}(S)$.

Proof. By lemma 1.5.18, we can write $V = \text{Im}(S) \oplus \ker(S)$. Let $w_1, \dots, w_m$ be a basis of $W := \text{Im}(S)$, and let $v_1, \dots, v_n$ be a basis of $U := \ker(S)$. Then, $\mathcal{B} := \{w_1, \dots, w_m, v_1, \dots, v_n\}$ is a basis for V , and

$$[S]_{\mathcal{B}} = \left(\begin{array}{c|c} I_m & 0 \\ \hline 0 & 0 \end{array} \right),$$

so $\text{Tr}(S) = m = \dim(\text{Im}(S))$. □

Lemma 2.5.3

Let (V, ρ_V) be a representation of a group G , Then, $\text{Hom}_G(\mathbb{C}_{\text{triv}}, V) \cong V^G$, with the isomorphism given by $\phi \mapsto \phi(1)$.

Proof. We first note that $\text{Hom}_G(\mathbb{C}_{\text{triv}}, V) \cong \text{Hom}(\mathbb{C}_{\text{triv}}, V)^G$. We also have an identification between $\text{Hom}(\mathbb{C}_{\text{triv}}, V)$ and V given by $\phi \mapsto \phi(1)$. This is an isomorphism of representations of G , since

$$(\rho_{\text{triv}}(g)(\phi))(1) = (\rho(g) \circ \phi \circ \rho_{\text{triv}}(g)^{-1})(1) = \rho_V(g)(\phi(1)).$$

□

Proof of Theorem 2.5.1. We break this down into several steps. The main idea is contained for when V is trivial, and then deduce the formula from there.

Step 1: We first prove this when V is trivial.

By lemma 2.5.3, $\text{Hom}_G(\mathbb{C}_{\text{triv}}, W) \cong W^G$. Recall that $S: W \rightarrow W$, $S := \frac{1}{|G|} \sum_{g \in G} \rho(g)$, is a projection operator with $\text{Im}(S) = W^G$. Thus, by lemma 2.5.2, $\dim W^G = \text{Tr}(S)$. We now calculate from the left:

$$\begin{aligned} \langle \chi_{\mathbb{C}_{\text{triv}}}, \chi_W \rangle &= \frac{1}{|G|} \sum_{g \in G} \chi_{\mathbb{C}_{\text{triv}}}(g) \overline{\chi_W(g)} \\ &= \frac{1}{|G|} \sum_{g \in G} \overline{\text{Tr}(\rho_W(g))} \\ &= \text{Tr} \left(\overline{\frac{1}{|G|} \sum_{g \in G} \rho_W(g)} \right) \\ &= \text{Tr}(\overline{S}) \\ &= \overline{\text{Tr}(S)}. \end{aligned}$$

Therefore, $\langle \chi_{\mathbb{C}_{\text{triv}}}, \chi_W \rangle = \overline{\dim \text{Hom}_G(\mathbb{C}_{\text{triv}}, W)}$, since dimensions is a real number, and its complex conjugate is itself: $\langle \chi_{\mathbb{C}_{\text{triv}}}, \chi_W \rangle = \dim \text{Hom}_G(\mathbb{C}_{\text{triv}}, W)$.

Step 2: Now deduce the general case.

Observe that for $f_1, f_2 \in \text{Fun}(G, \mathbb{C})$, we have, $\langle f_1, f_2 \rangle = \langle 1, \overline{f_1} f_2 \rangle$. Thus,

$$\begin{aligned}
 \langle \chi_V, \chi_W \rangle &= \langle 1, \overline{\chi_V} \chi_W \rangle \\
 &= \langle \chi_{\mathbb{C}_{\text{triv}}}, \chi_{\text{Hom}(V, W)} \rangle \\
 &= \dim \text{Hom}_G(\mathbb{C}_{\text{triv}}, \text{Hom}(V, W)) \\
 &= \dim \text{Hom}(V, W)^G \\
 &= \dim \text{Hom}_G(V, W).
 \end{aligned}$$

□

Remark 2.5.4 — In the proof of theorem 2.5.1, the reduction to $V = \mathbb{C}_{\text{triv}}$ was not required; instead, we can offer a direct proof as well. consider the G -linear projection $S: (V, W) \rightarrow \text{Hom}(V, W)$ given by

$$S(\phi) = \frac{1}{|G|} \sum_{g \in G} \overbrace{\rho_W(g) \circ \phi \circ \rho_V(g)^{-1}}^{= \rho_{\text{Hom}(V, W)}(g)(\phi)}.$$

We have, $\text{Im}(S) = \text{Hom}(V, W)^G = \text{Hom}_G(V, W)$. Therefore, $\dim \text{Hom}_G(V, W) = \dim \text{Im}(S) = \text{Tr}(S)$.

On the other hand,

$$\begin{aligned}
 \text{Tr}(S) &= \text{Tr} \left(\frac{1}{|G|} \sum_{g \in G} \rho_{\text{Hom}(V, W)}(g) \right) \\
 &= \frac{1}{|G|} \sum_{g \in G} \text{Tr} \left(\rho_{\text{Hom}(V, W)}(g) \right) \\
 &= \frac{1}{|G|} \sum_{g \in G} \chi_V(g) \overline{\chi_W(g)} \\
 &= \langle \chi_V, \chi_W \rangle.
 \end{aligned}$$

§2.5.i Applications of Theorem 2.5.1

Corollary 2.5.5

Let G be a finite group and $V_1, \dots, V_m$ a full set of irreducible representations. Then:

1. Let V and W be irreducible representations. Then

$$\langle \chi_V, \chi_W \rangle = \begin{cases} 1, & \text{if } V \cong W \\ 0, & \text{otherwise} \end{cases}.$$

2. The characters χ_{V_i} form an orthonormal basis of $\text{Fun}_{\text{cl}}(G, \mathbb{C}) \subseteq \text{Fun}(G, \mathbb{C})$.

3. Let V be a finite-dimensional representation. Then $V \cong \bigoplus_{i=1}^m V_i^{\langle \chi_{V_i}, \chi_V \rangle}$. Moreover, $\chi_V = \sum_{i=1}^m \langle \chi_{V_i}, \chi_V \rangle \chi_{V_i}$.

4. Let V be a finite-dimensional representation. Then,

$$\langle \chi_V, \chi_V \rangle = \sum_i r_i^2 = \sum_i \langle \chi_{V_i}, \chi_V \rangle^2.$$

5. A finite-dimensional representation V is irreducible if and only if $\langle \chi_V, \chi_V \rangle = 1$.

Proof. (1): By Schur's lemma, we have that

$$\dim \text{Hom}_G(V, W) = \begin{cases} 1, & (V, \rho_V) \cong (W, \rho_W) \\ 0, & (V, \rho_V) \not\cong (W, \rho_W) \end{cases}.$$

We combine this with theorem 2.5.1.

(2): By the first statement, the characters $(\chi_{V_i})_{1 \leq i \leq m}$ form an orthonormal set in $\text{Fun}(G, \mathbb{C})$. In fact, since characters are class functions (proposition 2.2.5), they form an orthonormal set in $\text{Fun}_{\text{cl}}(G, \mathbb{C})$. Since orthonormal sets are linearly independent, to check this is a basis it suffices to show that m is equal to the number of irreducible representations of G . This is equal to the number of conjugacy classes of G (theorem 1.15.15), which is equal to the dimension of $\text{Fun}_{\text{cl}}(G, \mathbb{C})$.

(3): The first statement is an immediate consequence of proposition 1.12.1 together with theorem 2.5.1. The second statement then follows from proposition 2.3.1. To explicitly write it out, we have the following: $r_i = \dim \text{Hom}_G(V_i, V) = \langle \chi_{V_i}, \chi_V \rangle$. Therefore, we have that

$$\chi_V = \sum_{i=1}^m r_i \chi_{V_i} = \sum_{i=1}^m \langle \chi_{V_i}, \chi_V \rangle \chi_{V_i}.$$

(4): This is an immediate consequence of the second and third statement. Alternatively this is proposition 1.12.5 together with theorem 2.5.1. Explicitly, we have

$$\langle \chi_V, \chi_V \rangle = \dim \text{Hom}_G(V, V) = \sum_{i=1}^m r_i^2 = \sum_{i=1}^m \langle \chi_{V_i}, \chi_V \rangle^2.$$

(5): This is an immediate consequence of the fourth statement: We have that $\langle \chi_V, \chi_V \rangle = 1$ if and only if there is some i such that $r_i = 1$ and for all other $j \neq i$, $r_j = 0$. In other words, this is equivalent to the fact that $V \cong V_i$. $\square$

Remark 2.5.6 — Note that the third statement implies theorem 2.1.11.

Corollary 2.5.7

Let V and W be finite dimensional irreducible representations of a finite group G . If $\chi_V = \chi_W$, then $V \cong W$.

Proof. Let $V_1, \dots, V_m$ be a full set of irreducible representations. Then corollary 2.5.5 tells us that

$$V \cong \bigoplus_{i=1} V_i^{\langle \chi_{V_i}, \chi_V \rangle}, \quad W \cong \bigoplus_{i=1} V_i^{\langle \chi_{V_i}, \chi_W \rangle}.$$

Since $\chi_V = \chi_W$, we can let $r_i := \langle \chi_{V_i}, \chi_V \rangle = \langle \chi_{V_i}, \chi_W \rangle$. Therefore, it follows that

$$V \cong \bigoplus_{i=1} V_i^{r_i} \cong W.$$

□

Definition 2.5.8. An **irreducible character** is a character of an irreducible representation.

Note that, in exercise 2.2.3, we used an obvious basis of the vector space of class functions: for each conjugacy class $\mathcal{C} \subseteq G$, we can consider the function $\delta_{\mathcal{C}}$ given by

$$\delta_{\mathcal{C}}(g) = \begin{cases} 1, & \text{if } g \in \mathcal{C} \\ 0, & \text{otherwise} \end{cases}.$$

Therefore, the content of character theory is *there are two natural bases of the vector space of class functions*, given by the Kronecker delta functions of conjugacy classes, or by the irreducible characters (corollary 2.5.5).

Remark 2.5.9 — The basis of Kronecker delta functions is not orthonormal: it is still true that $\langle \delta_{\mathcal{C}}, \delta_{\mathcal{C}'} \rangle = 0$, but $\langle \delta_{\mathcal{C}}, \delta_{\mathcal{C}} \rangle = |G|^{-1} \sum_{g \in \mathcal{C}} 1 = \frac{|\mathcal{C}|}{|G|}$. Thus, an orthonormal

basis is given by the functions $\sqrt{\frac{|G|}{|\mathcal{C}|}} \delta_{\mathcal{C}}$.

Remark 2.5.10 — There is an interpretation of the normalization coefficients in terms of the orbit-stabilizer theorem: note that $\mathcal{C}$ is an orbit of G under the conjugation action $G \times G, (g, h) \mapsto ghg^{-1}$. Let $h \in \mathcal{C}$; then, $\mathcal{C} = G \cdot h$ under this action. Hence, $\frac{|G|}{|\mathcal{C}|}$ is the size of the stabilizer of h . But the stabilizer is just the centralizer, $Z_G(h) = \{g \in G \mid gh = hg\}$. Thus, the orthonormal basis consists of functions $\sqrt{|Z_G(h)|} \delta_{\mathcal{C}}$, for $h \in \mathcal{C}$.

Aside from the normalization issue, the bases $\delta_{\mathcal{C}}$ are still quite different:

Example 2.5.11

Let $G = C_n = \{1, g, \dots, g^{n-1}\}$ be a cyclic group. Then, $\text{Fun}_{\text{cl}}(G, \mathbb{C}) = \text{Fun}(G, \mathbb{C})$, with dimension equal to $|G|$. The basis δ_g , $g \in G$ are just the functions which are one on a single element $g \in G$ and zero elsewhere (the ‘Kronecker delta functions’). On the other hand, the irreducible representations are of the form $(\mathbb{C}_\zeta, \rho_\zeta)$ where $\mathbb{C}_\zeta := \mathbb{C}$ and $\rho_\zeta(g^k) = \zeta^k \in \mathbb{C}^\times$, for $\zeta^n = 1$. This is also the same as the basis $\chi_{\mathbb{C}_\zeta} = \rho_\zeta$ of characters since taking the character of a one-dimensional matrix representation doesn’t do anything.

If we rather use the isomorphism $G \cong \mathbb{Z}/n = \{\bar{0}, \dots, \overline{n-1}\}$, then for $\zeta = e^{i\theta}$ (for $\theta = \frac{2\pi il}{n}$ for some $l \in \mathbb{Z}$), we get $\chi_{\mathbb{C}_\zeta}(\bar{k}) = e^{i\theta k}$, which is a basis of exponential functions. The change of basis matrix between the Kronecker delta functions and these exponential functions is the **discrete Fourier transform**, computed efficiently by the famous ‘fast Fourier transform’. The formula for this transform $f = (f(k))_{0 \leq k \leq n} \mapsto (r_\zeta)_{\zeta^n=1}$, with $f = \sum_\zeta r_\zeta \chi_{\mathbb{C}_\zeta}$, is given by corollary 2.5.5:

$$r_\zeta = \langle d, \chi_{\mathbb{C}_\zeta} \rangle = \frac{1}{n} \sum_{k=0}^{n-1} f(k) \zeta^{-k}.$$

Example 2.5.12

If G is a finite abelian group, almost the same thing as in the preceding example holds, except now we have to take a product of cyclic groups and exponential functions on a product of groups $\mathbb{Z}/n_i$. The change of basis is a product of discrete Fourier transforms.

With the preceding example(s) in mind, the general change-of-basis matrix between Kronecker delta functions and irreducible characters can be thought of as a **nonabelian Fourier transform**.

Example 2.5.13

Let’s consider the reflection representation (V, ρ_V) of S_n . Since this is irreducible, the formula in example 2.3.4 yields the identity

$$n! = n! \langle \chi_V, \chi_V \rangle = \sum_{\sigma \in S_n} (|\{1, \dots, n\}^\sigma| - 1)^2.$$

This is a purely combinatorial identity but does not seem very obvious from combinatorics alone! For example, when $n = 3$ we get

$$6 = (3 - 1)^2 + 3 \cdot (1 - 1)^2 + 2 \cdot (0 - 1)^2,$$

considering first the identity, then the two-cycles, then the three-cycles. When $n = 4$ we get

$$24 = (4 - 1)^2 + 6 \cdot (2 - 1)^2 + 8 \cdot (1 - 1)^2 + 3 \cdot (0 - 1)^2 + 6 \cdot (0 - 1)^2,$$

considering the identity, the two-cycles, the three-cycles, the products of two disjoint two-cycles, and finally the four cycles.

Example 2.5.14

We return to $G = C_n = \{1, g, \dots, g^{n-1}\}$ in the notation of example 2.5.11. Corollary 2.5.5 says that $\langle \chi_{\mathbb{C}_\zeta}, \chi_{\mathbb{C}_\xi} \rangle = \delta_{\zeta, \xi}$. We can verify this explicit:

$$\langle \chi_{\mathbb{C}_\zeta}, \chi_{\mathbb{C}_\xi} \rangle = \frac{1}{n} \sum_{l=0}^{n-1} \zeta^l (\xi)^{-l} = \frac{1}{n} \sum_{l=0}^{n-1} \left(\frac{\zeta}{\xi} \right)^l,$$

which is indeed $\delta_{\zeta, \xi}$ since $\frac{\zeta}{\xi}$ is an n -th root of unity.

Example 2.5.15

Let us keep the notation from the previous example. By exercise 1.2.11, an m -dimensional representation $(\mathbb{C}^m, \rho)$ is the same as a matrix $T \in GL_m(\mathbb{C})$ with $T^n = I$, by the correspondence $\rho(g^k) = T^k$. Let us decompose this as

$$(\mathbb{C}^m, \rho) \cong \bigoplus_{\zeta^n=1} (\mathbb{C}, \rho_\zeta)^{r_\zeta}.$$

By corollary 2.5.5,

$$r_\zeta = \overline{r_\zeta} = \langle \chi_\rho, \chi_{\mathbb{C}_\zeta} \rangle = \frac{1}{n} \sum_{l=0}^{n-1} \text{Tr}(T^l) \zeta^{-l}, \quad (1)$$

which is applying the discrete Fourier transform to χ_ρ .

Of course, we can also compute the r_ζ with linear algebra: it is the multiplicity of ζ as an eigenvalue of T , i.e., the multiplicity of ζ as a root of the characteristic polynomial $\det(xI - T)$ of T . This is equivalent to the above answer though: since T is diagonalizable with r_ξ entries of ξ on the diagonal for all ξ with $\xi^n = 1$, we have

$$\text{Tr}(T^l) = \sum_{\xi^n=1} r_\xi \xi^l. \quad (2)$$

Therefore, (1) becomes

$$r_\zeta = \frac{1}{n} \sum_{l=0}^{n-1} \zeta^{-l} \sum_{\xi^n=1} r_\xi \xi^l = \frac{1}{n} \sum_{\xi^n=1} r_\xi \sum_{l=0}^{n-1} \left(\frac{\xi}{\zeta} \right)^l,$$

which is again true since $\frac{\xi}{\zeta}$ is an n -th root of unity. (This is just reproving that $\chi_{\mathbb{C}_\zeta}$ and $\chi_{\mathbb{C}_\xi}$ are orthogonal for $\zeta \neq \xi$ as in example 2.5.14).

Remark 2.5.16 (Non-examinable) — Actually, (1) and (2) are almost the same transformations: this is saying (which we just proved) that the discrete Fourier transform is almost inverse to itself, up to negating the coefficient in the sum, as well as the extra normalization of $\frac{1}{n}$. To fix the second issue we can consider the normalized discrete Fourier transform, call it F , by multiplying the RHS of example 2.5.11 by $n^{-\frac{1}{2}}$; then $F^2(f)(x) = f(n-x)$, reverses the order of a sequence. Thus F^4 is the identity. (This is actually the same as the normalization of remark 2.5.9: it is

what is needed to make F unitary.)

§2.6 Character Tables

The idea of a character table is very simple: list the irreducible characters. For finite groups, this determines all of the characters by Maschke's theorem.

Definition 2.6.1. Let G be a finite group. Let $V_1, \dots, V_m$ be the irreducible representations of G , $\mathcal{C}_1, \dots, \mathcal{C}_m$ be the conjugacy classes of G , and $g_i \in \mathcal{C}_i$ be the representative. Then, the **character table** of G is the matrix whose (i, j) -th entry is $\chi_{V_i}(g_j)$.

Remark 2.6.2 — Recall that the number m equals the number of conjugacy classes and irreducible representations (theorem 1.15.15).

Example 2.6.3

Let $G = C_n = \{1, g, \dots, g^{n-1}\}$, and let $\xi = e^{\frac{2\pi i}{n}}$, the standard primitive n -th root of unity.

	1	g	$\dots$	g^{n-1}
$\chi_{\mathcal{C}_1}$	1	1	$\dots$	1
$\chi_{\mathcal{C}_\xi}$	1	ξ	$\dots$	ξ^{n-1}
$\chi_{\mathcal{C}_{\xi^2}}$	1	ξ^2	$\dots$	$\xi^{2(n-1)}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\chi_{\mathcal{C}_{\xi^{n-1}}}$	1	$\xi^{(n-1)^2}$	$\dots$	ξ

This is the Vandermonde matrix.

Example 2.6.4

For C_3 , we would have

	1	g	g^2
$\chi_{\mathcal{C}_1}$	1	1	1
$\chi_{\mathcal{C}_\xi}$	1	ξ	ξ^2
$\chi_{\mathcal{C}_{\xi^2}}$	1	ξ^2	ξ

Example 2.6.5

For C_4 , we would have

	1	g	g^2	g^3
$\chi_{\mathcal{C}_1}$	1	1	1	1
$\chi_{\mathcal{C}_\xi}$	1	i	-1	$-i$
$\chi_{\mathcal{C}_{\xi^2}}$	1	-1	i	-1
$\chi_{\mathcal{C}_{\xi^3}}$	1	$-i$	-1	i

Exercise 2.6.6. Denote the matrix above by $A_{kl} = \xi^{kl}$. Then, check that $A\bar{A}^T = n \cdot I$.

Example 2.6.7

Consider $G = C_2 \times C_2 = \{\pm 1\} \times \{\pm 1\}$. The elements of this group are $(1, 1)$, $(-1, 1)$, $(1, -1)$, $(-1, -1)$. Then, we have the table

	$(1, 1)$	$(-1, 1)$	$(1, -1)$	$(-1, -1)$
$\chi_{\mathbb{C}_{1,1}}$	1	1	1	1
$\chi_{\mathbb{C}_{-1,1}}$	1	-1	1	-1
$\chi_{\mathbb{C}_{1,-1}}$	1	1	-1	-1
$\chi_{\mathbb{C}_{-1,-1}}$	1	-1	-1	1

where $\chi_{a,b}(x, y) := x^a y^b$.

Example 2.6.8

Let G, H be groups and V, W be finite dimensional representations of G, H , respectively. Then,

$$\chi_{V \otimes W}(g, h) = \chi_V(g) \chi_W(h).$$

Exercise 2.6.9. Show that the character table of $G \times H$ is the Kronecker product of character tables of G, H .

Remark 2.6.10 — In fact, we see that this exercise is highlighted in the example above with $C_2 \times C_2$. Its character table is the Kronecker product of two copies of character table of C_2 .

Example 2.6.11

We compute the character table of S_3 . Recall that the irreducible representations S_3 are $\mathbb{C}_{\text{triv}}$ (trivial), $\mathbb{C}_-$ (sign), and V (reflection). We know that :

$$\begin{aligned} \chi_{\mathbb{C}_{\text{triv}}}(\sigma) &= 1, \quad \forall \sigma \\ \chi_{\mathbb{C}_-}(\sigma) &= \text{sign}(\sigma), \quad \forall \sigma \\ \chi_V(\sigma) &= |\{1, 2, 3\}^\sigma| - 1 \end{aligned}$$

Therefore,

	$()$	(12)	(123)
$\chi_{\mathbb{C}_{\text{triv}}}$	1	1	1
$\chi_{\mathbb{C}_-}$	1	-1	1
χ_V	2	0	-1

Example 2.6.12

Let us now calculate the $\langle \chi_V, \chi_W \rangle$ for any two irreducible subrepresentations V, W of S_3 .

Recall that corollary 2.5.5 says that for any two irreducible representation V, W , we

have $\langle \chi_V, \chi_W \rangle = \begin{cases} 1, & V \cong W \\ 0, & V \not\cong W \end{cases}$. Concretely:

$$\langle \chi_V, \chi_W \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_V(g) \overline{\chi_W(g)},$$

Let χ_V have row (a, b, c) in a character table and let χ_W have row (a', b', c') in the character table. That is,

$$\begin{aligned} \chi_V(()) &= a, & \chi_V((12)) &= b, & \chi_V((123)) &= c \\ \chi_W(()) &= a', & \chi_W((12)) &= b', & \chi_W((123)) &= c'. \end{aligned}$$

Thus, the sum would be

$$\begin{aligned} \langle \chi_V, \chi_W \rangle &= \frac{1}{|G|} \left(\chi_V(()) \overline{\chi_W(())} + \chi_V((12)) \overline{\chi_W((12))} + \chi_V((123)) \overline{\chi_W((123))} + \dots \right) \\ &= \frac{1}{6} (1 \cdot 1 + 3 \cdot 1 \cdot (-1) + 2 \cdot 1 \cdot 1) = 0. \end{aligned}$$

In fact, going through other calculations verifies corollary 2.5.5.

Exercise 2.6.13. Let G be a finite group. Show that $z \in G$ is central if and only if for all irreducible representations V of G , $|\chi_V(z)| = \dim(V)$.

Solution. First let (V, ρ) be a representation of G , and let us write $C = \bigoplus_{i=1}^m V_i$, where each V_i is irreducible. The key to this will be the group theoretic fact that every element in the center has a conjugacy class of size one consisting only itself. This is because we have that $\forall g \in G$ and $z \in Z(G)$, $gzg^{-1} = z$, where $Z(G)$ is the center of G .

First assume that $z \in Z(G)$. By the orthogonality relations, we have

$$|G| = \frac{|G|}{|\mathcal{C}_z|} = \sum_{i=1}^m |\chi_{V_i}(z)|^2,$$

where $\mathcal{C}_z$ denotes the conjugacy class $\{z\}$. Combining this with the sum of squares formula

$$|G| = \sum_{i=1}^m \dim(V_i)^2,$$

we attain that

$$\sum_{i=1}^m |\chi_{V_i}(z)|^2 - \dim(V_i)^2 = 0.$$

However, as $|\chi_{V_i}(z)| \leq \dim(V_i)$ in general, the above equality tells us that we must have the reverse inequality $|\chi_{V_i}(z)| \geq \dim(V_i)$ as well. Therefore, we attain that $|\chi_{V_i}(z)| = \dim(V_i)$.

Now for the converse. Let $z \in G$ and suppose that $|\chi_{V_i}(z)| = \dim(V_i)$, we can recover that

$$|G| = \sum_{i=1}^m \dim(V_i)^2 = \sum_{i=1}^m |\chi_{V_i}(z)|^2 = \frac{|G|}{|\mathcal{C}_z|},$$

where $\mathcal{C}_z$ denotes the conjugacy class containing z . The only way for this to hold is if $|\mathcal{C}_z| = 1$; in other words, $\mathcal{C}_z = \{z\}$. This implies that z is an element in the center, so we have proven both directions. $\square$

§2.6.i Row and Column Orthogonality; Unitarity

Let us continue from the idea in the previous example. In general, for a group G with irreducible representations $V_1, \dots, V_m$ and conjugacy classes $\mathcal{C}_1, \dots, \mathcal{C}_m$ with representations $g_1 \in \mathcal{C}_1, \dots, g_m \in \mathcal{C}_m$, corollary 2.5.5 tells us that

$$\begin{aligned} \delta_{ij} &= \langle \chi_{V_i}, \chi_{V_j} \rangle \\ &= \frac{1}{|G|} \sum_{g \in G} \chi_{V_i}(g) \overline{\chi_{V_j}(g)} \\ &= \frac{1}{|G|} \sum_{k=1}^m |\mathcal{C}_k| \chi_{V_i}(g_k) \overline{\chi_{V_j}(g_k)} \\ &= \frac{1}{|G|} \sum_{k=1}^m |\mathcal{C}_k| A_{ik} \overline{A_{jk}}. \end{aligned}$$

Here, $A_{ij} = \chi_{V_i}(g_j)$. Because we want to attain a sum without the *weights* $|\mathcal{C}_k|$, we try to normalize the rows so that we don't get this weight inside the sum. Define

$$U_{ij} := \sqrt{\frac{|\mathcal{C}_k|}{|G|}} A_{ij}.$$

Then,

$$\delta_{ij} = \sum_{k=1}^m U_{ik} \overline{U_{jk}}.$$

In other words, we're saying that $I = U \overline{U}^T$.

Definition 2.6.14. A **unitary matrix** is a matrix $U \in GL_n(\mathbb{C})$ s such that $U^{-1} = \overline{U}^T$.

Remark 2.6.15 — $U^{-1} = \overline{U}^T \iff U \overline{U}^T = I \iff \overline{U}^T U = I$.

Lemma 2.6.16

The following are equivalent:

1. U is unitary
2. The rows of U are orthogonal with respect to the dot product.
3. The columns of U are orthogonal with respect to the dot product.

Proof. This is essentially the remark above. $\square$

Now, by lemma 2.6.16, we deduce that $\bar{U}^T U = I$, where U is as defined above. Thus,

$$\begin{aligned}\delta_{ij} &= \sum_{k=1}^m U_{ki} \bar{U}_{kj} \\ &= \sum_{k=1}^m \left(\sqrt{\frac{|\mathcal{C}_i|}{|G|}} A_{ki} \right) \left(\sqrt{\frac{|\mathcal{C}_j|}{|G|}} A_{kj} \right) \\ &= \frac{\sqrt{|\mathcal{C}_i| |\mathcal{C}_j|}}{|G|} \sum_{k=1}^m A_{ki} \bar{A}_{kj}.\end{aligned}$$

To summarize, we have the following:

- Row Orthogonality: $\delta_{ij} |G| = \sum_{k=1}^m |\mathcal{C}_k| A_{ik} \bar{A}_{jk}$
- Column Orthogonality: $\delta_{ij} \frac{|G|}{\sqrt{|\mathcal{C}_i| |\mathcal{C}_j|}} = \sum_{k=1}^m A_{ki} \bar{A}_{kj}$

Example 2.6.17 (Examples for Column Orthogonality)

For $i = j$, we get

$$1 = \frac{|\mathcal{C}_i|}{|G|} \sum_{k=1}^m |A_{ki}|^2 \implies \frac{|G|}{|\mathcal{C}_i|} = \sum_{k=1}^m |A_{ki}|^2.$$

When $i \neq j$, we get

$$0 = \frac{|\mathcal{C}_i|}{|G|} \sum_{k=1}^m |A_{ki}|^2 \implies 0 = \sum_{k=1}^m \chi_{V_k}(g_i) \overline{\chi_{V_k}(g_j)}.$$

Example 2.6.18

In the example with S_3 , we can check some calculations:

$$\chi_{\mathcal{C}}((1)) \overline{\chi_{\mathcal{C}}((12))} + \chi_{\mathcal{C}_-}((1)) \overline{\chi_{\mathcal{C}_-}((12))} + \chi_V((1)) \overline{\chi_V((12))} = 1 \cdot \bar{1} + 1 \cdot \bar{1} + 2 \cdot \bar{0} = 0.$$

We also have

$$\frac{|S_3|}{|(12)|} = 2 = \chi_{\mathcal{C}_{\text{triv}}}((12)) \overline{\chi_{\mathcal{C}_{\text{triv}}}((12))} + \chi_{\mathcal{C}_-}((12)) \overline{\chi_{\mathcal{C}_-}((12))} + \chi_V((12)) \overline{\chi_V((12))}.$$

Remark 2.6.19 — Note that example 2.6.17 gave, for any g , that

$$\frac{|G|}{|[g]|} = \sum_{k=1}^m |\chi_{V_k}(g)|^2.$$

Particularly, letting $g = 1$, we have that $[g] = \{1\}$, and so

$$|G| = \sum_{k=1}^m |\chi_{V_k}(1)|^2 = \sum_{k=1}^m (\dim V_k)^2.$$

This recovers the sum of squares formula.

Remark 2.6.20 — Using the row and column orthogonality, we can fill in a single missing row or column. Indeed, in a unitary matrix, the condition that the last column v_n is orthogonal to the span of the preceding columns $v_1, \dots, v_{n-1}$ shows that $v_n \in (v_1, \dots, v_{n-1})^\perp$ which is one-dimensional; this fixes v_n up to scaling (by a number of absolute value one). In the case of the character table we know the first entry of the column is one (if we put the trivial representation first) and this fixes the scaling. Similarly for a missing row, the same argument applies except now the first entry is equal to the dimension, which we can compute from the sum of squares formula.

Concretely, the way to compute v_n up to scaling, for a unitary matrix, is to take any vector v not in the span of $v_1, \dots, v_{n-1}$; then by lemma 2.4.6, $v = \sum_{i=1}^n a_i v_i$ with $a_i = v \cdot \overline{v_i}$. Then v_n is a scalar multiple of $v - \sum_{i=1}^{n-1} a_i v_i$. The same technique works for a missing row. Applying this to the unitary matrix U obtained from the character table A by

$$U_{ij} = \sqrt{\frac{|C_j|}{|G|}} A_{ij},$$

we can get a missing row or column up to scaling, and then we simply rescale as indicated in the preceding paragraph.

In the case of a missing row, we can also use an easier technique: we know that

$$\chi_{C[G]} = |G|\delta_1 = \sum_{i=1}^m \dim V_i \cdot \chi_{V_i},$$

so if we know all of the χ_{V_i} , except one, we can get the remaining one (first computing the remaining dimension by the sum of squares formula). This yields concretely for $g_j \neq 1$:

$$\chi_{V_i}(g_k) = -(\dim V_i)^{-1} \sum_{j \neq i} \dim V_j \cdot \chi_{V_j}(g_k).$$

§2.6.ii More Examples

Note that we do not need to memorize any of this information for the exam.

Example 2.6.21

Let us compute the character table of D_8 .

Solution. Recall the description $D_8 = \{x^a y^b \mid 0 \leq a \leq 3, 0 \leq b \leq 1\}$ with x the anticlockwise 90° rotation and y the reflection about the x -axis. The conjugacy classes of rotations are $\{1\}$, $\{x, x^3\}$ and $\{x^2\}$ (since $xyx^{-1} = x^{-1} = x^3$ but x^2 is central), and the conjugacy classes of reflections are $\{y, x^2y\}$ and $\{xy, x^3y\}$ (since $xyx^{-1} = x^2y$ shows that in general $x^a y$ and $x^{a'} y$ are conjugate if and only if $2 \mid (a - a')$). This makes five conjugacy classes. As explained in example 1.14.10, there are four one-dimensional representations (by example 1.10.2) and one two-dimensional irreducible representation up to isomorphism (the one of example 1.1.6). Call the one dimensional representations

$(\mathbb{C}_{\pm,\pm}, \rho_{\pm,\pm})$), where $\rho_{+,+}$ is the trivial representation, $\rho_{-,+}$ is the representation sending reflections to -1 and rotations to 1 , $\rho_{+,-}$ sends the generating rotation x to -1 and the generating reflection y to 1 , and $\rho_{-,-}$ sends both x and y to -1 . Call the two-dimensional representation $\chi_{\mathbb{C}^2}$. The trace of a rotation matrix by angle θ is $2 \cos \theta$, so $\chi_{\mathbb{C}^2}(x) = 0$ whereas $\chi_{\mathbb{C}^2}(x^2) = -2$. Also the trace of a reflection matrix is always zero since it has one and minus one as eigenvalues (or from the explicit description). Putting this together, we get:

Size of class:	1	$\{x, x^3\}$	$\{x^2\}$	$\{y, x^2y\}$	$\{xy, x^3y\}$
	1	2	1	2	2
$\rho_{+,+}$	1	1	1	1	1
$\rho_{-,+}$	1	1	1	-1	-1
$\rho_{+,-}$	1	-1	1	1	-1
$\rho_{-,-}$	1	-1	1	-1	1
$\chi_{\mathbb{C}^2}$	2	0	-2	0	0

We can now go through some calculations of row and column orthogonality as a sanity check. For instance, we may check

$$1^2 + 1^2 + (-1)^2 + (-1)^2 + 0^2 = 4 = \frac{|D_8|}{|\{y, x^2y\}|}.$$

We can also check, for example, $\langle \rho_{+,-}, \rho_{-,-} \rangle$.

$$1 \cdot 1 + 2 \cdot (-1) \cdot (-1) + 1 \cdot 1 \cdot 1 + 2 \cdot (-1) \cdot (-1) + 2 \cdot (-1) \cdot 1 = 0.$$

□

Example 2.6.22

We can compute the character table of the **Quaternion group** $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with multiplication given by $i^2 = j^2 = k^2 = -1$, $ij = k$, $jk = i$, $ki = j$, $(-1)^2 = 1$. the conjugacy classes are $\{1\}$, $\{-1\}$, $\{\pm i\}$, $\{\pm j\}$, and $\{\pm k\}$. The one dimensional representations are $\rho_{\pm,\pm}$ determined by the values on i and j which are ± 1 (so the two subscripts of ρ give the signs of the images of i and j respectively). There is one two dimensional representation, $\rho(\pm 1) = \pm I$, $\rho(\pm i) = \pm \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$, $\rho(\pm j) = \pm \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, and $\rho(\pm k) = \pm \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}$. We get

Size of class:	1	$\{-1\}$	$\{\pm i\}$	$\{\pm j\}$	$\{\pm k\}$
	1	1	2	2	2
$\rho_{+,+}$	1	1	1	1	1
$\rho_{-,+}$	1	1	-1	1	-1
$\rho_{+,-}$	1	1	1	-1	-1
$\rho_{-,-}$	1	1	-1	-1	1
$\chi_{\mathbb{C}^2}$	2	-2	0	0	0

Remark 2.6.23 — Notice that this is the same table as for D_8 if we swap the columns accordingly: in the D_8 case we can instead order the columns as $\{1\}$, $\{x^2\}$, $\{y, x^2y\}$, $\{x, x^3\}$, $\{xy, x^2y\}$. This is remarkable since $Q_8 \not\cong D_8$: for instance, in Q_8 there are six elements of order four, whereas in D_8 there are only two. This shows

that the character table *does not* determine the group up to isomorphism.

Actually in this case, the two character tables have to coincide since 8 is a small number for a nonabelian group (e.g., the only possible sum of squares of dimensions is $1^2 + 1^2 + 1^2 + 1^2 + 2^2 = 8$ since 1 has to occur for the trivial representation but we can’t have only 1’s since the group is nonabelian). (See exercise ??)

Example 2.6.24

Let us compute the character table for A_4 .

Solution. As explained in exercise 1.14.13, there are three irreducible one-dimensional representations, $(\mathbb{C}, \rho_\zeta \circ q|_{A_4})$, for $\zeta^3 = 1$, defined as follows. Let $\mathbb{C}_\zeta = \mathbb{C}$. There is a surjection $q: S_4 \rightarrow S_3$, $q((12)) = (12)$, $q((23)) = (23)$, $q((34)) = (12)$, which restricts to a surjection $q|_{A_4}: A_4 \rightarrow A_3$. Then, $A_3 \cong C_3$, so its irreducible representations are ρ_ζ for $\zeta^3 = 1$ (corollary 1.9.7). Then, as outlined in exercise 1.14.13 it follows that $A_3 \cong (A_4)_{ab}$ and there is one three-dimensional irreducible representation. The three-dimensional irreducible representation can be given by restricting the reflection representation (V, ρ_V) of S_4 to the group A_4 , i.e., $(V, \rho_V \circ i)$ for $i: A_4 \rightarrow S_4$ the inclusion. Now there are four conjugacy classes in A_4 , given by the classes $[1] = \{1\}$, $[(12)(34)] = \{(12)(34), (13)(24), (14)(23)\}$, and the three-cycles must split into two conjugacy classes: $[(123)] = \{(123), (134), (142), (243)\}$, $[(132)] = \{(132), (124), (143), (234)\}$. For another way to compute the latter two conjugacy classes, see the following remark. Taking traces, we have, for $w := e^{\frac{2\pi i}{3}} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$,

	$[1]$	$[(12)(34)]$	$[(123)]$	$[(132)]$
Size of class:	1	3	4	4
$\mathbb{C}_1$	1	1	1	1
$\mathbb{C}_w$	1	1	w	w^2
$\mathbb{C}_{w^2}$	1	1	w^2	w
V	3	-1	0	0

We can also check that V is irreducible, as

$$3^2 + 3 \cdot (-1)^2 + 4 \cdot 0^2 + 4 \cdot 0^2 = 12 = |A_4|.$$

□

Example 2.6.25

Let us compute the character table for S_4 .

Solution. In example 1.14.5, we’ve already found the irreducible representations of S_4 . We have the trivial and sign representation, the reflection representation of S_3 , the reflection representation for S_4 , and the reflection times sign representation for S_4 .

	$[()]$	$[(12)]$	$[(123)]$	$[(12)(34)]$	$[(1234)]$
$\chi_{\mathbb{C}}$	1	1	1	1	1
$\chi_{\mathbb{C}_-}$	1	-1	1	1	-1
χ_U	2	0	-1	2	0
$\chi_{\mathbb{C}}$	3	1	0	-1	-1
χ_{V_-}	3	-1	0	-1	1

To fill the above for χ_U , we have taken the vertical dot products. For example,

$$1 \cdot 1 + 1 \cdot (-1) + 2 \cdot x + 3 \cdot 1 + 3 \cdot (-1) = 0 \implies x = 0.$$

□

Example 2.6.26

We search for the character table of A_5 .

Solution. We first decompose 60 as a sum of squares. We note that we have the trivial representation, and noting that A_5 is simple, this is the only one dimensional representation. We also have the restriction of four-dimensional representation reflection representation of S_5 . Thus, we have $60 - 1^2 - 4^2 = 43$ left to find. It can either be then

$$43 = 5^2 + 3^2 + 3^2 = 4^2 + 3^2 + 3^2 + 3^2.$$

We now find the number of conjugacy classes of A_5 to see how many squares there are. We recall the group theoretic fact that conjugacy classes from S_n may *split* in A_n (just as a slight remark, conjugates in A_n are always conjugate in S_n). The criterion for this is as follows: $\sigma \in A_n$ has $[\sigma]_{S_n}$ split into $[\sigma]_{A_n} \cup [(12)\sigma(12)]_{A_n}$ if and only if there exists no odd permutation commuting with σ . The conjugacy classes of S_5 are

$$(), (12), (12)(34), (123), (123)(45), (1234), (12345).$$

Amongst these, we only consider the even ones. We note that $(12)(34)$ commutes with (12) and $(123)(45)$ commutes with (45) (of course the identity doesn't split). Then, (12345) splits (it only commutes with powers of itself). This gives the conjugacy classes of A_5 as

$$[], [(12)(34)], [(123)], [(12345)], [(13524)].$$

Thus, our sum of squares is $60 = 1^2 + 3^2 + 3^2 + 4^2 + 5^2$. We now try to build the character table for A_5 . The column for the trivial representation is easy, and we have that the reflection representation S_5 restricts to an irreducible representation of A_5 (check this!).

Size of class:	$[\text{()}]$	$[(12)(34)]$	$[(123)]$	$[(12345)]$	$[(13524)]$
$\chi_{\mathbb{C}}$	1	15	20	12	12
$\chi_{\mathbb{C}^3_1}$	3				
$\chi_{\mathbb{C}^3_2}$	3				
$\chi_{\mathbb{C}^4}$	4	0	1	-1	-1
$\chi_{\mathbb{C}^5}$	5				

We now need to fill the remaining rows $\chi_{\mathbb{C}^3_1}, \chi_{\mathbb{C}^4}, \chi_{\mathbb{C}^5}$. We can use geometry to construct a three-dimensional irreducible representation since we know that A_5 acts as the group of rotations of the icosahedron or dodecahedron. We start with finding the characters of the three dimensional representations using the icosahedron. Note that the icosahedron is a regular solid in $\mathbb{R}^3$ with 12 vertices, 5 edges meeting at each, with 20 triangular faces. We use without given that the orientation-preserving symmetries of the icosahedron is a subgroup of $SO(3)$ isomorphic to A_5 , which we call G . In particular, any $G \leq SO(3) \leq GL(3)$ is, by definition, a faithful three-dimensional representation of $G \cong A_5$. This turns

out to be irreducible (won't prove). This gives a representation of A_5 on $\mathbb{R}^3$. Note that the elements of $SO(3)$ of order n are conjugate to the matrix

$$T_\zeta = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \operatorname{Re}(\zeta) & -\operatorname{Im}(\zeta) \\ 0 & \operatorname{Im}(\zeta) & \operatorname{Re}(\zeta) \end{pmatrix},$$

where ζ is a primitive n -th root of unity. This is because elements of order n must be fixed by some axis, and these must be rotations by some angle. In other words, a matrix $I \neq A \in SO(3)$ gives a splitting $\mathbb{R}^3 = \mathbb{R}_v \oplus v^\perp$, where $(v \in \mathbb{R}^3, v \neq 0)$, preserved by A , with

$$A|_{\mathbb{R}_v} = id_{\mathbb{R}_v}, \quad A|_{v^\perp} \text{ a rotation.}$$

Then, we get that:

- $(12)(34)$ has order 2, so is a negation (rotation of angle π)
- (123) has order 3, so is a rotation of angle $\pm \frac{2\pi}{3}$
- (12345) has order 5, so is a rotation of angle $\pm \frac{2\pi}{5}$ or $\pm \frac{4\pi}{5}$
- (13524) has order 5, so is a rotation of angle $\pm \frac{2\pi}{5}$ or $\pm \frac{4\pi}{5}$

We can now fill out the character table by focusing on the trace of these cases for T_ζ . For instance, we can attain the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \pm \frac{2\pi}{3} & \sin \pm \frac{2\pi}{3} \\ 0 & \sin \pm \frac{2\pi}{3} & \cos \pm \frac{2\pi}{3} \end{pmatrix}$$

for $\chi_{\mathbb{C}_1^3}([(12)(34)])$. This says that this matrix has trace $\chi((12)(34)) = 1 + 2 \cdot (-\frac{1}{2}) = 0$, so we can fill out this entry. Following similarly, (123) results in $\zeta = \frac{-1 \pm i\sqrt{3}}{2}$, and so $\chi((123)) = 1 + 2(\frac{-1}{2}) = 0$. Now for (12345) and (13524) , these are of order 5, so we need to find ζ . This requires a trick.

$$0 = \zeta^4 + \zeta^3 + \zeta^2 + \zeta + 1,$$

implying that

$$0 = \zeta^2(\zeta^2 + \zeta + 1 + \zeta^{-1} + \zeta^{-2}),$$

and setting $t = \zeta + \zeta^{-1}$ implies that we requires $t^2 + t - 1 = 0$. In fact, we note that $t = \zeta + \zeta^{-1} = \zeta + \bar{\zeta} = 2\operatorname{Re}(\zeta)$, which is a term present in the trace for T_ζ . Thus, solving this quadratic, we get $t = \frac{-1 \pm \sqrt{5}}{2}$, so it follows that

$$\chi(12345) = 1 + 2\operatorname{Re}(\zeta) = \frac{1 \pm \sqrt{5}}{2}.$$

In fact, there is some choice involved. We may let $\chi(12345) = \frac{1+\sqrt{5}}{2}$ or $\frac{1-\sqrt{5}}{2}$, which is what leads to the two cases as well. (in other words, this rotation process is a matter of labeling and from a combinatorial view, they're the same)

	$[()]$	$[(12)(34)]$	$[(123)]$	$[(13542)]$	$[(13524)]$
Size of class:	1	15	20	12	12
$\chi_{\mathbb{C}}$	1	1	1	1	1
$\chi_{\mathbb{C}_1^3}$	3	-1	0	$\frac{1+\sqrt{5}}{2}$	$\frac{1-\sqrt{5}}{2}$
$\chi_{\mathbb{C}_2^3}$	3	-1	0	$\frac{1-\sqrt{5}}{2}$	$\frac{1+\sqrt{5}}{2}$
$\chi_{\mathbb{C}^4}$	4	0	1	-1	-1
$\chi_{\mathbb{C}^5}$	5				

With this, we can fill everything out using column orthogonality. For instance,

$$5 \cdot (\chi_{\mathbb{C}^5}([(12)(34)])) + 1 \cdot 1 + 3 \cdot (-1) + 3 \cdot (-1) + 4 \cdot 0 = 0 \implies \chi_{\mathbb{C}^5}([(12)(34)]) = 1.$$

Continuing, we fill out the entire row.

Size of class:	$[()]$	$[(12)(34)]$	$[(123)]$	$[(13542)]$	$[(13524)]$
$\chi_{\mathbb{C}}$	1	1	1	1	1
$\chi_{\mathbb{C}_1^3}$	3	-1	0	$\frac{1+\sqrt{5}}{2}$	$\frac{1-\sqrt{5}}{2}$
$\chi_{\mathbb{C}_2^3}$	3	-1	0	$\frac{1-\sqrt{5}}{2}$	$\frac{1+\sqrt{5}}{2}$
$\chi_{\mathbb{C}^4}$	4	0	1	-1	-1
$\chi_{\mathbb{C}^5}$	5	1	-1	0	0

For completeness, let us find the five-dimensional representation. One thing to note is that all the entries are integers, unlike for the three-dimensional representations. An educated guess would be that this *missing* five-dimensional representation comes from an action of A_5 on a set of size 6. This comes from the fact that the trivial representation is always present on the representation associated to the action of G on a set X . Then, note that if $\mathbb{C}[X] \cong \mathbb{C} \oplus (\mathbb{C}^5, \rho)$, then $\chi_{\mathbb{C}[X]} = \chi_{\mathbb{C}} + \chi_{\mathbb{C}^5}$. By this, the number of fixed points can be calculated by adding 1 to the bottom row. Therefore,

- $(12)(34)$ fixing two points
- (123) fixing zero points.
- (12345) and (13524) fixing one point.

There are now two constructions.

Construction 1: The set of opposite pairs of vertices of the icosahedron. This has size 6.

Construction 2: Let $x = (12345)$ and $y = (12)(35)$. Then, $x^5 = y^2 = 1$ and

$$yxy^{-1} = (12)(35)(12345)(12)(35) = (15432) = x^{-1}.$$

Thus $H = \langle x, y \rangle \cong D_5$, so $|H| = |D_5| = 10$. Therefore, $|A_5/H| = 6$. Therefore, the left coset action of A_5 on A_5/H is an action on a set of size 6 (check that it has the right properties). $\square$

Remark 2.6.27 — A representation associated to the action of G on a set X always has a trivial 1-dimensional subrepresentation $\mathbb{C}(1, \dots, 1)$. This is used in the final part about finding the five-dimensional representation.

§2.7 Kernels of Representations and Normal Subgroups

The next proposition shows that the kernel of a representation can be read off from the character.

Proposition 2.7.1

Let (V, ρ_V) be a representation of a finite group. Then $\ker \rho_V = \{g \in G \mid \chi_V(g) = \dim V\}$.

Proof. We need to show that $\rho_V(g) = I_V$ if and only if $\text{Tr}(\rho_V(g)) = \dim V$. It is clear that $g \in \ker \rho_V$ implies $\chi_V(g) = \dim V$. We only need to show the converse. Suppose $\chi_V(g) = \dim(V)$. By proposition 2.2.8, $|\chi_V(g)| = \dim V$ if and only if $\chi_V(g)$ is a scalar matrix; so $\rho_V(g) = \lambda I_V$ for some $\lambda \in \mathbb{C}$. But then $\chi_V = \lambda \dim(V)$, so that $\lambda = 1$. Therefore, $g \in \ker \rho_V$. $\square$

Remark 2.7.2 — This is one of the few places we needed proposition 2.2.8. Thus, maybe you should review it: the result followed by applying the triangle inequality (viewing $\mathbb{C}$ as $\mathbb{R}^2$) to $\chi_V(g) = \zeta_1 + \cdots + \zeta_n$, where ζ_i are the eigenvalues of $\rho_V(g)$ with multiplicity (which are roots of unity, hence of absolute value one).

Remark 2.7.3 — We can read off the commutator subgroup from the character table due to this theorem. In particular, if $g \in [G, G]$, then it is in the kernel of the abelianization map q_{ab} . Since the one dimensional representations of a group are exactly those given by q_{ab} , we have the $[G, G]$ is in the intersection of the kernels of one-dimensional representations.

Lemma 2.7.4

If a representation V of G decomposes as $V \cong \bigoplus_{i=1}^r V_i^{r_i}$, then $\ker(V) = \bigcap_{j \in J} K_j$, where $K_j = \ker(V_j)$, where $J = \{j \mid r_j \geq 1\}$.

Proof. We have that $\rho_V(g) = I$ if and only if for all $i \in J$, $\rho_{V_i}(g) = I$. $\square$

The next lemma is a generalization of part of the argument of theorem 1.15.2 (there we used the statement below for the case $H = \{1\}$). For $H \leq G$ a subgroup, recall that the left translation action $G \times G/H \rightarrow G/H$ is given by $g \cdot g'H := gg'H$.

Lemma 2.7.5

Let $N \triangleleft G$ be a normal subgroup. Let $(\mathbb{C}[G/N], \rho_{\mathbb{C}[G/N]})$ be the representation associated to the left translation action. Then, $\ker \rho_{\mathbb{C}[G/N]} = N$.

Proof. We first prove for the special case when $N = \{1\}$. Let g be such that $\rho_{\mathbb{C}[G]}(g) = \text{Id}_{\mathbb{C}[G]}$. Then, for all $h \in G$, $gh = h$. In other words, $g = 1$. For general N , the kernel of $\rho_{\mathbb{C}[G/N]}$ (considered as a representation of G) is

$$\{g \in G \mid q(g) \in \ker \rho_{\mathbb{C}[G/N]}\} = \{g \in G \mid q(g) = 1\} = \ker(q) = N.$$

$\square$

Remark 2.7.6 — Caution that the hypothesis that H is normal is necessary (obviously, kernels are always normal).

Remark 2.7.7 — Note the same notation is used for the regular representation of the group G/N , but these two mean different things. However, we may relate these two constructions: pull back the second by the homomorphism $q: G \rightarrow G/N$ defined by $g \mapsto [g]$ to get the first.

Now we show that every normal subgroup is an intersection of kernels of irreducible representations. Together with proposition 2.7.1, this shows that we can read off all normal subgroups (in terms of unions of conjugacy classes) directly from the character table. Let G be a finite group and $V_1, \dots, V_m$ be a full set of irreducible representations, and let $K_i := \ker \rho_{V_i}$.

Proposition 2.7.8

Let $N \triangleleft G$ be any normal subgroup. Let $J := \{j \in \{1, \dots, m\} \mid N \leq K_j\}$. Then, $N = \bigcap_{j \in J} K_j$.

Proof. By lemma 2.7.5, $\ker \rho_{\mathbb{C}[G/N]} = N$. Therefore, by lemma 2.7.4, it follows that $N = \bigcap_{j \in J'} K_j$, where J' is as defined there. Note that $K_j \leq N$ for $j \in J$, hence $J' \subseteq J$. Therefore, $N = \bigcap_{j \in J'} K_j \supseteq \bigcap_{j \in J} K_j$. Letting $j \in J'$, then $N \leq K_j$ for all $j \in J$, so $j \in J$. $\square$

From the above, we can give an explicit way to read off the normal subgroups from the character table:

Corollary 2.7.9

The normal subgroups of G are precisely the subgroups N_J of the form

$$N_J := \{n \in G \mid \chi_{V_j}(n) = \chi_{V_j}(1), \forall j \in J\}, \quad J \subseteq \{1, \dots, m\}.$$

Proof. Via proposition 2.7.1, since $\dim V_j = \chi_{V_j}(1)$, we have that $N_J = \bigcap_{j \in J} K_j$. Thus, K_j is normal, so is N_J . Conversely, by proposition 2.7.8, every normal subgroup is obtained in this way. $\square$

Example 2.7.10

Let $\{1\} = N_{\{1, \dots, m\}}$. If for all $j \in \{1, \dots, m\}$, $\chi_{V_j}(n) = \chi_{V_j}(1)$, then in fact for every finite dimensional representation V_j , $\chi_V(n) = \chi_V(1)$. Thus, in particular, for the regular representation $\chi_{\mathbb{C}[G]}(n) = \chi_{\mathbb{C}[G]}(1)$, so $n = 1$.

Example 2.7.11

Let $G = N_{\{1\}}$. Then, for all $n \in G$, $\chi_{\mathbb{C}}(n) = \chi_{\mathbb{C}}(1)$.

Solution. By proposition 2.7.5, the normal subgroups of G are precisely the sets

$$\{n \mid \forall j \in J, n \in \ker V_j\},$$

as J ranges over all subsets of $\{1, \dots, n\}$. Thus, by proposition 2.7.1, $n \in \ker V_j$ if and only if $\chi_{V_j}(n) = \dim V_j = \chi_{V_k}(1)$. $\square$

Example 2.7.12

Let's take the example of D_4 (note that Q_8 has the same character table). By example 2.6.21, the proper nontrivial normal subgroups of the form K_i are $\{1\} \cup \{x^2\} \cup \mathcal{C}$, where $\mathcal{C} \in \{\{x, x^3\}, \{y, x^2y\}, \{xy, x^3y\}\}$. There are three of these, all of size four. The intersection of any two of them is $\{1, x^2\} = Z(D_4)$.

Example 2.7.13

Next, consider Q_8 . Of course, the character table is the same as that of D_4 so the answer has to be the same, but the elements are written by different labels. By example 2.6.22, the proper nontrivial normal subgroups K_i are $\{1\} \cup \{-1\} \cup \mathcal{C}$ for $\mathcal{C} \in \{\{\pm i\}, \{\pm j\}, \{\pm k\}\}$. There are again three of these, of size four. The intersection of any two of them is $\{\pm 1\} = Z(Q_8)$.

Example 2.7.14

Consider now A_4 . Looking at example 2.6.24, the only proper nontrivial K_i are both equal to $\{1, (12)(34), (13)(24), (14)(23)\}$, which is the commutator subgroup $[A_4, A_4]$ (i.e., the kernel of $q|_{A_4}: A_4 \rightarrow A_3 \cong (A_4)_{ab}$).

Example 2.7.15

Take the example of S_4 . By example 2.6.25, we can get two nontrivial subgroups N_j : $K_2 = A_4$ for $V_2 = \mathbb{C}_-$ and $K_3 = \{1, (12)(34), (13)(24), (14)(23)\}$ for $V_4 = U$. Their intersection is again K_3 , so these are all of the proper normal subgroups.

Example 2.7.16

Finally, let's look at A_5 . Here, we already know this group is simple so has no proper nontrivial normal subgroups, and indeed from example 2.6.26 we see that all the K_j are trivial (except $K_1 = A_5$ of course).

3 Algebras and Modules

§3.1 Definitions and Examples

Definition 3.1.1. A $\mathbb{C}$ -algebra is a tuple of

- A a vector space
- $m_A: A \times A \rightarrow A$ (multiplication), also notated $m_A(a, b) = a \cdot b$ or ab
- 1_A (we may just write 1)

satisfying:

- multiplication is bilinear:
 - $\forall a, b \in A$ and $\forall \lambda \in \mathbb{C}$, $\lambda m_A(a, b) = m_A(\lambda a, b) = m_A(a, \lambda b)$
 - $\forall a, b, c \in A$, $m_A(a, b + c) = m_A(a, b) + m_A(a, c)$ and $m_A(a + b, c) = m_A(a, c) + m_A(b, c)$
- multiplication is associative: $\forall a, b, c \in A$, $a \cdot (b \cdot c) = (a \cdot b) \cdot c$, i.e., $m_A(a, m_A(b, c)) = m_A(m_A(a, b), c)$
- identity: $\forall a \in A$, $a \cdot 1_A = a = 1_A \cdot a$, i.e., $m_A(a, 1_A) = a = m_A(1_A, a)$

Remark 3.1.2 — We have two ‘multiplication-like’ operations: $m_A: A \times A \rightarrow A$ and scalar multiplication $\mathbb{C} \times A \rightarrow A$. There is a relationship between these. For $\lambda \in \mathbb{C}$, $a \in A$, $\lambda a = \lambda(1_A a) = (\lambda 1_A) a$.

Example 3.1.3

$\mathbb{C}$ with the usual multiplication is a $\mathbb{C}$ -algebra.

Example 3.1.4

Given $\mathbb{C}$ -algebras A, B , the **direct sum algebra** $A \oplus B$, with the algebra structure on $A \times B$, with, for $(a, b), (a', b') \in A \oplus B$, $(a, b) \cdot (a', b') = (aa', bb')$.

Example 3.1.5

Some concrete examples of the above are $\mathbb{C} \oplus \mathbb{C}$ or $\mathbb{C} \oplus \mathbb{C} \oplus \dots \oplus \mathbb{C}$

Example 3.1.6 (Algebra of (1-variable) polynomials over $\mathbb{C}$)

We have $\mathbb{C}[x] = \left\{ \sum_{i=0}^m a_i x^i \mid a_0, \dots, a_m \in \mathbb{C}, a_m \neq 0 \right\} \cup \{0\}$. Multiplication is the usual multiplication of polynomials.

Remark 3.1.7 — In the above, we note that the condition $a_m \neq 0$ is so that we don't *double-list* polynomials.

Example 3.1.8

Consider $A := \mathbb{C}[\epsilon]/(\epsilon^2)$, i.e., $\{a + b\epsilon \mid a, b \in \mathbb{C}\}$ with multiplication

$$(a + b\epsilon)(c + d\epsilon) = ac + \epsilon(ad + bc).$$

Definition 3.1.9. An **algebra homomorphism** $\phi: A \rightarrow B$ is a linear map which is multiplicative and sends 1_A to 1_B . It is an **isomorphism** if there is an inverse algebra homomorphism $\phi^{-1}: B \rightarrow A$.

Exercise 3.1.10. Let $\phi: A \rightarrow B$ be an algebra homomorphism. Then, it is an isomorphism if and only if it is bijective.

Example 3.1.11

Given a $\mathbb{C}$ -algebra A , we have already seen an algebra homomorphism $\mathbb{C} \rightarrow A$ defined by $\lambda \mapsto \lambda 1_A$.

Example 3.1.12

The vector space $M_n(\mathbb{C})$ with matrix multiplication is a noncommutative $\mathbb{C}$ -algebra.

Example 3.1.13

Let V be a vector space over $\mathbb{C}$. Then, $\text{End}(V)$ is a $\mathbb{C}$ -algebra, with composition as the multiplication.

Example 3.1.14

Let V be a finite dimensional vector space over $\mathbb{C}$, and let $\mathcal{B}$ be a basis for V , with $n = |\mathcal{B}|$. Then, we have an isomorphism of $\mathbb{C}$ -algebras from $\text{End}(V) \xrightarrow{\sim} M_n(\mathbb{C})$ given by $S \mapsto [S]_{\mathcal{B}}$. From previous courses, we know this is an isomorphism of matrices, and it preserves multiplication since

$$[S \circ T]_{\mathcal{B}} = [S]_{\mathcal{B}}[T]_{\mathcal{B}}.$$

Example 3.1.15

Let G be a finite group. Then, $\mathbb{C}[G]$ is an algebra, with the multiplication

$$\left(\sum_{g \in G} a_g g \right) \left(\sum_{g \in G} b_g g \right) = \sum_{h \in G} \sum_{g' \in G} a_h b_{g'} h g'.$$

Writing $h g' = g$, we have $g' = h^{-1}g$. Then, we can rewrite the above as

$$\sum_{g \in G} \sum_{h \in G} a_h b_{h^{-1}g} g.$$

Remark 3.1.16 — This also works for G infinite, so long as we specify that we consider (a_g) is finitely supported.

Example 3.1.17

Let X be a topological space; then, the continuous function $f: X \rightarrow \mathbb{C}$ form a $\mathbb{C}$ -algebra (multiplication, addition, etc. being defined pointwise). We may also let X be a smooth manifold; then, the differentiable functions $f: X \rightarrow \mathbb{C}$ form a $\mathbb{C}$ -algebra (multiplication etc defined pointwise).

Example 3.1.18

If $f_1, \dots, f_m$ are polynomials in n variables $x_1, \dots, x_n$, then the quotient ring $\mathbb{C}[x_1, \dots, x_n]/(f_1, \dots, f_m)$ has a natural $\mathbb{C}$ -algebra. This can be interpreted (by analogy with example 3.1.17) as the algebra of polynomial functions on the variety $\{(x_1, \dots, x_n) \in \mathbb{C}^n \mid f(x_1, \dots, x_n) = \dots = f_m(x_1, \dots, x_n) = 0\}$. Note that example 3.1.8 is a special case of this.

Definition 3.1.19. Let A be an algebra; then we can define another algebra called the **opposite algebra**, denoted A^{op} , on the same underlying vector space and same identity, but by reversing the order of multiplication: $m_{A^{\text{op}}}(a, b) := m_A(b, a)$.

Example 3.1.20

$\iota: \mathbb{C}[G] \xrightarrow{\sim} \mathbb{C}[G]^{\text{op}}$, given by

$$\iota \left(\sum_{g \in G} a_g g \right) := \sum_{g \in G} a_g g^{-1}.$$

This is an isomorphism of algebras.

Remark 3.1.21 — In the above, ι is not inverting all elements of $\mathbb{C}[G]$. As a trivial example, 0 has no inverse: $0 \cdot v = 0 \neq 1$ for all $v \in \mathbb{C}[G]$ (and indeed for any

algebra). For a less trivial example, consider $e := \frac{1}{|G|} \sum_{g \in G} g$, which has the property that $e\iota(e) = e^2 = e \neq 1$ if $G \neq \{1\}$.

Example 3.1.22

We have $\iota: M_n(\mathbb{C})^{\text{op}} \xrightarrow{\sim} M_n(\mathbb{C})$ given by $\iota(M) := M^T$.

Example 3.1.23

For a finite dimensional vector space V , $\iota: \text{End}(V)^{\text{op}} \xrightarrow{\sim} \text{End}(V^*)$.

§3.2 Modules

Definition 3.2.1. A **left module** over an algebra A is a pair (V, ρ_V) where V is a vector space and $\rho_V: A \rightarrow \text{End}(V)$ is an algebra homomorphism.

Definition 3.2.2. A **right module** over an algebra A is a left module over A^{op} .

Remark 3.2.3 — We will usually just deal with left modules.

Remark 3.2.4 — While this new definition may be phrased differently to the characterization of R -modules (for a ring R), we can see that this is actually equivalently if we take the definition of an R -module to be an abelian group equipped with a ring homomorphism $\phi: R \rightarrow \text{End}_{\mathbb{Z}}(M)$.

Definition 3.2.5. A **homomorphism of A -modules** or an **A -linear map** $T: V \rightarrow W$ is a linear map compatible with the action, in the sense that $\forall a \in A, T \circ \rho_V(a) = \rho_W(a) \circ T$.

Notation 3.2.6. $\text{Hom}_A(V, W)$ is the vector space of A -module homomorphisms from V to W . Likewise, $\text{End}_A(V)$ is the vector space of A -module homomorphisms from V to V .

Remark 3.2.7 — Here is another point of view on the left A -module definition: a map $A \times V \rightarrow V$ given by $(a, v) \mapsto \rho_V(a)(v)$. Likewise, for the right module definition: a map $W \times A \rightarrow W$ given by $(w, a) \mapsto \rho_W(a)(w)$.

Definition 3.2.8. A **submodule** W of a module V is a linear subspace of V , such that $\forall a \in A, \rho_V(a)W \subseteq W$. A submodule is **simple** or **irreducible**, if it has no proper nonzero submodule. There is also a natural **quotient construction** V/W for W a submodule defined by $\rho_{V/W}(a)(v + W) = \rho_V(a)(v) + W$.

Proposition 3.2.9

Let $(V, \rho), (V', \rho')$ be left modules over an algebra A and let $T: (V, \rho) \rightarrow (V', \rho')$ be A -linear. Then, $\ker(T) \subseteq (V, \rho)$, $\text{Im}(T) \subseteq (V', \rho')$ are submodules. Further, the first isomorphism theorem holds: $\text{Im}(T) \cong V / \ker(T)$.

Proof. If $v \in \ker(T)$, then $T_v = 0$, so for $a \in A$,

$$T(\rho(a)v) = \rho'(a)(T_v) = \rho'(a)(0) = 0,$$

so $\rho(a)v \in \ker(T)$. For the second, if $v \in V$ and $a \in A$ with $\rho'(a)T(v) = T(\rho(a)v) \in \text{Im}(T)$, so $\rho'(a)(\text{Im}(T)) \subseteq \text{Im}(T)$. $\square$

Remark 3.2.10 — In many cases, we won't prove the result for modules since it will be analogous to the case of representations, as can be seen above.

Lemma 3.2.11 (Schur's Lemma)

Let (V, ρ) and (W, ρ_W) be two A -modules be simple (corresponding definition for representations). Then,

1. Given $T: (V, \rho_V) \rightarrow (W, \rho_W)$ is A -linear, either T is an isomorphism or T is 0.
2. Suppose V is finite-dimensional. Given $T: (V, \rho_V) \rightarrow (V, \rho_V)$ is A -linear, $\exists \lambda \in \mathbb{C}$ such that $T = \lambda \cdot id_V$.

Remark 3.2.12 — For the proof, refer to lemma 1.8.1.

Remark 3.2.13 — To say that an A -module is simple is the analogy of irreducibility for representations.

Definition 3.2.14. The **(external) direct sum** of A -modules $V_1, \dots, V_n$ is the vector space direct sum $\bigoplus_{i=1}^n V_i$, equipped with the A -module structure

$$a(v_1, \dots, v_n) = (av_1, \dots, av_n).$$

Remark 3.2.15 — Note that in the above, we really mean

$$\rho(a)(v_1, \dots, v_n) = (\rho(a)v_1, \dots, \rho(a)v_n)$$

Definition 3.2.16. An A -module is **decomposable** if it is isomorphic to a direct sum of two nonzero modules and otherwise, it is **indecomposable**.

Definition 3.2.17. An A -module is **semisimple** if it is isomorphic to a direct sum of simple modules.

Example 3.2.18

Let $A = M_n(\mathbb{C})$. Then, $V = \mathbb{C}^n$ is a left module, with $\rho: M_n(\mathbb{C}) \rightarrow \text{End}(\mathbb{C}^n)$ as the induced action of $M_n(\mathbb{C})$ on $\mathbb{C}^n$: $\rho(M)v = Mv$. This is just multiplication of a vector by a matrix.

$\mathbb{C}^n$, considered as a $M_n(\mathbb{C})$ -module, is simple. Indeed, let $W \subseteq \mathbb{C}^n$ be a nonzero submodule; then, $\exists v \in W$ such that $v \neq 0$. For any $v' \in \mathbb{C}^n$, $\exists M \in M_n(\mathbb{C})$ such that $Mv = v'$. Therefore, $\mathbb{C}^n \subseteq M_n(\mathbb{C})W \subseteq W$. Therefore, $W = \mathbb{C}^n$.

Here is now the basis-free version of the example above.

Example 3.2.19

Let V be a vector space and $A = \text{End}(V)$. Then, we have $\rho: \text{End}(V) \rightarrow \text{End}(V)$ (the identity map) giving a linear action of $\text{End}(V)$ on V , i.e., a map

$$\begin{aligned} \text{End}(V) \times V &\rightarrow V \\ (T, v) &\mapsto Tv \end{aligned}$$

This is ‘isomorphic’ to the previous example (after defining a notion of isomorphism).

Example 3.2.20

Let G be a group and V be a representation of G . Then we get a left $\mathbb{C}[G]$ -module $(V, \tilde{\rho}_V)$, by extending ρ_V linearly. That is,

$$\tilde{\rho}_V \left(\sum_{g \in G} a_g g \right) = \sum_{g \in G} a_g \rho_V(g).$$

Conversely, given $(V, \tilde{\rho}_V)$ a left $\mathbb{C}[G]$ -module, define a representation of G by restriction of $\mathbb{C}[G]$ to G . We now need to check whether we can find an inverse or not. However, this is well-defined, since

$$\tilde{\rho}_V(gg^{-1}) = \tilde{\rho}_V(g)\tilde{\rho}_V(g^{-1}),$$

so $\tilde{\rho}_V(g)$ is invertible, so $\tilde{\rho}_V(g) \in GL(V)$.

Remark 3.2.21 — Informally speaking, this tells us that representations of groups are equivalent to left modules of their group algebras.

Example 3.2.22

Let $A = \mathbb{C}[\epsilon]/(\epsilon^2)$ (from example 3.1.8). a typical element of this is $a + b\epsilon$ (for $a, b \in \mathbb{C}$) and multiplication is defined as

$$(a + b\epsilon)(c + d\epsilon) = ac + (ad + bc)\epsilon.$$

This is a two dimensional algebra over $\mathbb{C}$. Let us consider A itself as an A -module. We shall analyze the proper nonzero submodules of A . This is necessarily one dimensional, which we'll call $\mathbb{C}(a + b\epsilon)$ for $a, b \in \mathbb{C}$. For this to be a submodule, we need that for all $x \in \mathbb{C}$, there exist a $\lambda \in \mathbb{C}$ such that

$$(\epsilon + x)(a + b\epsilon) = \lambda(a + b\epsilon).$$

The LHS is equal to $ax + (a + bx)\epsilon$, so we must have

$$ax + (a + bx)\epsilon = \lambda a + (\lambda b)\epsilon.$$

Thus, $ax = a\lambda$ and $a + bx = \lambda b$. We have that $a = 0$ or $\lambda = x$, but then $a + bx = bx$, so $a = 0$ in either case. Thus, $\mathbb{C}\epsilon = \{b\epsilon \mid b \in \mathbb{C}\}$ is the only proper nonzero submodule. Therefore, $A = \mathbb{C}[\epsilon]/(\epsilon^2)$ is not simple and it is indecomposable (since we need two different submodules).

Remark 3.2.23 — Viewing an algebra as a module is a construction we will see again.

Example 3.2.24

Let (V, ρ_V) be a left A -module and (W, ρ_W) be a left B -module. Then, $V \oplus W$ is a left $A \oplus B$ -module with $\rho_{V \oplus W}(a, b) := \rho_V(a) + \rho_W(b)$.

Exercise 3.2.25. Every left module U over $A \oplus B$ is of the form from example 3.2.24, for unique $V, W \subseteq U$. In particular, $V := (1, 0)U$ and $W := (0, 1)U$, which gives $U \cong V \oplus W$.

§3.3 Semisimple Algebras

Definition 3.3.1. Let V be a left A -module and $U \subseteq V$ be a submodule. We say $W \subseteq V$ is **complementary** to U if $V = U \oplus W$.

Proposition 3.3.2

Let A be an algebra and let $V_1, \dots, V_m$ be simple left A -modules, and let $V := V_1 \oplus \dots \oplus V_m$. Then, for any $U \subseteq V$ a submodule, there exists a set $I \subseteq \{1, \dots, m\}$ such that (considering V_i inside V),

$$V_I := \bigoplus_{i \in I} V_i$$

is complementary to U .

Proof. Let $I \subseteq \{1, \dots, m\}$ be a set which is maximal with $U \cap V_I = (0)$. We claim that $U + V_I = V$. If not, then there exists an i such that $V_i \not\subseteq U + V_I$. Then, since V_i is simple, $V_i \cap (U + V_I) = (0)$. However, that also means $U \cap (V_i + V_I) = (0)$, contradicting the maximality of our choice of I . $\square$

Corollary 3.3.3

Every submodule and quotient module of a semisimple left module is also semisimple. In the notation of the proposition, they are all isomorphic to V_I for some I .

Proof. Let us use the notation of the proposition. We need to show that U and V/U are semisimple. The composition $V_I \rightarrow V \rightarrow V/U$ is an isomorphism, since $V = V_I \oplus U$. But V_I is semisimple. Hence V/U is semisimple. By the same reasoning, $U \rightarrow V \rightarrow V/V_I$ is an isomorphism. By the first statement, V/V_I is semisimple, and isomorphic to V_J for some J (we can take $J = I^c$, the complement of I). $\square$

Remark 3.3.4 — In fact, the proof shows that if $V \cong \bigoplus_{i=1}^m V_i$, then quotient modules and submodules of V are all isomorphic to V_I for some $I \subseteq \{1, 2, \dots, m\}$.

Using this, we can strengthen our old uniqueness result for decompositions into nonisomorphic irreducible subrepresentations:

Corollary 3.3.5

Suppose that $V = V_1 \oplus \dots \oplus V_n$ for V_i simple submodules which are pairwise nonisomorphic. Then every submodule of V is equal to $V_{i_1} \oplus \dots \oplus V_{i_m}$ for some $1 \leq i_1 \leq \dots \leq i_m \leq n$.

Proof. Let $W \subseteq V$ be a submodule. By the preceding corollary, there exists $I \subseteq \{1, \dots, m\}$ such that $V_I \subseteq W$. Now consider the injective homomorphism $V_I \cong W \hookrightarrow V = V_1 \oplus \dots \oplus V_m$, which we'll call $\Psi = (\psi_{ij})_{i \in I, j \in \{1, \dots, m\}}$. By Schur's lemma, $\psi_{ij} = 0$ for $i \neq j$ and $\psi_{ij} = \lambda \cdot \text{Id}_{V_i}$. Since Ψ is injective, for all $i \in I$, $\lambda_i \neq 0$. Therefore,

$$W = \Psi(V_I) = V_I \subseteq V.$$

$\square$

Lemma 3.3.6

Let V be an n -dimensional left module over an algebra A (with n finite). Then V is a quotient module of A^n .

Proof. Let $v_1, \dots, v_n$ be a basis. Then we can define homomorphism of A -modules $A^n \rightarrow V$ by $(a_1, \dots, a_n) \mapsto \sum_{i=1}^n \rho_V(a_i)v_i$, where each $a_i \in A$. We now show that this is surjective. if $v \in V$, we may write it as $\sum c_i v_i$ for $c_i \in \mathbb{C}$. Then,

$$T(c_1, \dots, c_n) = T(c_1 \cdot 1_A, \dots, c_n \cdot 1_A) = \sum \rho_V(c_i \cdot 1)v_i = \sum c_i v_i.$$

Note here that as $c_i \in \mathbb{C}$, $\rho_V(c_i 1)v_i = c_i \rho_V(1)v_i = c_i v_i$. Therefore, $V = \text{Im}(T) \cong A^n / \ker(T)$. $\square$

Remark 3.3.7 — Note that (from problem set 1), if G is a finite group and V is an irreducible representation of G , then V is isomorphic to a quotient of $\mathbb{C}[G]$.

Proof. Given $v \in V$, we have a homomorphism $T_v: \mathbb{C}[G] \rightarrow V$. Choosing $v \neq 0$, T_v is nonzero. Thus, since V is irreducible, $\text{Im}(T_v) = V$. This implies that $V \cong \text{Im}(v) \cong \mathbb{C}[G]/\ker(T_v)$. $\square$

Remark 3.3.8 — Note that this argument is similar to the method used in the lemma above.

Lemma 3.3.9

All left simple left modules of A are quotients A/V , for some submodule $V \subseteq A$.

Proof. Let W be a simple left A -module, and let $0 \neq w \in W$. Define $T: A \rightarrow W$, with $T(a) := aw$. Then, T is a homomorphism of A -modules, and $w \in \text{Im}(T)$, so $\text{Im}(T) \neq (0)$. Since W is simple, $\text{Im}(T) = W$, we get that $W = \text{Im}(T) \cong A/\ker(T)$. $\square$

Theorem 3.3.10 (Semisimple Conditions)

The following are equivalent. For a finite dimensional algebra A ,

1. A is semisimple as a left module over itself.
2. Every finite dimensional left A -module is semisimple.

Proof. Note that $\Leftarrow$ is obvious, so we work the direction $\Rightarrow$. Let V be a finite dimensional left A -module. Then, by lemma 3.3.6, there exists an n such that V is isomorphic to a quotient of A^n . By assumption, A is semisimple, so A^n is, by corollary 3.3.3, every quotient of A^n is semisimple. $\square$

Definition 3.3.11. A finite dimensional algebra A is **semisimple**, if it semisimple as a module over itself.

Remark 3.3.12 — Note that by theorem 3.3.10, then every finite dimensional left A -module is semisimple.

Lemma 3.3.13

The matrix algebra $M_n(\mathbb{C})$, as a module over itself, is isomorphic to $(\mathbb{C}^n)^n$.

Proof. Define $V_i \subseteq M_n(\mathbb{C})$ to be the $n \times n$ matrices which are zero outside the i -th column. Then, the $(V_i)_{1 \leq i \leq n}$ are submodules and are each isomorphic (as modules) to $\mathbb{C}^n$. Furthermore, $M_n(\mathbb{C}) = \oplus_{i=1}^n V_i \cong (\mathbb{C}^n)^n$. $\square$

Remark 3.3.14 — This is using the structure of $\mathbb{C}^n$ as a left $M_n(\mathbb{C})$ -module introduced in example 3.2.18.

Theorem 3.3.15

Every finite dimensional left module over $M_n(\mathbb{C})$ is isomorphic to $(\mathbb{C}^n)^r$, for some r .

Proof. Let V be a finite-dimensional left module over $A = M_n(\mathbb{C})$. By lemma 3.3.6, V is a quotient of $M_n(\mathbb{C})^k$, for some k . By lemma 3.3.13, $M_n(\mathbb{C})^k \cong (\mathbb{C}^n)^{nk}$. By the remark after corollary 3.3.3, a quotient of $(\mathbb{C}^n)^{nk}$ is isomorphic to $(\mathbb{C}^n)^r$, for some r . $\square$

Lemma 3.3.16

Let A be an algebra. Then, there is an isomorphism of algebras $\iota: \text{End}_A(A) \cong A^{\text{op}}$ defined by

$$\iota(\phi) := \phi(1_A).$$

Proof. Every $\phi \in \text{End}_A(A)$ satisfies $\phi(a) = \phi(a \cdot 1_A) = a\phi(1_A)$ for all $a \in A$ and hence ϕ is uniquely determined by $\phi(1_A)$; conversely, the formula $a \mapsto av$ defines a valid endomorphism of A , for any a . Therefore, ι is an isomorphism of vector spaces. Now, for $\phi, \psi \in \text{End}_A(A)$, we have

$$\begin{aligned} \iota(\phi \circ \psi) &= (\phi \circ \psi)(1_A) = \phi(\psi(1_A)) = \phi(\psi(1_A)1_A) = \psi(1_A)\phi(1_A) \\ &= \iota(\psi)\iota(\phi) = m_{A^{\text{op}}}(\iota(\phi), \iota(\psi)). \end{aligned}$$

$\square$

Remark 3.3.17 — $\text{End}_A(A)$ is the homomorphisms of A -modules to itself.

Remark 3.3.18 — Note that Schur's lemma says that for V a simple left A -module, $\text{End}_A(V) \cong \mathbb{C}$. Therefore, if an algebra A is a simple module over itself, then $\mathbb{C} \cong \text{End}_A(A) \cong A^{\text{op}}$, so $A \cong \mathbb{C}$ (the trivial algebra).

Lemma 3.3.19

For left A -modules, (V_1, ρ_1) , (V_2, ρ_2) , (W, ρ_2) , we have

$$\begin{aligned} \text{Hom}_A(V_1 \oplus V_2, W) &\cong \text{Hom}_A(V_1, W) \oplus \text{Hom}_A(V_2, W) \\ \text{Hom}_A(W, V_1 \oplus V_2) &\cong \text{Hom}_A(W, V_1) \oplus \text{Hom}_A(W, V_2). \end{aligned}$$

Proof. We omit the proof. $\square$

Theorem 3.3.20 (Artin-Wedderburn)

Let A be a finite dimensional algebra. Then, the following are equivalent.

1. A is semisimple.
2. A is a direct sum of matrix algebras, $A \cong \bigoplus_{i=1}^m M_{n_i}(\mathbb{C})$ for some $n_1, n_2, \dots, n_m$.
3. There exist a full set $V_1, \dots, V_m$ of simple left A -modules, such that there is an isomorphism

$$\Phi: A \xrightarrow{\sim} \bigoplus_{i=1}^m \text{End}_A(V_i).$$

Proof. (3 $\implies$ 2): clear.

(2 $\implies$ 1): clear via lemma 3.3.13.

(1 $\implies$ 2): Suppose that A is semisimple. Then, $A \cong \bigoplus_{i=1}^m V_i^{r_i}$ as an A -module, where $V_1, \dots, V_m$ are non-isomorphic simple A -modules. Consider $\text{End}_A(\cdot)$ of both sides of $\text{End}_A(A) \cong \text{End}_A\left(\bigoplus_{i=1}^m V_i^{r_i}\right)$. The LHS is isomorphic to A^{op} by lemma 3.3.16. In the RHS, lemma 3.3.19 tells us that

$$\begin{aligned} \text{End}_A\left(\bigoplus_{i=1}^m V_i^{r_i}\right) &\cong \bigoplus_{i=1}^m \bigoplus_{j=1}^m \text{Hom}_A(V_i^{r_i}, V_j^{r_j}) \quad (\text{lemma 3.3.19}) \\ &\cong \bigoplus_{i=1}^m \text{End}_A(V_i^{r_i}) \quad (\text{Schur's lemma}) \\ &\cong \bigoplus_{i=1}^m M_{r_i}(\mathbb{C}) \quad (\text{Schur's lemma}) \end{aligned}$$

Therefore, $A \cong \bigoplus_{i=1}^m M_{r_i}(\mathbb{C})^{\text{op}} \cong \bigoplus_{i=1}^m M_{r_i}(\mathbb{C})$.

(2 $\implies$ 3): Suppose A is $\bigoplus_{i=1}^m M_{n_i}(\mathbb{C})$. By exercise 3.2.25, a module V over $\bigoplus_{i=1}^m M_{n_i}(\mathbb{C})$ is a direct sum of V_i s, where each V_i is a $M_{n_i}(\mathbb{C})$ -module. If V is simple, then this ‘direct sum’ has all components zero except one (since simple modules have no submodules). If V_i is the nonzero component, then V_i is a simple $M_{n_i}(\mathbb{C})$ -module; thus, $V_i \cong \mathbb{C}^{n_i}$. Then, we have $\text{End}_A(V_i) = \text{End}(\mathbb{C}^{n_i}) = M_{n_i}(\mathbb{C})$. $\square$

Corollary 3.3.21

Let A be a finite dimensional semisimple algebra, and let $V_1, \dots, V_m$ be a full set of its simple left modules. Then,

$$\dim A = \sum_{i=1}^m \dim \text{End}_A(V_i) = \sum_{i=1}^m (\dim V_i)^2.$$

Example 3.3.22

Let G be a finite group. From theorem 1.15.2, we have an isomorphism of vector spaces

$$\Phi: \mathbb{C}[G] \xrightarrow{\sim} \bigoplus_{i=1}^m \text{End}(V_i),$$

where $V_1, \dots, V_m$ are the simple left $\mathbb{C}[G]$ -modules. In fact, this is an isomorphism of algebras. This shows that $\mathbb{C}[G]$ satisfies the equivalent conditions of the Artin Wedderburn theorem.

§3.4 Character Theory

Definition 3.4.1. Let (V, ρ_V) be a finite dimensional left module of an algebra A . The **character** $\chi_A: A \rightarrow \mathbb{C}$ is $\chi_V(a) = \text{Tr}(\rho_V(a))$, for $a \in A$.

Definition 3.4.2. Define $A_{\text{Tr}}^* = \{f: A \rightarrow \mathbb{C} \mid f \text{ is linear and for all } a, b \in A, f(ab - ba) = 0\}$, the **Hochschild traces** of A .

Remark 3.4.3 — We note that for all $a, b \in A$, $\text{Tr}(\rho_V(a)\rho_V(b)) = \text{Tr}(\rho_V(b)\rho_V(a))$. Therefore, $\text{Tr}(\rho_V(ab)) = \text{Tr}(\rho_V(ba))$, and so $\chi_V(ab) = \chi_V(ba)$. Thus, for all (V, ρ) , $\chi_V \in A_{\text{Tr}}^*$. This is the analogue for representations that a character is a class function.

Example 3.4.4

Let $A = \mathbb{C}[G]$. Then,

$$\begin{aligned} A_{\text{Tr}}^* &= \{f: \mathbb{C}[G] \rightarrow \mathbb{C} \mid f \text{ linear and for all } a, b \in \mathbb{C}[G], f(ab) = f(ba)\} \\ &= \{f: G \rightarrow \mathbb{C} \mid \forall a, b \in G, f(ab) = f(ba)\} \\ &= \{\text{class functions on } G\} \end{aligned}$$

Note that from the second to third line, we've made the substitution ab^{-1} for a .

Exercise 3.4.5. For $A = M_n(\mathbb{C})$, $A_{\text{Tr}}^* = \mathbb{C} \cdot \text{Tr}$, i.e., the only $f: M_n(\mathbb{C}) \rightarrow \mathbb{C}$ linear such that $\forall a, b \in M_n(\mathbb{C})$, $f(ab) = f(ba)$ are multiple of the trace.

Proposition 3.4.6

$V \cong W$ implies $\chi_V = \chi_W$.

Proof. It is a consequence of the fact that trace is invariant under conjugation (by an invertible transformation). Let $T: V \xrightarrow{\sim} W$. Then, $\forall a \in A$, $\rho_W(a) \circ T = T \circ \rho_V(a)$, so

$$\begin{aligned} \text{Tr} \rho_W(a) &= \text{Tr}(T^{-1} \circ \rho_V(a) \circ T) \\ &= \text{Tr}(T^{-1} \circ T \circ \rho_V(a)) \\ &= \text{Tr}(\rho_V(a)). \end{aligned}$$

□

Theorem 3.4.7

Let A be a finite dimensional semisimple algebra and $(V_1, \rho_1), \dots, (V_m, \rho_m)$ be a full set of simple A -modules. Then, the (χ_{V_i}) are linearly independent.

Proof. WLOG, by Artin-Wedderburn, we write $A = \bigoplus_{i=1}^m M_{n_i}(\mathbb{C})$, for some $n_1, \dots, n_m$. The simple left A -modules are $(V_i)_{1 \leq i \leq m}$, where $V_i = \mathbb{C}^{n_i}$, and $\rho_{V_i}(a_1, \dots, a_m)v = a_i v$, where $a_i \in M_{n_i}(\mathbb{C})$ (note that this is a side effect of the Artin-Wedderburn theorem). The character χ_{V_i} has

$$\chi_{V_i}(a_1, \dots, a_m) = \text{Tr}(a_i).$$

Let $\lambda_1, \dots, \lambda_m \in \mathbb{C}$ and suppose that

$$0 = \lambda_1 \chi_{V_1} + \dots + \lambda_m \chi_{V_m}.$$

We want to show that $\lambda_i = 0$, for all i . For each i , we have

$$\begin{aligned} 0 &= 0(0, \dots, 0, id_{V_i}, 0, \dots, 0) = (\lambda_1 \chi_{V_1} + \dots + \lambda_m \chi_{V_m})(0, \dots, 0, \underbrace{I_{n_i}}_{i\text{-th}}, 0, \dots, 0) \\ &= 0 + \dots + 0 + \text{Tr}(I_{n_i}) + 0 + \dots + 0 \\ &= \lambda_i \cdot n_i. \end{aligned}$$

This implies $\lambda_i = 0$, and so $(\chi_{V_1}, \dots, \chi_{V_m})$ are linearly interdependent. $\square$

Theorem 3.4.8

Let A be a finite dimensional semisimple algebra, and let V, W be finite dimensional left A -module with $\chi_V = \chi_W$. Then, $V \cong W$.

Proof. Let $(V_1, \rho_1), \dots, (V_m, \rho_m)$ be a full set of simple left A -modules. Since A is semisimple, there exists $r_1, \dots, r_m$ and $s_1, \dots, s_m$ such that

$$V \cong \bigoplus_{i=1}^m V_i^{r_i}, \quad W \cong \bigoplus_{i=1}^m V_i^{s_i}.$$

We have

$$\sum_{i=1}^m r_i \chi_{V_i} = \chi_V = \chi_W = \sum_{i=1}^m s_i \chi_{V_i}.$$

By theorem 3.4.7, the (χ_{V_i}) are linearly independent, so for all i , $r_i = s_i$. Therefore,

$$V \cong \bigoplus_{i=1}^m V_i^{r_i} = \bigoplus_{i=1}^m V_i^{s_i} \cong W.$$

$\square$

Theorem 3.4.9

Let A be a finite dimensional semisimple algebra and let $(V_1, \rho_1), \dots, (V_m, \rho_m)$ be a full set of simple left A -modules. Then, $\{\chi_{V_1}, \dots, \chi_{V_m}\}$ forms a basis for A_{Tr}^* .

Proof. By remark 3.4.3, for all i , $\chi_{V_i} \in A_{\text{Tr}}^*$ and by theorem 3.4.7, $\{\chi_{V_1}, \dots, \chi_{V_m}\}$ are linearly independent. We just need to show this set spans A_{Tr}^* . As in the proof of theorem 3.4.7, WLOG $A = \bigoplus_{i=1}^m M_{n_i}(\mathbb{C})$. In this case,

$$\left(\bigoplus_{i=1}^m M_{n_i}(\mathbb{C}) \right)_{\text{Tr}}^* = \left\{ \sum_i \lambda_i t_i \mid \lambda_1, \dots, \lambda_m \in \mathbb{C} \right\},$$

where $t_i: \bigoplus_{i=1}^m M_{n_i}(\mathbb{C}) \rightarrow \mathbb{C}$ is defined by $t_i(a_1, \dots, a_m) = \text{Tr}(a_i)$. In particular, $\left(\bigoplus_{i=1}^m M_{n_i}(\mathbb{C}) \right)_{\text{Tr}}^*$ has dimension m , so (by counting dimensions) $\{\chi_{V_1}, \dots, \chi_{V_m}\}$ is a spanning set. $\square$

Remark 3.4.10 — For a non-semisimple algebra, it is no longer true that the necessarily linearly independent characters of simple left A -modules form a basis for A_{Tr}^* . Indeed, suppose that A is commutative: $ab = ba$ for all $a, b \in A$. Then, $A_{\text{Tr}}^* = A^*$. However, in general there are fewer than $\dim A^*$ simple left A -modules. For instance, consider $A = \mathbb{C}[\epsilon]/(\epsilon^2)$. There is only one simple left module up to isomorphism, $\mathbb{C} \cdot \epsilon$, but $\dim A = 2 > 1$.

Remark 3.4.11 — We cannot form a character table for a general algebra (even if it is finite-dimensional and semisimple algebra) since we don’t have a collection of elements to take traces of. Indeed, for many finite groups G, H , the character tables can be different (so $G \not\cong H$), but nonetheless there exists an isomorphism $\phi: \mathbb{C}[G] \cong \mathbb{C}[H]$ that doesn’t send G to H . Through ϕ , the characters $\mathbb{C}[G] \rightarrow \mathbb{C}$, $\mathbb{C}[H] \rightarrow \mathbb{C}$ of irreducible representations of G are the same, but since ϕ doesn’t send G to H this is no contradiction with the character tables being different.

Remark 3.4.12 — As in proposition 2.3.1, we can apply the same proof to say that for V, W finite dimensional left modules over a general algebra A ,

$$\chi_{V \oplus W} = \chi_V + \chi_W.$$

However, we can’t quite do the same thing for V^* and $V \otimes W$, since these are *not* in general left modules over A when V and W are.

V^* is a right module over A , given by dualizing endomorphisms: the map $\rho_{V^*}: A^{\text{op}} \rightarrow \text{End}(V^*)$ is defined by $\rho_{V^*}(a) = \rho_V(a)^*$. This is an algebra homomorphism since, as in example 3.1.23, the dualization map $\text{End}(V) \rightarrow \text{End}(V^*)$ is anti-multiplicative; that is, $(S \circ T)^* = T^* \circ S^*$. In other words, we have an algebra isomorphism $\text{End}(V)^{\text{op}} \xrightarrow{\sim} \text{End}(V^*)$, and therefore we get an algebra homomorphism $A^{\text{op}} \rightarrow \text{End}(V)^{\text{op}} \rightarrow \text{End}(V^*)$. In any case, we attain $\text{Tr}(S^*) = \text{Tr}(S)$, so if we think of V^* as a right module over A , then we attain

$$\chi_{V^*} = \chi_V,$$

rather than getting complex conjugates as we did in the group case. The reason why we had complex conjugates in the case for groups is because we also used the inversion map $G^{\text{op}} \xrightarrow{\sim} G$, which induced $\mathbb{C}[G]^{\text{op}} \xrightarrow{\sim} \mathbb{C}[G]$, to turn V^* back into a left

module over $\mathbb{C}[G]$.

As for $V \otimes W$, there is no way to make this into any sort of module over A in general. However, if (V, ρ_V) is a module over A and (W, ρ_W) is a module over B , then $V \boxtimes W$ is a module over $A \otimes B$ with

$$\rho_{V \boxtimes W}(a \otimes b) = \tilde{\rho}_V(a) \otimes \tilde{\rho}_W(b).$$

Then, as before, $\chi_{V \boxtimes W}(a \otimes b) = \chi_V(a)\chi_W(b)$.

Remark 3.4.13 — If V is a finite-dimensional left module and W is a submodule, then we can take a basis $\mathcal{C}$ of W and extend it to a basis $\mathcal{B}$ of V . In this basis, $\rho_V^{\mathcal{B}}(a)$ is a block upper-triangular of the form:

$$\begin{pmatrix} \rho_W^{\mathcal{C}}(a) & * \\ 0 & \rho_{V/W}^{\mathcal{B} \setminus \mathcal{C}}(a) \end{pmatrix},$$

where we view $\mathcal{B} \setminus \mathcal{C}$ as a basis for V/W by adding W . As a consequence, we can deduce the more general identity

$$\chi_V = \chi_W + \chi_{V/W}.$$

The same identity is valid for group representations (by the same argument, or by letting the algebra be the group algebra).